

# ANALYTICITY OF ALTERNATING-MAP FUNCTORS AND VECTOR BUNDLES

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**ABSTRACT.** The alternating-map bifunctor on normed spaces over a nontrivially normed field  $K$  is always  $C^\infty$ , and its maps on morphism spaces are analytic in degree  $k$  exactly when  $k! \neq 0$  in  $K$ . This criterion remains sharp for Banach spaces. In positive characteristic, spherical completeness of the target gives an analytic restriction in every degree. Nevertheless, for  $K = \mathbb{F}_p(t)$  and  $k \geq p$  there is no largest full analytic subcategory: two individually analytic objects can have a nonanalytic cross-action, even when all spaces have finite algebraic dimension and nonarchimedean norms.

A different picture emerges when regularity is tested on families of morphisms. Every analytic family parametrized by an open subset of  $K^d$  induces an analytic alternating-map family, in every characteristic and degree, without completeness assumptions on the fibers. Consequently alternating maps preserve analytic normed vector bundles and their operator-valued analytic morphisms over analytic manifolds with open  $K^d$  charts. We also prove positive results for  $c_0$  and ordinary  $\ell^1$  parameter spaces, explain the obstruction for invertible families over general Banach bases, and characterize universal analytic reflection through closed Banach subspaces by projections onto diagonal tensor powers with exponential norm bounds. The sharp Banach counterexample and the principal operator positive results are formalized in Lean with Mathlib; the verification scope of the subsequent extensions is stated separately.

## 1. INTRODUCTION

The passage from vector spaces to alternating maps is contravariant in the source and covariant in the target. We study both its regularity on whole spaces of morphisms and its behavior on analytic families of morphisms. These are different questions in positive characteristic. Analyticity on all morphism spaces has a sharp degree restriction, whereas every analytic family with standard finite-dimensional parameters is preserved. This distinction gives a construction of alternating-map bundles over bases with open coordinate charts in  $K^d$ , despite the failure of an analytic bifunctor on the entire category.

**Categories and regularity.** Fix a nontrivially normed field  $K$ . Let  $\mathbf{Vec}_K$  be the category of normed  $K$ -spaces and bounded linear maps, and let  $\mathbf{Ban}_K$  be its full subcategory of complete spaces. When  $K$  has positive characteristic, write  $\mathbf{Vec}_K^\circ$  for the full subcategory of nonarchimedean, spherically complete spaces. The field itself need not be complete.

For an integer  $k \geq 0$ , let  $\mathbf{Alt}^k(E; F)$  be the normed space of bounded alternating  $k$ -linear maps  $E^k \rightarrow F$ . Here *alternating* means vanishing whenever two arguments coincide, including

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The original counterexample, its informal proof and the original write-up were produced with Anthropic’s Claude (in Claude Code). The subsequent categorical and analytic-family results, and the present revision, were developed with OpenAI’s Codex. Informal proofs were audited by independent AI agents. The original operator results were formalized in Lean with Mathlib using Codex and Autoform; §14 distinguishes that checked development from the later extensions. Neither the manuscript nor its formalization has yet received a human expert review.

in characteristic 2. Pullback in  $E$  and pushforward in  $F$  define the bifunctor

$$\mathrm{Alt}^k: \mathrm{Vec}_K^{\mathrm{op}} \times \mathrm{Vec}_K \longrightarrow \mathrm{Vec}_K, \quad (E, F) \longmapsto \mathrm{Alt}^k(E; F).$$

Its action on a pair of morphisms  $u: E \rightarrow E'$  and  $v: F \rightarrow F'$  is

$$\mathrm{Alt}^k(u, v): m \longmapsto v \circ m \circ (u, \dots, u).$$

The opposite category in the first variable records pullback. Write  $\mathcal{L}(E, E')$  for the hom space of bounded linear maps, equipped with the operator norm. We call the bifunctor *analytic*, or  $C^\infty$ , when every induced map on hom spaces

$$\mathcal{L}(E, E') \times \mathcal{L}(F, F') \longrightarrow \mathcal{L}(\mathrm{Alt}^k(E'; F), \mathrm{Alt}^k(E; F'))$$

has that regularity. The product carries the maximum norm. This definition concerns the joint action on both variables, rather than separate regularity. It applies to the restrictions below as well.

### Main results.

**Main Theorem.** Let  $K$  be a nontrivially normed field and let  $k \geq 0$  be an integer.

- (1) The bifunctor

$$\mathrm{Alt}^k: \mathrm{Vec}_K^{\mathrm{op}} \times \mathrm{Vec}_K \longrightarrow \mathrm{Vec}_K$$

is  $C^\infty$ . It is analytic if and only if  $k! \neq 0$  in  $K$ . Both assertions hold also for its restriction

$$\mathrm{Alt}^k: \mathrm{Ban}_K^{\mathrm{op}} \times \mathrm{Ban}_K \longrightarrow \mathrm{Ban}_K.$$

- (2) If  $K$  has positive characteristic, the restriction

$$\mathrm{Alt}^k: \mathrm{Vec}_K^{\mathrm{op}} \times \mathrm{Vec}_K^\circ \longrightarrow \mathrm{Vec}_K^\circ$$

is analytic in every degree.

- (3) With no restriction on  $K$  or  $k$ , the bifunctor  $\mathrm{Alt}^k$  preserves every analytic family of morphisms parametrized by an open subset of  $K^d$ . Consequently it induces a bifunctor on analytic normed vector bundles over any analytic manifold with open  $K^d$  charts, preserving operator-valued analytic bundle morphisms. No completeness or dimension hypothesis on the fibers is required.
- (4) For every prime  $p$  and every  $k \geq p$ , over the  $t$ -adically normed field  $K = \mathbb{F}_p(t)$  there is no largest full subcategory of  $\mathrm{Vec}_K^{\mathrm{op}} \times \mathrm{Vec}_K$  on which  $\mathrm{Alt}^k$  is analytic. This remains true for subcategories closed under isomorphisms and, within the corresponding ambient category, when all objects have finite algebraic dimension and nonarchimedean norms.

Part (1) gives the exact analytic range  $k < p$  in characteristic  $p > 0$ ; in characteristic zero, every finite degree is analytic. Part (2) removes the restriction on the degree by restricting the target category. Its codomain is part of the assertion: alternating maps into a spherically complete nonarchimedean space form a spherically complete nonarchimedean space. No completeness assumption on  $K$  is needed. In the analytic cases in parts (1) and (2), the maps on hom spaces are in fact given by finite power series. Part (3) permits arbitrary normed fibers over bases with open  $K^d$  charts, without requiring analyticity on the entire operator space. Its proof is analytic reflection through a closed linear isometry for finite-coordinate parameters (Theorem 20). Over incomplete fields, finite algebraic dimension alone does not supply continuous coordinates; this is precisely what permits the examples in part (4). The latter assertion does not decide the corresponding existence question over complete fields or among Banach pairs.

**Operator form and sharpness.** The proofs work with the action of a fixed-target functor  $\text{Alt}^k(-; F)$  on hom spaces. We now write this action explicitly and record the stronger counterexample underlying part (1) of the Main Theorem. For normed  $K$ -spaces  $E, E', F$ , the pullback action is

$$(1.1) \quad \begin{aligned} Q &= Q_{E, E', F}^K: \mathcal{L}(E, E') \longrightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)), \\ Q(f)(m) &= m \circ (f, \dots, f). \end{aligned}$$

For multilinear maps without the alternation condition, precomposition is the diagonal of a bounded multilinear map and is therefore analytic. Requiring the values to remain alternating changes the problem. The usual polarization argument divides by  $k!$  and does not apply when that scalar vanishes. We show that this obstruction is unavoidable if analyticity is required for all Banach spaces over  $K$ .

**Theorem 1** (Counterexample theorem). *Let  $K$  be a nontrivially normed field of characteristic  $p > 0$ , and let  $k \geq p$ . There are Banach spaces  $E$  and  $F$  over  $K$  such that the degree- $k$  map*

$$Q: \mathcal{L}(E, E) \rightarrow \mathcal{L}(\text{Alt}^k(E; F), \text{Alt}^k(E; F)), \quad Q(f)(m) = m \circ (f, \dots, f),$$

*is analytic at no point of  $\mathcal{L}(E, E)$ . Equivalently, there is no bounded  $k$ -linear map  $P$  on  $\mathcal{L}(E, E)$  with  $P(f, \dots, f) = Q(f)$  for all  $f$ . The Banach space  $F$  admits no equivalent nonarchimedean norm. The same  $E, F$  work for any reindexing of the  $k$  arguments.*

**Corollary 2.** *Let  $K$  be any nontrivially normed field and  $k \geq 0$  any integer. Then  $Q_{E, E', F}^K$  is analytic for all  $K$ -Banach spaces  $E, E', F$  if and only if  $k! \neq 0$  in  $K$ . More precisely:*

- (1) *if  $k! \neq 0$  in  $K$ , then  $Q_{E, E', F}^K$  is analytic, indeed given at every point by a finite power series, for all normed spaces  $E, E', F$ , complete or not;*
- (2) *if  $k! = 0$  in  $K$ , then there are Banach spaces  $E = E'$  and  $F$ , with  $F$  admitting no equivalent nonarchimedean norm, such that  $Q_{E, E', F}^K$  is analytic at no point.*

Thus the factorial condition is exact, even when the source and target of the linear maps are required to coincide. The failure is also local in the strongest sense: in the examples, analyticity fails at every base point. By the lift criterion of Proposition 8, it is enough to prove that no bounded multilinear lift exists.

Finite-dimensional source spaces give a second positive case: over a complete field,  $Q$  is analytic in every degree if either  $E$  or  $E'$  is finite-dimensional, with no restriction on  $F$  (Corollary 13). We give explicit determinant formulas for these lifts in §4. Finite-dimensional targets require a different analysis, given in Remark 81.

**From operators to functors.** The translation between the two formulations uses only bounded bilinear postcomposition. Write  $v_*a = v \circ a$ . The map  $(v, T) \mapsto v_* \circ T$  is bounded bilinear, with  $\|v_* \circ T\| \leq \|v\| \|T\|$ , and the bifunctor's joint action is

$$(1.2) \quad \text{Alt}^k(u, v) = v_* \circ Q_{E, E', F}^K(u).$$

Consequently smoothness or analyticity of every  $Q$  gives the corresponding regularity of the bifunctor. Conversely, restriction to  $v = \text{id}_F$  recovers  $Q$ . Part (1) of the Main Theorem therefore follows from Corollary 2 and the universal smoothness statement in Proposition 16. Part (2) is proved in §2. A bounded  $k$ -linear lift of  $Q$  gives a bounded  $(k+1)$ -linear lift of the joint action by placing the pushforward variable in the last slot. Thus the finite-power-series conclusion also passes from the operator statements to the bifunctor.

For completeness of the target category,  $\text{Alt}^k(E; F)$  is complete whenever  $F$  is complete: an operator-norm Cauchy sequence converges pointwise in  $F$ , and its boundedness, multilinearity

and alternation pass to the limit. Proposition 4 gives the stronger closure property required in Main Theorem (2). In degree zero,  $\text{Alt}^0(E; F)$  is naturally isometric to  $F$ ; pullback is constant and the joint action is linear in the pushforward variable.

**Motivation: vector bundles.** The categorical formulation is motivated by the construction of vector bundles; compare Lang's functors of class  $C^p$  [5, Ch. III, §4]. Let  $M$  be a  $C^n$  manifold over  $K$ , where  $n \in \mathbb{N} \cup \{\infty, \omega\}$  and  $C^\omega$  means analytic. A  $C^n$  vector bundle  $\mathcal{V} \rightarrow M$  with typical fiber  $V$  can be described by an open cover  $(U_i)$  of  $M$  and  $C^n$  transition maps

$$g_{ij}: U_i \cap U_j \longrightarrow \text{Aut}_K(V) \subseteq \text{End}_K(V).$$

Here  $\text{Aut}_K(V)$  consists of bounded linear isomorphisms with bounded inverse, and  $C^n$  regularity is measured in the operator-norm space  $\text{End}_K(V) = \mathcal{L}(V, V)$ . Our convention is that  $g_{ij}$  converts coordinates in chart  $j$  to coordinates in chart  $i$ . The transition maps satisfy

$$g_{ii}(x) = \text{id}_V, \quad g_{ij}(x) \circ g_{j\ell}(x) = g_{i\ell}(x)$$

on the relevant overlaps. Conversely, such a cocycle glues the products  $U_i \times V$  into a  $C^n$  vector bundle: the coordinate changes  $(x, v) \mapsto (x, g_{ij}(x)v)$  are  $C^n$  because operator evaluation is bounded bilinear. In particular,  $g_{ji} = g_{ij}^{-1}$ , and both directions are  $C^n$  by the cocycle data.

Now let

$$\mathcal{F}: \underbrace{\text{Vec}_K \times \cdots \times \text{Vec}_K}_{r \text{ factors}} \longrightarrow \text{Vec}_K$$

be a  $C^n$  multifunctor: its induced maps on products of hom spaces are jointly  $C^n$ . Given  $C^n$  vector bundles  $\mathcal{V}_1, \dots, \mathcal{V}_r$  over  $M$ , choose a common trivializing cover, write  $V_a$  for their typical fibers, and let  $g_{ij}^a$  be their transition maps. Set  $\mathbf{g}_{ij} = (g_{ij}^1, \dots, g_{ij}^r)$  and  $W = \mathcal{F}(V_1, \dots, V_r)$ . The induced transition maps  $h_{ij}(x) = \mathcal{F}(\mathbf{g}_{ij}(x))$  are  $C^n$  as maps into  $\text{End}_K(W)$ . Functoriality gives

$$\begin{aligned} h_{ij}(x) \circ h_{j\ell}(x) &= \mathcal{F}(\mathbf{g}_{ij}(x) \circ \mathbf{g}_{j\ell}(x)) \\ &= \mathcal{F}(\mathbf{g}_{i\ell}(x)) = h_{i\ell}(x), \quad h_{ii}(x) = \text{id}_W. \end{aligned}$$

Thus the  $h_{ij}$  form a  $C^n$  cocycle and define a vector bundle  $\mathcal{F}(\mathcal{V}_1, \dots, \mathcal{V}_r)$  with fiber  $\mathcal{F}((\mathcal{V}_1)_x, \dots, (\mathcal{V}_r)_x)$  over  $x$ . Applying  $\mathcal{F}$  to changes of trivialization identifies the resulting bundles, so the construction is independent of the chosen cover and trivializations. In particular, one may apply a multifunctor repeatedly to the same bundle. Functoriality supplies the cocycle identity, while regularity on hom spaces supplies the regularity of the new bundle.

The same construction works for mixed variance, using  $g_{ji}$  in each contravariant argument. These inverse transitions are already  $C^n$ , so no separate regularity theorem for inversion is needed. For example, if  $\mathcal{E}, \mathcal{H} \rightarrow M$  have typical fibers  $E, F$  and transition maps  $g_{ij}, h_{ij}$ , then the bundle with fibers  $\text{Alt}^k(\mathcal{E}_x; \mathcal{H}_x)$  has transition maps

$$a \longmapsto h_{ij}(x) \circ a \circ (g_{ji}(x), \dots, g_{ji}(x)).$$

The Main Theorem therefore constructs this as a  $C^n$  vector bundle for every  $n \leq \infty$  whenever the input bundles are  $C^n$ . For analytic input bundles, it gives an analytic bundle when  $k! \neq 0$  in  $K$ , and in every degree when  $K$  is nonarchimedean and the typical target fiber  $F$  is nonarchimedean and spherically complete. Part (3) supplies a further conclusion independent of these degree and fiber conditions: if  $M$  has open  $K^d$  charts, the alternating-map bundle is analytic for all analytic input bundles, in every characteristic and degree. The proof applies to transition families themselves rather than to the entire operator space. It also applies to bundle morphisms, so this is a functorial construction independent of the chosen analytic atlas (Theorem 25). Taking  $\mathcal{E} = TM$  and  $\mathcal{H} = M \times K$  recovers analytic differential  $k$ -forms in this setting. The additional content is that the input bundles may have infinite-dimensional fibers.

The parameter-space hypothesis matters. An affine analytic family of invertible operators on a Banach fiber can induce a nonanalytic pullback family on alternating maps (Proposition 24). Thus an arbitrary analytic change of frame over a general Banach base need not be preserved. This is a failure of the natural construction under changes of trivialization, not a claim that the underlying pointwise bundle admits no analytic structure.

**Related results.** Mathlib treats manifolds and vector bundles over arbitrary nontrivially normed fields, with the same regularity indices  $n \in \mathbb{N} \cup \{\infty, \omega\}$ . Over every such field,  $Q$  is of class  $C^n$  for every  $n \in \mathbb{N} \cup \{\infty\}$ , with no completeness or dimension assumptions; this is proved in the draft Mathlib contribution [8]. In characteristic 0, Gouëzel proves analyticity by polarization [2] and uses it in the bundle construction [3]. The same polarization argument works whenever  $k! \neq 0$  in  $K$ ; we prove it in §4. Theorem 1 shows that the regularity range in [8] is sharp over every field of characteristic  $p > 0$  once  $k \geq p$ : in these examples, Mathlib’s `ContDiffAt K ω` fails at every point.

The answer is different if the target norm is required to be nonarchimedean. If  $K$  is nonarchimedean and  $F$  is nonarchimedean and spherically complete, then  $Q$  is analytic for all normed  $E, E'$  (Main Theorem). The proof in §2 constructs a contracting retraction from multilinear maps to alternating maps by spherical completeness and extension of bounded linear maps. This positive result is independent of the counterexample theorem and Corollary 2. A special case appears in [7, Remark 16]. Over a discretely valued base, the Main Theorem in fact rules out a Banach counterexample whenever  $F$  has an equivalent nonarchimedean norm (Proposition 82).

The earlier characteristic-2 construction [7] uses a nonarchimedean target and gives non-analyticity at 0 for  $E = E' = \ell^\infty$  and  $k = 2$  over every complete nonarchimedean field of characteristic 2 that is not spherically complete. Its formal hypothesis is a descending sequence of closed balls with empty intersection, equivalent to failure of spherical completeness (Remark 68). The higher-characteristic cases were left open there. The present construction works over every nontrivially normed field of positive characteristic, including spherically complete fields, by allowing targets with no equivalent nonarchimedean norm. The remaining questions for nonarchimedean targets are discussed in §13.

**Proofs and verification scope.** The counterexample theorem and factorial classification, including their supporting finite-field and functional-analytic arguments, are formalized in Lean 4 with Mathlib. The checked development also contains the operator form of Main Theorem (2), the discrete-target result and the sorted-basis lift. The analytic-family and no-largest-domain results are later extensions. Their mathematical proofs are included here; §14 and the repository’s theorem map give their separate formalization status.

The paper is organized to separate the positive theory from the two distinct obstructions. §2 proves the spherically complete theorem independently. §3 supplies the analytic lift criterion, and §4 gives factorial, finite-coordinate fiber and smoothness results. §5 introduces admissible families and proves finite-coordinate analytic reflection; §6 treats bundles and infinite-dimensional parameter classes. §7 proves that no largest full analytic domain exists in the stated generality. §8 then gives the tensor criterion for the stronger universal reflection problem. The original sharp Banach counterexample occupies §§9–12: the finite-field obstruction, its Laurent series realization, scalar extension and the final assembly. §13 concludes with further target classes and the remaining classification problems.

## 2. SPHERICALLY COMPLETE TARGETS

This section proves part (2) of the Main Theorem. Spherical completeness allows us to project bounded multilinear maps onto alternating maps without increasing the norm. Applying this projection to the ordinary multilinear precomposition formula produces an analytic lift in every characteristic and degree. The argument is independent of the counterexample construction. The retraction strategy reflects the classical role of spherically complete spaces as injective objects in nonarchimedean Banach-space theory; see Schikhof [14, §2.8] for this background over complete fields. A special case is stated in [7, Remark 16]; the proof here follows the extension and retraction arguments in the Lean development. Related Lean formalizations of compatible extension and spherical completion in the nonarchimedean setting appear in Yuan [15, Prop. 6.6 and Thm. 7.3].

**2.1. Statement and strategy.** A norm is *nonarchimedean* if  $\|x + y\| \leq \max(\|x\|, \|y\|)$ . A metric space is *spherically complete* if every nonempty family of closed balls that meet pairwise has a common point. We write  $\overline{B}(a, r) = \{x : \|x - a\| \leq r\}$ , with  $r \geq 0$ . In a nonarchimedean space, two such balls meet if and only if the distance between their centres is at most the larger radius; in that case the ball of smaller radius is contained in the other.

As in the introduction,  $\mathcal{L}(X, Y)$  denotes bounded linear maps. Write  $\text{Mult}^k(X; Y)$  for the bounded  $k$ -linear maps and  $\text{Alt}^k(X; Y)$  for their alternating subspace, with the operator norms determined by

$$\|m(x_1, \dots, x_k)\| \leq \|m\| \prod_{i=1}^k \|x_i\|.$$

When the target norm is nonarchimedean, both operator spaces are nonarchimedean, regardless of the norm on  $X$ .

Throughout this section,  $K$  is a nonarchimedean nontrivially normed field and  $k \geq 0$  is an integer. Positive characteristic implies the nonarchimedean hypothesis by Lemma 6. We prove the following operator statement, which also records every smoothness order. For normed  $K$ -spaces  $E, E'$  and a nonarchimedean, spherically complete  $F$ ,

$$Q_{E, E', F}^K \text{ is } C^n \quad (n \in \mathbb{N} \cup \{\infty, \omega\}).$$

Together with spherical completeness of  $\text{Alt}^k(E; F)$  and (1.2), this proves Main Theorem (2).

No completeness assumption on  $K, E$  or  $E'$  is needed, and no nonarchimedean hypothesis is imposed on  $E$  or  $E'$ . For  $k \geq 1$ , put  $A = \text{Alt}^k(E; F)$  and  $M = \text{Mult}^k(E; F)$ . The desired projection is a bounded linear map  $\rho: M \rightarrow A$  of norm at most one that fixes every alternating map. Once it is available, the formula

$$P(f_1, \dots, f_k)(m) = \rho(m \circ (f_1, \dots, f_k))$$

is bounded and multilinear in the operators  $f_i$ , and has diagonal  $Q$ . This is the required analytic lift: expansion around any operator gives a finite power series. To construct  $\rho$ , we first prove an extension lemma that preserves a family of bounds, then use it to show that  $A$  is spherically complete.

**2.2. Extension with simultaneous bounds.** The following form of the nonarchimedean extension argument keeps a prospective extension in several operator balls at once. We allow a seminorm on the domain, since the free space used in the next step may have nonzero vectors of seminorm zero.

**Lemma 3** (compatible extension). *Let  $U$  be a nonarchimedean seminormed  $K$ -space and let  $G$  be a nonarchimedean, spherically complete normed  $K$ -space. Suppose a nonempty family  $(T_\alpha, r_\alpha)$ , with  $T_\alpha \in \mathcal{L}(U, G)$  and  $r_\alpha \geq 0$ , satisfies*

$$\|T_\alpha - T_\beta\| \leq \max(r_\alpha, r_\beta) \quad \text{for all } \alpha, \beta.$$

*Let  $D \subseteq U$  be a linear subspace and let  $T_0: D \rightarrow G$  be linear with*

$$\|T_0 d - T_\alpha d\| \leq r_\alpha \|d\| \quad (d \in D, \text{ all } \alpha).$$

*Then  $T_0$  extends to a bounded linear map  $T: U \rightarrow G$  such that  $\|T - T_\alpha\| \leq r_\alpha$  for every  $\alpha$ . In particular, every contraction from a subspace of  $U$  to  $G$  extends to a contraction on  $U$ .*

*Proof.* It suffices to extend across one vector  $x \notin D$  while preserving every bound. The required value  $Tx$  must belong to the balls

$$(2.1) \quad \overline{B}(T_\alpha(d+x) - T_0 d, r_\alpha \|d+x\|) \quad (\alpha, d \in D).$$

We check that these balls meet pairwise. Write  $c_{\alpha,d} = T_\alpha(d+x) - T_0 d$ , and suppose  $r_\alpha \leq r_\beta$ . Then

$$c_{\alpha,d} - c_{\beta,e} = T_\alpha(d-e) - T_0(d-e) + (T_\alpha - T_\beta)(e+x).$$

Since  $d-e \in D$  and  $\|d-e\| \leq \max(\|d+x\|, \|e+x\|)$ , the hypotheses give

$$\begin{aligned} \|c_{\alpha,d} - c_{\beta,e}\| &\leq \max(r_\alpha \|d-e\|, r_\beta \|e+x\|) \\ &\leq \max(r_\alpha \|d+x\|, r_\beta \|e+x\|). \end{aligned}$$

This is exactly the condition for the two balls to meet. The other ordering of the radii is symmetric.

By spherical completeness, choose  $c$  in all the balls (2.1). Define

$$T(d+\lambda x) = T_0 d + \lambda c.$$

The decomposition is unique because  $x \notin D$ . For  $\lambda \neq 0$ , apply the ball condition with  $\lambda^{-1}d$  in place of  $d$  and multiply by  $|\lambda|$ ; this gives  $\|T(d+\lambda x) - T_\alpha(d+\lambda x)\| \leq r_\alpha \|d+\lambda x\|$ . For  $\lambda = 0$  the original bound applies.

Partially order the extensions satisfying all these inequalities by inclusion of their domains. Unions of chains remain linear and preserve the bounds, so Zorn's lemma gives a maximal extension. The one-vector argument forces its domain to be all of  $U$ . Fixing any one index  $\alpha$  gives  $\|Tu\| \leq \max(\|T_\alpha\|, r_\alpha)\|u\|$ , so the extension is bounded. For the last assertion, take the single pair  $(T_\alpha, r_\alpha) = (0, 1)$ .  $\square$

**2.3. Spherical completeness of alternating-map spaces.** Ordinary completeness alone would not let us apply the extension lemma with target  $A$ . We prove the required stronger property by presenting alternating maps as linear maps on a free space subject to linear relations. The nonarchimedean seminorm on that free space is chosen so that this presentation preserves the operator norm, even when  $E$  itself is not nonarchimedean.

**Proposition 4.** *For a nonarchimedean nontrivially normed field  $K$ , a normed  $K$ -space  $E$  and a nonarchimedean, spherically complete  $F$ , the space  $\text{Alt}^k(E; F)$  is a nonarchimedean, spherically complete normed space for every  $k \geq 0$ .*

*Proof.* For  $k = 0$ , the space identifies isometrically with  $F$ . Suppose  $k \geq 1$ . Let  $U$  be the free  $K$ -vector space on the set  $E^k$ , with generators  $[x]$  and seminorm

$$\left\| \sum_x c_x [x] \right\|_U = \max_x |c_x| \prod_{i=1}^k \|x_i\|.$$

The maximum is over the finite support, with empty maximum zero. This seminorm is nonarchimedean because the absolute value on  $K$  is nonarchimedean. Let  $R \subseteq U$  be the linear span of the relations expressing additivity and scalar homogeneity in each slot, together with  $[x]$  whenever two coordinates of  $x$  coincide.

An alternating map  $a \in A$  defines a linear map vanishing on  $R$  by

$$\hat{a}\left(\sum_x c_x[x]\right) = \sum_x c_x a(x).$$

The nonarchimedean inequality in  $F$  gives

$$\left\|\hat{a}\left(\sum_x c_x[x]\right)\right\| \leq \max_x |c_x| \|a(x)\| \leq \|a\| \left\|\sum_x c_x[x]\right\|_U.$$

Conversely, a bounded linear map  $T: U \rightarrow F$  vanishing on  $R$  defines an alternating map  $a(x) = T([x])$  with  $\|a(x)\| \leq \|T\| \prod_i \|x_i\|$ . Thus these two constructions are inverse linear isometries. In particular, the equal-input relations impose strong alternation in every characteristic.

Take a nonempty family of pairwise-meeting closed balls in  $A$ , with centres  $a_\alpha$  and radii  $r_\alpha$ . Their corresponding linear maps satisfy  $\|\hat{a}_\alpha - \hat{a}_\beta\| \leq \max(r_\alpha, r_\beta)$  and vanish on  $R$ . Apply Lemma 3 with  $D = R$ ,  $T_0 = 0$  and  $G = F$ . It produces a bounded linear map  $T: U \rightarrow F$  that vanishes on  $R$  and satisfies  $\|T - \hat{a}_\alpha\| \leq r_\alpha$  for every  $\alpha$ . The associated alternating map belongs to every original ball. This proves spherical completeness; no closure assumption on  $R$  is needed.  $\square$

**2.4. The retraction and the analytic lift.** We now apply the extension lemma a second time, using the alternating-map space itself as the target. This produces the projection needed to turn ordinary precomposition into a polynomial with alternating values.

*Proof of Main Theorem (2).* In degree zero,  $\text{Alt}^0(-; F)$  is naturally  $F$  and the bifunctor acts by the pushforward morphism, so the conclusion holds. Assume  $k \geq 1$  and retain  $A = \text{Alt}^k(E; F)$  and  $M = \text{Mult}^k(E; F)$ . Their operator norms are nonarchimedean, and  $A$  is spherically complete by Proposition 4. The inclusion  $j: A \rightarrow M$  is a linear isometry. Its inverse on  $j(A)$  is therefore a contraction. Lemma 3 extends this inverse to a contraction  $\rho: M \rightarrow A$  with  $\rho j = \text{id}_A$ .

For  $f_1, \dots, f_k \in \mathcal{L}(E, E')$  and  $m \in \text{Alt}^k(E'; F)$ , define

$$P(f_1, \dots, f_k)(m) = \rho(m \circ (f_1, \dots, f_k)).$$

This is linear in  $m$  and multilinear in the  $f_i$ . The estimate

$$\|P(f_1, \dots, f_k)(m)\| \leq \|m\| \prod_{i=1}^k \|f_i\|$$

shows that  $P$  is a bounded  $k$ -linear map into  $\mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F))$ , with  $\|P\| \leq 1$ . On the diagonal,  $m \circ (f, \dots, f)$  is already alternating, so  $P(f, \dots, f) = Q(f)$ .

For completeness, expand at an arbitrary  $f_0 \in \mathcal{L}(E, E')$ . For  $0 \leq r \leq k$ , define a bounded  $r$ -linear map  $q_r$  by summing over the  $r$ -element subsets  $S$  of  $\{1, \dots, k\}$ : insert its  $r$  arguments in the slots of  $S$ , in increasing order, and put  $f_0$  in all remaining slots of  $P$ . Multilinearity gives the finite identity

$$Q(f_0 + h) = \sum_{r=0}^k q_r(h, \dots, h), \quad \|q_r\| \leq \binom{k}{r} \|P\| \|f_0\|^{k-r}.$$

The binomial coefficient in the bound is an ordinary nonnegative integer counting the summands. With  $q_r = 0$  for  $r > k$ , this is a power series of infinite radius at every  $f_0$ . Consequently  $Q$  is continuously polynomial, hence  $C^\omega$  and  $C^n$  for all  $n \in \mathbb{N} \cup \{\infty\}$ . Equation (1.2) makes the full bifunctor analytic on hom spaces, and Proposition 4 shows that its values are nonarchimedean and spherically complete. This proves the categorical statement.  $\square$

The retraction supplies the alternating values directly, with no division by  $k!$ . The source spaces need no nonarchimedean hypothesis because the extension argument takes place in the free presentation and in operator spaces with nonarchimedean targets. The following sections develop the other positive cases and then construct counterexamples whose targets admit no equivalent nonarchimedean norm.

### 3. ANALYTIC PRELIMINARIES

The lift criterion reduces the analytic question to a bounded multilinear identity. Its proof uses uniqueness on scalar lines and applies in every characteristic, without completeness assumptions.

**3.1. Conventions.** We write  $\mathbb{N} = \{0, 1, 2, \dots\}$ . Throughout,  $K$  denotes the base field and  $k$  the degree. A *normed field*  $K$  carries a multiplicative absolute value  $|\cdot|$  satisfying the triangle inequality. It is *nontrivially normed* if some  $c \in K$  has  $0 < |c| < 1$ , and *nonarchimedean* if  $|x + y| \leq \max(|x|, |y|)$ . A *normed  $K$ -space*  $X$  satisfies  $\|\lambda x\| = |\lambda| \|x\|$ ; it is *Banach* if it is complete.

We write  $\mathcal{L}(X, Y)$  for the bounded linear maps,  $\text{Mult}^k(X; Y)$  for the bounded  $k$ -linear maps  $X^k \rightarrow Y$ , and  $\text{Alt}^k(X; Y) \subseteq \text{Mult}^k(X; Y)$  for the closed subspace of alternating maps, all with the operator norm. A subscript  $K$  records the field when needed. An alternating map  $m$  is antisymmetric: expanding  $0 = m(\dots, y + z, \dots, y + z, \dots)$  gives  $m(\dots, y, \dots, z, \dots) = -m(\dots, z, \dots, y, \dots)$ , and hence  $m(x_{\sigma(1)}, \dots, x_{\sigma(k)}) = \text{sgn}(\sigma) m(x_1, \dots, x_k)$  for  $\sigma \in S_k$ . In characteristic 2 the converse fails.

We index the  $k$  arguments by  $\{1, \dots, k\}$  throughout. Reindexing by any finite set with  $k$  elements is an isometric isomorphism of alternating-map spaces and intertwines precomposition. Thus the results also apply to the arbitrary finite index types used in Lean, with the same counterexample spaces. For each fixed degree we write

$$Q = Q_{E, E', F}^K: \mathcal{L}(E, E') \rightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)),$$

where  $Q(f)(m)(x_1, \dots, x_k) = m(fx_1, \dots, fx_k)$ .

**Definition 5.** A *bounded  $k$ -linear lift* of  $Q$  is a bounded  $k$ -linear map  $P: \mathcal{L}(E, E')^k \rightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F))$  with  $P(f, \dots, f) = Q(f)$  for all  $f \in \mathcal{L}(E, E')$ .

A map  $g: X \rightarrow Z$  between normed spaces is *analytic at  $x_0$*  if there are bounded  $n$ -linear maps  $p_n: X^n \rightarrow Z$  ( $n \geq 0$ ) and a radius  $R > 0$  with  $\sup_n \|p_n\| R^n < \infty$  such that  $g(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h)$ , a convergent series, whenever  $\|h\| < R$ . Up to shrinking  $R$  this is Mathlib's `AnalyticAt` [6], and in Mathlib `ContDiffAt`  $K \omega$  at a point implies `AnalyticAt` there (`ContDiffAt.analyticAt`).

**Lemma 6.** *A normed field of characteristic  $p > 0$  is nonarchimedean.*

*Proof.* Frobenius and the triangle inequality give, for every  $m \geq 0$ ,

$$|x + y|^{p^m} = |x^{p^m} + y^{p^m}| \leq 2 \max(|x|, |y|)^{p^m}.$$

Taking  $p^m$ -th roots and letting  $m \rightarrow \infty$  proves the assertion.  $\square$

**3.2. The lift criterion.** A power series for  $Q$  at any base point determines a lift through its degree- $k$  term. The only uniqueness statement needed is one-dimensional, so the argument does not use a polarization formula or division by  $k!$ .

**Lemma 7** (one-variable uniqueness). *Let  $K$  be nontrivially normed and  $Z$  a normed  $K$ -space, not necessarily complete. Let  $z_0, z_1, \dots \in Z$ ,  $\rho > 0$  and  $M \geq 0$  with  $\|z_n\|\rho^n \leq M$  for all  $n$ . If  $\sum_n t^n z_n$  converges to 0 for every  $t \in K$  with  $|t| < \rho$ , then  $z_n = 0$  for all  $n$ .*

*Proof.* Otherwise let  $N$  be least with  $z_N \neq 0$ . For  $0 < |t| < \rho/2$  the series  $\sum_{n \geq N} t^{n-N} z_n$  converges, being  $t^{-N}$  times the convergent tail  $\sum_{n \geq N} t^n z_n$ , and  $t^N(z_N + \sum_{n \geq N} t^{n-N} z_n) = 0$ . Hence

$$\|z_N\| = \left\| \sum_{n \geq N} t^{n-N} z_n \right\| \leq \sum_{n \geq N} M \rho^{-n} |t|^{n-N} = M \rho^{-N} \frac{|t|/\rho}{1 - |t|/\rho},$$

where the norm of a convergent series is bounded by the sum of the norms by continuity of the norm on partial sums. Since  $K$  is nontrivially normed, there are  $t \neq 0$  with  $|t|$  arbitrarily small, so  $\|z_N\| = 0$ , a contradiction.  $\square$

**Proposition 8** (lift criterion). *Let  $K$  be any nontrivially normed field,  $E, E', F$  normed  $K$ -spaces and  $k \geq 1$ .*

- (1) *If  $Q$  has a bounded  $k$ -linear lift  $P$ , then at every  $f_0 \in \mathcal{L}(E, E')$  the map  $Q$  is given by a finite power series of infinite radius; in particular  $Q$  is analytic at every point.*
- (2) *If  $Q$  is analytic at some point  $f_0$ , with power series  $(p_n)_n$  there, then  $p_k$  is a bounded  $k$ -linear lift of  $Q$ .*
- (3) *Hence the following are equivalent:  $Q$  has a bounded  $k$ -linear lift;  $Q$  is analytic at every point;  $Q$  is analytic at some point. If they fail, then Mathlib's `ContDiffAt`  $K \omega Q f_0$  fails at every  $f_0$ .*

*Proof.* (1) For  $y \in \mathcal{L}(E, E')$ , multilinearity gives

$$Q(f_0 + y) = P(f_0 + y, \dots, f_0 + y) = \sum_{r=0}^k q_r(y, \dots, y),$$

where  $q_r(y_1, \dots, y_r)$  is the sum, over the  $r$ -element subsets  $S \subseteq \{1, \dots, k\}$ , of  $P$  evaluated with  $y_1, \dots, y_r$  in the slots of  $S$  in increasing order and  $f_0$  in the other slots. Each  $q_r$  is bounded  $r$ -linear, with  $\|q_r\| \leq \binom{k}{r} \|P\| \|f_0\|^{k-r}$ , so  $(q_r)_{r \leq k}$  is a power series of infinite radius for  $Q$  at  $f_0$ .

(2) We compare the given power series along each line with the polynomial obtained by expanding precomposition. The expansion is taken in the larger space of all multilinear maps, where each coefficient is visibly defined. Let  $\text{incl}: \text{Alt}^k(E; F) \hookrightarrow \text{Mult}^k(E; F)$  be the isometric inclusion, and put

$$Z := \mathcal{L}(\text{Alt}^k(E'; F), \text{Mult}^k(E; F)), \quad \mathcal{I}(T) := \text{incl} \circ T.$$

Then  $\mathcal{I}$  is a linear isometry into  $Z$ , hence injective. Fix  $h \in \mathcal{L}(E, E')$ ,  $h \neq 0$ . For  $m \in \text{Alt}^k(E'; F)$ ,  $x \in E^k$  and  $t \in K$ , multilinearity of  $m$  gives the *ambient polynomial identity*

$$(3.1) \quad \mathcal{I}(Q(f_0 + th)) = \sum_{r=0}^k t^r A_r(h), \quad A_r(h)(m)(x) := \sum_{|S|=r} m(u_1^S, \dots, u_k^S),$$

where  $u_i^S = hx_i$  for  $i \in S$  and  $u_i^S = f_0 x_i$  for  $i \notin S$ . Here  $A_r(h) \in Z$  is bounded, with  $\|A_r(h)\| \leq \binom{k}{r} \|h\|^r \|f_0\|^{k-r}$ , and  $A_k(h) = \mathcal{I}(Q(h))$ . (For  $r < k$  we do not need to know whether  $A_r(h)(m)$  is alternating; this is why (3.1) is written in  $Z$ .) Let  $R$  and  $M' := \sup_n \|p_n\| R^n$  be as in the definition of analyticity, and put  $\rho := R/\|h\|$ . For  $|t| < \rho$  we have  $\|th\| < R$ ,

so  $Q(f_0 + th) = \sum_n t^n p_n(h, \dots, h)$ , and applying  $\mathcal{I}$  and (3.1) gives  $\sum_n t^n z_n = 0$  in  $Z$ , with  $z_n := \mathcal{I}(p_n(h, \dots, h)) - A_n(h)$  and  $A_n(h) := 0$  for  $n > k$ . Since  $\|z_n\| \rho^n \leq M' + \max_{r \leq k} \|A_r(h)\| \rho^r$ , Lemma 7 gives  $z_k = 0$ , that is,  $\mathcal{I}(p_k(h, \dots, h)) = \mathcal{I}(Q(h))$ , hence  $p_k(h, \dots, h) = Q(h)$ . For  $h = 0$  both sides vanish since  $k \geq 1$ .

(3) follows from (1) and (2), and from `ContDiffAt.analyticAt` for the last clause.  $\square$

*Remark 9.* The degree- $k$  term of *every* power series representing  $Q$  is a lift, regardless of the base point. This does not assert uniqueness of the multilinear lift: in positive characteristic, different multilinear maps can have the same diagonal.

#### 4. POSITIVE RESULTS

Alongside the spherical-completeness argument of §2, two further constructions give analytic lifts: dividing by  $k!$ , when this scalar is nonzero, or choosing coordinates under suitable finite-dimensional hypotheses. We also prove smoothness in every characteristic. It descends from the space of all multilinear maps through a closed isometric inclusion and therefore requires neither hypothesis.

##### 4.1. Invertible factorial: in particular, $p > k$ .

**Proposition 10.** *Let  $K$  be a nontrivially normed field and let  $E, E', F$  be normed  $K$ -spaces. If  $k! \neq 0$  in  $K$ , then  $Q$  is given at every point by a finite power series of infinite radius. In particular, it is analytic whenever  $\text{char } K = 0$ , or whenever  $\text{char } K = p > k$ . No completeness or dimension hypothesis is needed.*

*Proof.* For  $k = 0$ , precomposition is constant. Suppose  $k \geq 1$  and define

$$P(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \frac{1}{k!} \sum_{\sigma \in S_k} m(f_{\sigma(1)}x_1, \dots, f_{\sigma(k)}x_k).$$

This is multilinear in the  $f_i$  and in the  $x_j$ , linear in  $m$ , and satisfies

$$\|P(f_1, \dots, f_k)(m)(x_1, \dots, x_k)\| \leq |(k!)^{-1}| k! \|m\| \prod_i \|f_i\| \prod_j \|x_j\|,$$

where the second  $k!$  on the right is a real number counting the terms. To check alternation, suppose  $x_a = x_b$  and let  $\tau$  exchange  $a$  and  $b$ . The summands indexed by  $\sigma$  and  $\sigma \circ \tau$  cancel by antisymmetry of  $m$ . This pairing is a fixed-point-free involution, so it proves vanishing also in characteristic 2. Thus  $P$  takes values in the required space of alternating maps and is a bounded  $k$ -linear lift: on the diagonal, all  $k!$  summands equal  $Q(f)(m)(x)$ . Proposition 8 gives the claimed power series.

In a field of characteristic  $p > 0$ , the scalar  $k!$  is nonzero precisely when  $k < p$ . This gives the stated positive-characteristic case.  $\square$

**4.2. Finite-coordinate operator domains and codomains.** The finite-dimensional argument uses continuous coordinate functionals. We state this hypothesis explicitly so that the result also applies over incomplete fields whenever the chosen finite-dimensional space has its usual coordinate topology.

**Proposition 11** (finite coordinate bases). *Let  $K$  be a nontrivially normed field and let  $E, E', F$  be normed  $K$ -spaces. If either  $E$  or  $E'$  has a finite algebraic basis whose coordinate functionals are continuous, then  $Q_{E, E', F}^K$  has a bounded  $k$ -linear lift in every degree  $k \geq 1$ . In particular, it is continuously polynomial, and hence analytic, at every point. No completeness or nonarchimedean hypothesis on  $F$  is required.*

*Proof.* Let  $H$  denote the space with a finite coordinate basis, write that basis as  $e_1, \dots, e_n$ , and let  $\varepsilon_1, \dots, \varepsilon_n \in \mathcal{L}_K(H, K)$  be its coordinate functionals. For an increasing  $k$ -tuple  $s = (s_1 < \dots < s_k)$  in  $\{1, \dots, n\}$ , put

$$c_s := \prod_{a=1}^k \|\varepsilon_{s_a}\| \|e_{s_a}\|, \quad C_H := k! \sum_{s_1 < \dots < s_k} c_s.$$

Here the factor  $k!$  in  $C_H$  is the positive real integer counting the terms in a determinant, not its image in  $K$ . In particular,  $C_H \geq 0$  even when  $k! = 0$  in  $K$ .

Suppose first that  $H = E$ . For  $f_1, \dots, f_k \in \mathcal{L}_K(E, E')$ ,  $m \in \text{Alt}_K^k(E'; F)$  and  $x \in E^k$ , define

$$P(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \sum_{s_1 < \dots < s_k} \det(\varepsilon_{s_a}(x_b))_{a,b} m(f_1 e_{s_1}, \dots, f_k e_{s_k}).$$

The determinant makes this expression alternating and multilinear in  $x$ . It is linear in  $m$  and multilinear in  $f_1, \dots, f_k$ . Expanding the determinant over permutations gives

$$\left| \det(\varepsilon_{s_a}(x_b))_{a,b} \right| \leq k! \prod_a \|\varepsilon_{s_a}\| \prod_b \|x_b\|.$$

Together with the operator bound for  $m$ , the ordinary triangle inequality therefore yields

$$\|P(f_1, \dots, f_k)(m)(x)\| \leq C_E \|m\| \prod_a \|f_a\| \prod_b \|x_b\|.$$

Thus the displayed formula defines a bounded  $k$ -linear map into  $\mathcal{L}_K(\text{Alt}_K^k(E'; F), \text{Alt}_K^k(E; F))$ , with  $\|P\| \leq C_E$ . On the diagonal, the finite exterior-basis expansion of  $x_1 \wedge \dots \wedge x_k$  gives

$$m(fx_1, \dots, fx_k) = \sum_{s_1 < \dots < s_k} \det(\varepsilon_{s_a}(x_b))_{a,b} m(fe_{s_1}, \dots, fe_{s_k}),$$

so  $P(f, \dots, f) = Q(f)$ .

Now suppose that  $H = E'$ . Define instead

$$P(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \sum_{s_1 < \dots < s_k} \det(\varepsilon_{s_a}(f_a x_b))_{a,b} m(e_{s_1}, \dots, e_{s_k}).$$

Each row of the determinant is linear in its corresponding operator  $f_a$ , and its columns make the result alternating and multilinear in  $x$ . The expression is again linear in  $m$ . The determinant estimate now reads

$$\left| \det(\varepsilon_{s_a}(f_a x_b))_{a,b} \right| \leq k! \prod_a (\|\varepsilon_{s_a}\| \|f_a\|) \prod_b \|x_b\|.$$

It follows that the same operator bound holds with  $C_{E'}$  in place of  $C_E$ . Expanding  $f x_1 \wedge \dots \wedge f x_k$  in the exterior basis of  $E'$  proves the diagonal identity.

Both formulas are finite sums. Strong alternation follows from equal columns in a determinant, so the proof works in every characteristic and involves no division by  $k!$ . Proposition 8 gives continuous polynomiality and analyticity. If  $k > n$ , the sums are empty and the relevant alternating-map space is zero; the same argument gives the zero lift. Degree zero, excluded from the lift criterion's statement, is immediate because  $Q$  is constant.  $\square$

**Lemma 12.** *Let  $\mathbb{K}$  be a complete nontrivially normed field and  $U$  a finite-dimensional normed  $\mathbb{K}$ -space. Then every linear functional on  $U$  is bounded, and  $U$  is complete.*

*Proof.* We prove both assertions by induction on  $n = \dim U$ , the case  $n = 0$  being immediate. Suppose  $n \geq 1$  and let  $g \neq 0$  be a linear functional on  $U$ . Its kernel  $H$  has dimension  $n - 1$ , so the induction hypothesis makes  $H$  complete and therefore closed in  $U$ . Choose  $v$  with  $g(v) = 1$ . Since  $v \notin H$ ,  $\delta := \inf_{h \in H} \|v - h\| > 0$ . Write  $x = g(x)v + h$  with  $h \in H$ . If  $g(x) \neq 0$ , then

$$\|x\| = |g(x)| \|v + g(x)^{-1}h\| \geq |g(x)|\delta.$$

Thus  $|g(x)| \leq \delta^{-1}\|x\|$  for every  $x$ , proving boundedness.

Choose a basis of  $U$ . Its coordinate functionals are now bounded, so the coordinate map  $U \rightarrow \mathbb{K}^n$ , with the max norm on  $\mathbb{K}^n$ , is bounded. The inverse map  $(c_i) \mapsto \sum_i c_i e_i$  is bounded by  $\sum_i \|e_i\|$ . Hence  $U$  is bi-Lipschitz equivalent to the complete space  $\mathbb{K}^n$ , and is itself complete.  $\square$

**Corollary 13.** *Let  $K$  be a complete nontrivially normed field, and let  $E, E', F$  be normed  $K$ -spaces. If either  $E$  or  $E'$  is finite-dimensional over  $K$ , then  $Q_{E, E', F}^K$  is analytic at every point in every degree. No completeness assumption on  $E, E'$  or  $F$  is needed in the statement.*

*Proof.* Lemma 12 gives continuous coordinates on whichever of  $E, E'$  is finite-dimensional. Proposition 11 then applies.  $\square$

*Remark 14* (the coordinate hypothesis over an incomplete field). Finite algebraic dimension alone does not imply continuity of coordinate functionals over an incomplete field. For example, choose  $a \in \widehat{K} \setminus K$  and give  $H := K + Ka \subseteq \widehat{K}$  the induced norm. The vectors  $1, a$  form a  $K$ -basis. If  $a_j \in K$  tends to  $a$ , then  $a - a_j \rightarrow 0$  in  $H$ , whereas its  $a$ -coordinate is always 1. Thus that coordinate functional is discontinuous. This prevents dropping the coordinate hypothesis from the preceding proof; it does not by itself give a counterexample to analyticity of  $Q$  for such spaces. Over an incomplete field the proposition still applies, for example, to finite-dimensional spaces whose norm is equivalent to a coordinate maximum norm.

Finite-dimensionality of the target  $F$  leads to a different question. Remark 81 reduces it to scalar-valued forms and gives a positive result over a spherically complete nonarchimedean base.

**4.3. Smoothness in every characteristic.** We first record why finite-order differentiability descends through a closed isometric inclusion. The argument also explains why smoothness survives when a multilinear analytic lift does not.

**Lemma 15** (smoothness through a closed isometric inclusion). *Let  $U, V, W$  be normed spaces over a nontrivially normed field  $K$ , and let  $j: V \rightarrow W$  be a linear isometry with closed range. For a map  $g: U \rightarrow V$  and  $n \in \mathbb{N} \cup \{\infty\}$ , the map  $g$  is  $C^n$  if and only if  $j \circ g$  is  $C^n$ .*

*Proof.* One direction follows by composition with a continuous linear map. For the converse, continuity follows from the isometry. Suppose  $j \circ g$  has Fréchet derivative  $A: U \rightarrow W$  at  $x$ . For each  $h \in U$ ,

$$Ah = \lim_{t \rightarrow 0, t \neq 0} \frac{j(g(x + th)) - j(g(x))}{t}.$$

The quotients lie in  $j(V)$ , and nontriviality of the norm supplies nonzero scalars tending to zero. Since  $j(V)$  is closed,  $Ah \in j(V)$ . Thus  $A = j \circ B$  for a unique linear map  $B: U \rightarrow V$ . The identity  $\|Bh\| = \|Ah\|$  makes  $B$  bounded, and the isometry transfers the Fréchet remainder estimate from  $A$  to  $B$ . Consequently  $g$  is differentiable and

$$D(j \circ g) = j_* \circ Dg, \quad j_*: \mathcal{L}(U, V) \longrightarrow \mathcal{L}(U, W), \quad T \longmapsto j \circ T.$$

The map  $j_*$  is again a linear isometry with closed range. Indeed, an operator belongs to its range exactly when all its values lie in  $j(V)$ ; this is a closed condition, being an intersection of

inverse images under continuous evaluation maps. Applying the same argument to  $j_*$  proves the assertion for  $C^{n+1}$  from the assertion for  $C^n$ . Induction proves every finite order, and hence  $C^\infty$ .  $\square$

**Proposition 16** (smoothness of precomposition). *For every nontrivially normed field  $K$ , all normed  $K$ -spaces  $E, E', F$ , and every integer  $k \geq 0$ , the map  $Q_{E, E', F}^K$  is  $C^n$  for every  $n \in \mathbb{N} \cup \{\infty\}$ . In particular, precomposition is always  $C^\infty$ , without completeness, finite-dimensionality, or characteristic assumptions.*

*Proof.* The inclusion  $\text{Alt}^k(E; F) \hookrightarrow \text{Mult}^k(E; F)$  is isometric and has closed range: alternation is the intersection of the closed conditions  $m(x_1, \dots, x_k) = 0$  for tuples with two equal entries. Postcomposition therefore gives a linear isometry with closed range

$$\mathcal{I}: \mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)) \longrightarrow \mathcal{L}(\text{Alt}^k(E'; F), \text{Mult}^k(E; F)).$$

In the larger target, precomposition is the diagonal of the bounded  $k$ -linear map

$$A(f_1, \dots, f_k)(m)(x_1, \dots, x_k) = m(f_1 x_1, \dots, f_k x_k),$$

whose norm is at most 1. Hence  $\mathcal{I} \circ Q$  is a continuous polynomial and is  $C^\infty$ . Lemma 15 gives the result. The degree-zero case is again constant.  $\square$

This is the argument formalized in [8]. It uses closedness to recover derivatives, rather than a projection onto alternating maps. The induction proves all finite orders but does not recover analytic coefficients in positive characteristic. Theorem 1 shows that the resulting  $C^\infty$  conclusion cannot in general be strengthened to  $C^\omega$ .

## 5. ANALYTIC FAMILIES OF MORPHISMS

Regularity on an entire operator space is stronger than regularity along a prescribed family of operators. This distinction gives a useful replacement for restricting the objects of the alternating-map bifunctor. We retain every object and every individual morphism, and ask which analytic families the bifunctor preserves. The resulting family structure is canonical. Its usefulness comes from the concrete parameter classes for which the additional condition is automatic.

**5.1. The canonical family structure.** For objects  $X = (E, F)$  and  $Y = (E', F')$  of  $\text{Vec}_K^{\text{op}} \times \text{Vec}_K$ , write

$$V_X = \text{Alt}^k(E; F), \quad H_{XY} = \mathcal{L}(E', E) \times \mathcal{L}(F, F'),$$

and denote the action on morphisms by

$$A_{XY}: H_{XY} \longrightarrow \mathcal{L}(V_X, V_Y), \quad A_{XY}(u, v)m = v \circ m \circ (u, \dots, u).$$

The product defining  $H_{XY}$  has the maximum norm. These conventions make the contravariance in  $E$  explicit: the first component of a morphism  $X \rightarrow Y$  is a map  $E' \rightarrow E$ .

**Definition 17.** Let  $U$  be open in a normed  $K$ -space. A map  $\gamma: U \rightarrow H_{XY}$  is *admissible* if both  $\gamma$  and  $A_{XY} \circ \gamma$  are analytic. Families on analytic manifolds are admissible when their expressions in parameter charts are admissible.

**Proposition 18.** *Admissible families contain constants and are closed under analytic reparameterization, local gluing, and categorical composition. Composition is allowed to depend on either a common parameter or independent parameters. They form the largest collection of ordinary analytic families for which the action of  $\text{Alt}^k$  is analytic.*

*Proof.* Constants, reparameterization, and gluing follow from the corresponding properties of analytic maps. For composable families  $\gamma: U \rightarrow H_{XY}$  and  $\eta: U \rightarrow H_{YZ}$ , composition in  $H$  is bounded bilinear, and functoriality gives

$$A_{XZ}(\eta \circ \gamma) = A_{YZ}(\eta) \circ A_{XY}(\gamma).$$

Composition of bounded operators is bounded bilinear as well, so both required maps are analytic. For independent parameter spaces, pull the two families back to their product and apply the same argument. Finally, any ordinary analytic family preserved analytically by  $\text{Alt}^k$  satisfies exactly the two conditions of Definition 17, and hence belongs to this collection.  $\square$

Equivalently, identify the set  $H_{XY}$  with the graph

$$\Gamma_{A_{XY}} = \{(h, A_{XY}(h)) : h \in H_{XY}\} \subseteq H_{XY} \times \mathcal{L}(V_X, V_Y)$$

and declare a family analytic when its map to the ambient product is analytic. No assertion that this graph is an analytic submanifold is involved. In the language of plots, this is a category whose hom sets are analytic plot spaces: the parameter domains are open subsets of normed  $K$ -spaces, the reparameterizations are analytic maps, and covers are open covers. The smooth concrete-sheaf framework of Baez and Hoffnung [1] motivates this formulation; the analytic family structure used here is specified directly by Definition 17 and Proposition 18.

This structure need not preserve the vector-space operations on hom sets. Suppose  $k > 0$  and  $u: U \rightarrow \mathcal{L}(E', E)$  is analytic but its pullback action is not. With  $F' = F$ , both  $(u, 0)$  and  $(0, \text{id}_F)$  are admissible because their induced actions vanish. Their sum  $(u, \text{id}_F)$  is not admissible. Thus addition of admissible families can fail. Multiplication by an analytic scalar family does preserve admissibility, since  $A_{XY}(\lambda u, \lambda v) = \lambda^{k+1} A_{XY}(u, v)$ .

**5.2. Reflection through a closed subspace.** The range condition behind admissibility can be separated from the operator construction. Set

$$W = \mathcal{L}(V_X, \text{Alt}^k(E'; F')), \quad Z = \mathcal{L}(V_X, \text{Mult}^k(E'; F')),$$

and let  $j: W \hookrightarrow Z$  be the map induced by forgetting alternation. It is a linear isometry with closed image. Indeed, an element of  $Z$  belongs to its image precisely when every value vanishes on every tuple with two equal entries. These are closed linear evaluation conditions; the operator norms before and after forgetting alternation agree. This argument does not require completeness.

Although  $A_{XY}$  can fail to be analytic, its composite with  $j$  is always a bounded homogeneous polynomial of degree  $k + 1$ . A multilinear representative is obtained by using  $k$  independent input operators and one independent output operator:

$$(5.1) \quad B((u_1, v_1), \dots, (u_{k+1}, v_{k+1}))(m)(x_1, \dots, x_k) = v_{k+1}(m(u_1 x_1, \dots, u_k x_k)).$$

Its norm is at most one, and its diagonal is  $jA_{XY}$ . For pullback alone, the corresponding degree is  $k$ . Thus an analytic family always gives an analytic ambient map  $jA_{XY}\gamma$ . The remaining question is whether this map is analytic into  $W$ .

We first record how closedness detects homogeneous coefficients. The conclusion concerns their diagonals, not the values of a chosen multilinear representative on arbitrary tuples.

**Lemma 19** (closedness of homogeneous terms). *Let  $W$  be a closed linear subspace of a normed space  $Z$ , and suppose*

$$f(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h)$$

*is an analytic expansion with values in  $Z$ . If  $f$  takes values in  $W$  near  $x_0$ , then  $p_n(x, \dots, x) \in W$  for every  $n$  and every parameter vector  $x$ .*

*Proof.* For  $n = 0$  this follows by evaluating at  $x_0$ . Inductively, for each fixed  $x$  and sufficiently small nonzero  $t \in K$ , the vector

$$t^{-n} \left( f(x_0 + tx) - \sum_{r < n} t^r p_r(x, \dots, x) \right)$$

belongs to  $W$ . It tends to  $p_n(x, \dots, x)$  as  $t \rightarrow 0$ . To see the convergence directly, choose  $\rho > 0$  and  $M$  such that  $\|p_r\| \rho^r \leq M$  for all  $r$ . The norm of the remaining tail is at most

$$M \left( \frac{\|x\|}{\rho} \right)^n \frac{|t| \|x\| / \rho}{1 - |t| \|x\| / \rho}$$

when  $|t| \|x\| < \rho$ . Nontriviality of the norm provides arbitrarily small nonzero  $t$ . Closedness now proves the induction step.  $\square$

**Theorem 20** (finite-coordinate analytic reflection). *Let  $j: W \hookrightarrow Z$  be a linear isometry with closed image, let  $U \subseteq K^d$  be open, and let  $f: U \rightarrow W$ . If  $jf$  is analytic, then  $f$  is analytic. The spaces  $W, Z$  and the field  $K$  need not be complete. The same statement holds when the parameter space is continuously linearly isomorphic to  $K^d$ .*

*Proof.* Identify  $W$  with its image, use maximum-norm coordinates on  $K^d$ , and expand  $jf$  at a parameter point. Lemma 19 shows that every homogeneous polynomial  $x \mapsto p_n(x, \dots, x)$  has its values in  $W$ . For  $d \geq 1$  and  $n \geq 1$ , let  $e_1, \dots, e_d$  be the coordinate vectors. For each multi-index  $\alpha = (\alpha_1, \dots, \alpha_d)$  of total degree  $n$ , put

$$c_\alpha = \sum_{\substack{(i_1, \dots, i_n) \in \{1, \dots, d\}^n \\ \#\{r: i_r = j\} = \alpha_j \ (1 \leq j \leq d)}} p_n(e_{i_1}, \dots, e_{i_n}).$$

Then  $p_n(x, \dots, x) = \sum_{|\alpha|=n} c_\alpha x^\alpha$ . All  $c_\alpha$  belong to  $W$ : pass to the algebraic quotient  $Z/W$  and use polynomial uniqueness over the infinite field  $K$ . This vector-valued uniqueness follows by applying scalar coordinate functionals on the finite-dimensional span of the coefficients; no topology on the quotient is needed.

Choose, for each  $\alpha$ , the nondecreasing word  $(i_1(\alpha), \dots, i_n(\alpha))$  with those multiplicities, and define the  $W$ -valued multilinear map

$$q_n(x_1, \dots, x_n) = \sum_{|\alpha|=n} c_\alpha \prod_{r=1}^n (x_r)_{i_r(\alpha)}.$$

Its diagonal agrees with that of  $p_n$ . Moreover,

$$(5.2) \quad \|q_n\| \leq \sum_{|\alpha|=n} \|c_\alpha\| \leq d^n \|p_n\|,$$

because the original expansion has exactly  $d^n$  ordered words. The degree-zero coefficient is  $f(x_0)$ . The bounds give a common positive radius for the  $q_n$ . Their partial sums have images under  $j$  converging to  $jf(x)$ , so the isometry shows that they converge to the already given vector  $f(x)$  inside  $W$ . Completeness is therefore unnecessary. The case  $d = 0$  is immediate, and continuous linear coordinates give the last assertion.  $\square$

*Remark 21* (Preservation of the coefficient bound). If  $Z$  has a nonarchimedean norm, both finite sums in the construction satisfy maximum estimates. Thus  $\|c_\alpha\| \leq \|p_n\|$  and  $\|q_n\| \leq \|p_n\|$  for maximum-norm coordinates on  $K^d$ . The reflected expansion has at least the radius of the ambient expansion, without completeness of  $W$  or  $Z$ . The  $c_0$  theorem below extends this same

estimate to infinitely many coordinates; there completeness supplies the infinite coefficient sums.

**Corollary 22.** *Every analytic morphism family parametrized by an open subset of  $K^d$  is admissible for  $\text{Alt}^k$ , in every characteristic and every degree, with arbitrary normed fibers.*

*Proof.* Apply Theorem 20 to the closed isometric inclusion  $j$  above and the analytic map  $jA_{XY}\gamma$ , using (5.1).  $\square$

Over an incomplete field, finite algebraic dimension alone does not imply the existence of continuous coordinates. The coordinate hypothesis in Theorem 20 is consequently essential to its stated scope. When  $K$  is complete, every finite-dimensional normed parameter space has such coordinates. In particular, standard finite-dimensional analytic probes cannot detect the failure of analyticity of the full operator action. Arbitrary normed parameter spaces, including the operator spaces themselves, are needed to recover the regularity notion used in the Main Theorem.

**5.3. The exact local coefficient problem.** The same argument identifies what remains to be proved for a general parameter space.

**Proposition 23.** *Let  $\gamma$  be analytic near  $x_0$  and choose an ambient expansion*

$$jA_{XY}\gamma(x_0 + h) = \sum_{n \geq 0} p_n(h, \dots, h).$$

*The induced map  $A_{XY}\gamma$  is analytic at  $x_0$  if and only if there are bounded  $n$ -linear maps  $q_n$  into  $W$  and  $r > 0$  such that*

$$jq_n(h, \dots, h) = p_n(h, \dots, h) \quad \text{for all } n, h, \quad \sup_n \|q_n\| r^n < \infty.$$

*Proof.* If the induced map is analytic at  $x_0$ , take an expansion there in  $W$ , apply  $j$ , and compare the two expansions on every scalar line. Lemma 7 identifies their homogeneous diagonals. Conversely, shrink  $r$  to lie within the radius of the ambient expansion. The image under  $j$  of the partial sums of the proposed series converges to  $jA_{XY}\gamma(x_0 + h)$ , because its homogeneous diagonals agree term by term with the ambient series. Since  $j$  is an isometry, the partial sums converge inside  $W$  to  $A_{XY}\gamma(x_0 + h)$ .  $\square$

Admissibility on an open parameter domain means that this criterion holds at every point. For an incomplete target, the existence of an expansion at a single point need not give analyticity on a neighborhood. It is therefore not necessary that the chosen maps  $p_n$  themselves take values in  $W$  away from their diagonals. Nor can the condition be replaced by ordinary derivative jets in positive characteristic: Frobenius terms have vanishing derivatives but need not vanish as homogeneous power-series terms. The relevant issue is a bounded multilinear representative for each homogeneous diagonal, with compatible growth across the degrees.

**5.4. The obstruction persists for invertible families.** Restricting to changes of frame does not remove the obstruction on unrestricted parameter spaces. The following elementary construction transfers any bad pullback map to a family of automorphisms.

**Proposition 24** (invertible shear family). *Let  $E, D, F$  be normed  $K$ -spaces, and suppose*

$$Q: \mathcal{L}(E, D) \longrightarrow \mathcal{L}(\text{Alt}^k(D; F), \text{Alt}^k(E; F))$$

*is not analytic at  $u_0$ . Put  $H = D \oplus E$ , with the maximum norm, and define*

$$g(u) = \begin{pmatrix} \text{id}_D & u \\ 0 & \text{id}_E \end{pmatrix}.$$

Then  $g$  and  $g^{-1}$  are affine analytic operator families, but  $u \mapsto g(u)^*$  on  $\text{Alt}^k(H; F)$  is not analytic at  $u_0$ . In particular, the counterexample can occur at  $u_0 = 0$ , where  $g(0) = \text{id}_H$ .

*Proof.* The inverse is  $g(-u)$ . Let

$$J: \text{Alt}^k(D; F) \rightarrow \text{Alt}^k(H; F), \quad Jm = m \circ \pi_D^{\times k},$$

and let  $R: \text{Alt}^k(H; F) \rightarrow \text{Alt}^k(E; F)$  be restriction along  $i_E: E \rightarrow H$ . Both maps are bounded linear, and  $\pi_D g(u) i_E = u$ , so

$$(5.3) \quad R \circ g(u)^* \circ J = Q(u).$$

If  $g(u)^*$  were analytic at  $u_0$ , bounded linear composition with  $R$  and  $J$  would make  $Q$  analytic there. For the last assertion, take the counterexample of Theorem 1, which is nonanalytic at zero.  $\square$

Theorem 1 makes  $E, D, F$ , and hence the parameter space  $\mathcal{L}(E, D)$ , Banach. Thus even an analytic family of invertible operators, with analytic inverse, can induce a nonanalytic family on alternating maps over a Banach base. Equation (5.3) also makes the obstruction precise: it is already visible after bounded linear extension and restriction of forms.

## 6. ANALYTIC VECTOR BUNDLES AND POSITIVE PARAMETER CLASSES

The finite-coordinate theorem gives a bundle construction in every characteristic, even when the fibers are infinite-dimensional. For certain infinite-dimensional bases, summability gives further positive results. These are conditions on the parameter space of a family, and should be distinguished from the conditions on source or target fibers in the earlier positive theorems.

**6.1. Bundles over finite-coordinate bases.** Here an analytic normed vector bundle has operator-valued analytic transition maps in its local trivializations. Bundle morphisms are required to have operator-valued analytic expressions as well. For bundles  $\mathcal{E}, \mathcal{F}$  over the same analytic manifold  $M$ , take common trivializing neighborhoods with fiber transitions  $g_{\alpha\beta}$  and  $h_{\alpha\beta}$ . The candidate transition maps on alternating forms are

$$(6.1) \quad m \longmapsto h_{\alpha\beta}(x) \circ m \circ (g_{\alpha\beta}(x)^{-1}, \dots, g_{\alpha\beta}(x)^{-1}).$$

The inverse  $g_{\alpha\beta}^{-1} = g_{\beta\alpha}$  is an analytic operator family by the bundle atlas. Functoriality gives the cocycle identity for (6.1).

**Theorem 25** (alternating-map bundles). *Let  $M$  be an analytic  $K$ -manifold modeled on  $K^d$ , or on a normed space with continuous linear coordinates in  $K^d$ . For every  $k \geq 0$ , fiberwise alternating maps define a bifunctor*

$$\text{Alt}^k: \text{VB}_K^\omega(M)^{\text{op}} \times \text{VB}_K^\omega(M) \longrightarrow \text{VB}_K^\omega(M),$$

where  $\text{VB}_K^\omega(M)$  denotes analytic normed vector bundles over  $M$  with the operator-valued analytic bundle morphisms just specified. There is no restriction on the characteristic, the degree, or the dimensions and completeness of the fibers.

*Proof.* In each base chart, the pair  $(g_{\alpha\beta}^{-1}, h_{\alpha\beta})$  is an analytic family with a finite-coordinate parameter space. Corollary 22 makes (6.1) analytic. The cocycle identity therefore glues the local products with fiber  $\text{Alt}^k(E; F)$  to an analytic normed vector bundle. The same argument applies to a change of trivialization and to the local expressions of a pair of bundle morphisms, with the first morphism directed contravariantly. The identities and composition laws hold fiberwise by functoriality of alternating maps, giving the stated bifunctor.  $\square$

Together with the unconditional  $C^\infty$  result, this supplies the alternating-map bundle construction for every smoothness order over such finite-coordinate bases. The analytic conclusion does not assert that the action on arbitrary operator spaces is analytic; it uses the base charts as parameter spaces. Over a complete field, the hypothesis includes every finite-dimensional normed model space.

The shear of Proposition 24 explains the limit of this conclusion. On the trivial bundle with base  $\mathcal{L}(E, D)$  and fiber  $H$ , the map  $(u, z) \mapsto (u, g(u)z)$  is an analytic change of trivialization with analytic inverse. Its induced change of trivialization on the alternating-map bundle is not operator-valued analytic. Thus the construction is not preserved by every analytic change of frame over arbitrary Banach bases. This does not assert that the underlying pointwise alternating-map bundle has no analytic structure: the counterexample concerns compatibility with the original analytic bundle atlas.

**6.2. Nonarchimedean  $c_0$  parameters.** For an arbitrary index set  $I$ , write  $c_0(I, K)$  for the space of scalar families tending to zero outside finite sets, with its supremum norm. Completeness is imposed on the output of the coefficient construction, not on the field or the parameter space.

**Theorem 26.** *Let  $K$  be nonarchimedean, let  $j: W \hookrightarrow Z$  be a linear isometry, and assume that  $W$  is complete and  $Z$  has a nonarchimedean norm. If  $U \subseteq c_0(I, K)$  is open and  $jf: U \rightarrow Z$  is analytic for a map  $f: U \rightarrow W$ , then  $f$  is analytic.*

*Proof.* The image of  $j$  is closed. Identify it with  $W$  and expand  $jf$  at a point of  $U$ . Lemma 19, followed by finite-coordinate polynomial uniqueness, puts every grouped coordinate coefficient of  $p_n(x, \dots, x)$  in  $W$ . For  $n \geq 1$ , choose a total order on  $I$ . For a nondecreasing word  $i_1 \leq \dots \leq i_n$ , denote its grouped coefficient by  $c_{i_1, \dots, i_n}$ . It is a sum of values of  $p_n$  on coordinate vectors. The nonarchimedean inequality gives

$$\|c_{i_1, \dots, i_n}\| \leq \|p_n\|.$$

For  $n \geq 1$ , define

$$q_n(x_1, \dots, x_n) = \sum_{i_1 \leq \dots \leq i_n} c_{i_1, \dots, i_n}(x_1)_{i_1} \cdots (x_n)_{i_n}.$$

This sum converges in  $W$ . Indeed, outside a sufficiently large finite set of words at least one coordinate lies outside a chosen finite support approximation for the corresponding  $x_r$ . That factor is arbitrarily small, while all other factors are bounded. Thus the summands tend to zero outside finite sets, which is the summability condition in a complete nonarchimedean normed space. The same estimate gives

$$\|q_n(x_1, \dots, x_n)\| \leq \|p_n\| \prod_{r=1}^n \|x_r\|, \quad \|q_n\| \leq \|p_n\|.$$

Summation preserves multilinearity. On finitely supported vectors the diagonal is the original grouped polynomial, and continuity and density extend this equality to every vector in  $c_0(I, K)$ . Use  $q_0 = f(x_0)$ . The coefficient bounds preserve the ambient radius, and the isometry identifies the sum with  $f$  as in Theorem 20.  $\square$

**Corollary 27.** *Over a nonarchimedean field, every analytic morphism family parametrized by an open subset of  $c_0(I, K)$  is admissible for  $\text{Alt}^k$  when its final value-space target  $F'$  is complete and nonarchimedean. The same holds after a bounded linear isomorphism of parameter spaces.*

*Proof.* When  $F'$  is complete and nonarchimedean, so are  $\text{Alt}^k(E'; F')$  and the operator space  $W$ ; the ambient operator space  $Z$  is nonarchimedean as well. Apply Theorem 26 to  $A_{XY}\gamma$ .  $\square$

**6.3. Ordinary  $\ell^1$  parameters.** For this subsection,  $\ell^1(I, K)$  has the ordinary sum norm  $\|x\|_1 = \sum_{i \in I} |x_i|$ , even when  $K$  is nonarchimedean. The proof uses the fixed degree of the outer polynomial. It does not require a uniform lifting bound for all Taylor degrees of an arbitrary ambient analytic map.

**Lemma 28.** *Let  $W$  be a closed subspace of a Banach space  $Z$ , and let  $B: \ell^1(J, K)^d \rightarrow Z$  be bounded multilinear with diagonal values in  $W$ . Then its diagonal has a bounded  $W$ -valued  $d$ -linear representative  $C$  with*

$$\|C\| \leq d! \|B\|.$$

Here  $d!$  denotes a real bound for a number of terms, not a scalar to be inverted in  $K$ .

*Proof.* For  $d = 0$  the assertion is immediate. Otherwise, choose a total order on  $J$  and group the coordinate coefficients of the diagonal polynomial by nondecreasing words. Finite-coordinate polynomial uniqueness puts each grouped coefficient in  $W$ . Each group contains at most  $d!$  ordered words, so its norm is at most  $d! \|B\|$ . Define  $C$  by the sorted-word formula used above. Its summands are absolutely summable, since

$$\sum_{j_1 \leq \dots \leq j_d} |(x_1)_{j_1} \cdots (x_d)_{j_d}| \leq \prod_{r=1}^d \|x_r\|_1.$$

The sum exists in the complete space  $W$ , is multilinear, and has the asserted bound. Diagonal equality follows first on finite supports and then by density.  $\square$

**Theorem 29** (fixed-degree reflection along  $\ell^1$  families). *Let  $W$  be a closed subspace of a Banach space  $Z$ , let  $H$  be normed, and suppose that  $P: H \rightarrow W$  is the diagonal, in  $Z$ , of a bounded  $d$ -linear map  $B: H^d \rightarrow Z$ . For every analytic map  $\gamma: U \rightarrow H$ , where  $U$  is open in  $\ell^1(I, K)$ , the map  $P\gamma: U \rightarrow W$  is analytic. The field  $K$  need not be complete.*

*Proof.* The case  $d = 0$  is immediate. Extend  $B$  continuously to  $\widehat{H}^d$ , using completeness of  $Z$ . Since  $H$  is dense in its completion and  $W$  is closed, the diagonal of this extension takes values in  $W$ ; call it  $\widehat{P}$ . For every bounded linear  $T: \ell^1(J, K) \rightarrow \widehat{H}$ , Lemma 28, applied to  $B(T, \dots, T)$ , gives a bounded multilinear representative for  $\widehat{P}T$  of norm at most  $d! \|B\| \|T\|^d$ . In particular,  $\widehat{P}T$  is analytic.

It remains to factor  $\gamma$  locally through one such  $T$ . Translate the parameter point to zero and write

$$\gamma(x) = \sum_{n \geq 0} a_n(x, \dots, x).$$

Choose  $s \in K$  with  $0 < |s|$  smaller than a radius of this expansion, so that  $M = \sup_n |s|^n \|a_n\| < \infty$ . Let  $J = \coprod_{n \geq 0} I^n$ , including the single empty word when  $n = 0$ . On the coordinate vectors of  $\ell^1(J, K)$  set

$$T(e_{n, i_1, \dots, i_n}) = s^n a_n(e_{i_1}, \dots, e_{i_n}).$$

These vectors have norm at most  $M$ , so absolute summation in  $\widehat{H}$  gives a bounded linear map  $T: \ell^1(J, K) \rightarrow \widehat{H}$  with norm at most  $M$ .

For  $\|x\|_1 < |s|$ , define a vector in  $\ell^1(J, K)$  by

$$g(x)_{n, i_1, \dots, i_n} = s^{-n} x_{i_1} \cdots x_{i_n}.$$

The sum of the absolute values of its degree- $n$  coordinates is  $|s|^{-n} \|x\|_1^n$ . The geometric tail estimate proves that  $g(x)$  belongs to  $\ell^1(J, K)$  and that its series of homogeneous coordinate blocks converges there. Each block is the diagonal of a bounded  $n$ -linear map of norm at most

$|s|^{-n}$ , so  $g$  is analytic at zero. These are explicitly  $K$ -valued coordinates, and convergence is verified against the explicit vector  $g(x)$ ; completeness of  $K$  is not being used.

Expanding each  $a_n$  in  $\ell^1$  coordinates gives

$$Tg(x) = \sum_{n \geq 0} a_n(x, \dots, x) = \gamma(x)$$

inside  $\widehat{H}$ . Consequently  $P\gamma = (\widehat{P}T)g$  is analytic at the chosen point. Repeating the construction at every parameter point proves the theorem. The factorial estimate used the fixed outer degree  $d$  throughout, rather than the varying Taylor degree  $n$ .  $\square$

**Corollary 30.** *Every analytic morphism family parametrized by an open subset of  $\ell^1(I, K)$  is admissible for  $\text{Alt}^k$  when the final value-space target  $F'$  is complete. The same holds when the parameter space is a bounded linear retract of an  $\ell^1(I, K)$  space. No completeness is required of the other fibers.*

*Proof.* The ambient space  $Z = \mathcal{L}(V_X, \text{Mult}^k(E'; F'))$  is Banach, and  $W$  is closed in  $Z$ . Apply Theorem 29 to the degree- $(k+1)$  polynomial (5.1). For pullback alone one may instead use degree  $k$ . If  $i: P \rightarrow \ell^1(I, K)$  and  $r: \ell^1(I, K) \rightarrow P$  are bounded linear with  $ri = \text{id}_P$ , apply the result to  $\gamma r$  on the open set  $r^{-1}(U)$ , and then restrict along  $i$ .  $\square$

**Corollary 31** (Failure of descent through quotient parameters). *Let  $K$  be a complete nontrivially normed field of characteristic  $p > 0$ , and let  $k \geq p$ . There are Banach spaces  $E, F$ , an index set  $J$ , and a bounded open linear surjection*

$$q: \ell^1(J, K) \longrightarrow H := \mathcal{L}(E, E)$$

*such that  $Q_{E,E,F}^K \circ q$  is analytic, whereas  $Q_{E,E,F}^K$  is analytic at no point. The map  $q$  has no local right inverse on a nonempty open subset of  $H$  that is Fréchet differentiable at even one point of that subset.*

*Proof.* Choose the Banach counterexample  $E, F$  from Theorem 1. Let  $J = \{h \in H : \|h\| \leq 1\}$  and define

$$q((a_h)_{h \in J}) = \sum_{h \in J} a_h h.$$

Absolute summability and completeness of  $H$  make this a bounded linear map of norm at most one. Fix  $t \in K$  with  $0 < |t| = \theta < 1$ . For every nonzero  $h \in H$ , choose an integer  $m$  such that

$$\|h\| \leq |t^m| < \theta^{-1}\|h\|.$$

Then  $v = t^{-m}h$  belongs to  $J$ , and the vector having the single nonzero coordinate  $t^m$  at  $v$  maps to  $h$  and has norm less than  $\theta^{-1}\|h\|$ . Hence  $q$  is surjective, and this uniform bound also proves that it is open. Both its domain and codomain are Banach.

Corollary 30, applied to the analytic family  $a \mapsto (q(a), \text{id}_F)$ , makes  $Q_{E,E,F}^K \circ q$  analytic. The original map is nowhere analytic by construction.

If a local section  $s: U \rightarrow \ell^1(J, K)$  were differentiable at  $h_0 \in U$ , differentiating  $q \circ s = \text{id}_U$  would give

$$q \circ Ds(h_0) = \text{id}_H.$$

The bounded linear map  $Ds(h_0)$  would therefore be a global right inverse of  $q$ . But then

$$Q_{E,E,F}^K = (Q_{E,E,F}^K \circ q) \circ Ds(h_0)$$

would be analytic, a contradiction.  $\square$

Thus an analytic family can become admissible after a bounded linear quotient reparameterization even though it was not admissible originally. The open-cover locality of admissible families does not imply descent through such quotient maps. In particular, the bad parameter space  $H$  cannot be a bounded linear retract of an ordinary  $\ell^1$  space.

The bundle argument of Theorem 25 now gives two further forms of the construction: over bases modeled on  $c_0(I, K)$  when the output fibers are complete and nonarchimedean, and over bases modeled on  $\ell^1(I, K)$  or its bounded linear retracts when the output fibers are complete. For a statement about an entire category of bundles, one can restrict all fibers respectively to nonarchimedean Banach spaces or to Banach spaces; alternating-map fibers remain in the same category.

On arbitrary analytic bases one can instead specify admissible transition families and require admissibility of changes of atlas and bundle morphisms. Proposition 18 ensures compatibility with composition. When the alternating-map spaces at both endpoints are Banach (in particular, when the value fibers are complete), an admissible family of isomorphisms with analytic underlying inverse family has an admissible inverse family. Its alternating action is the inverse of the original action. Locally identify the two endpoint spaces by one fixed isomorphism; inversion is then analytic by the convergent Neumann series in the Banach endomorphism algebra. The shear example shows why arbitrary analytic changes of atlas cannot simply be substituted for admissible ones in this formulation.

## 7. ANALYTIC DOMAINS AND THE FAILURE OF A LARGEST FULL SUBCATEGORY

The positive cases suggest restricting the objects on which the alternating-map bifunctor acts. There is, however, an obstruction to obtaining a single best restriction: two objects can each have analytic endomorphism action while the action on morphisms between them is not analytic. We first identify a canonical class of objects that is compatible with every analytic restriction, and then construct incompatible objects.

### 7.1. A core compatible with every analytic domain.

**Definition 32.** A full subcategory  $\mathcal{D} \subseteq \mathbf{Vec}_K^{\text{op}} \times \mathbf{Vec}_K$  is an *analytic domain* for  $\text{Alt}^k$  if the restriction of  $\text{Alt}^k$  to  $\mathcal{D}$  is analytic on every hom space. Let  $\mathsf{T}_k(K)$  be the full subcategory of normed spaces  $F$  such that  $Q_{E,D,F}^K$  is analytic for every pair of normed spaces  $E, D$ .

The definition of an analytic domain includes morphisms between distinct objects. Requiring only that each endomorphism action be analytic is strictly weaker, as Theorem 37 will show.

**Proposition 33** (The universal-target core). *The full subcategory  $\mathbf{Vec}_K^{\text{op}} \times \mathsf{T}_k(K)$  is an analytic domain. It is the largest analytic domain of the form  $\mathbf{Vec}_K^{\text{op}} \times \mathcal{T}$ . Moreover, adjoining all its objects to any analytic domain again gives an analytic domain. The category  $\mathsf{T}_k(K)$  is closed under bounded linear isomorphisms, bounded linear retracts, and finite products.*

*Proof.* Consider a morphism from  $(E, F)$  to  $(D, G)$ , represented by  $u: D \rightarrow E$  and  $v: F \rightarrow G$ . Writing  $v_*$  for postcomposition, its action has the two factorizations

$$(7.1) \quad \text{Alt}^k(u, v) = v_* \circ Q_{D,E,F}^K(u) = Q_{D,E,G}^K(u) \circ v_*.$$

Operator composition is bounded bilinear. The first expression is therefore jointly analytic if  $F \in \mathsf{T}_k(K)$ , and the second is jointly analytic if  $G \in \mathsf{T}_k(K)$ . Thus a universally analytic target at either endpoint suffices. This proves both analyticity of the core and its compatibility with every analytic domain. Conversely, analyticity on a product with unrestricted first variable, restricted to  $v = \text{id}_F$ , is exactly the defining property of  $\mathsf{T}_k(K)$ .

Suppose  $i: F \rightarrow G$  and  $r: G \rightarrow F$  are bounded linear maps with  $ri = \text{id}_F$ . Then

$$Q_{E,D,F}^K(u) = r_* \circ Q_{E,D,G}^K(u) \circ i_*.$$

Consequently the property passes from  $G$  to its retract  $F$ , and in particular is invariant under bounded linear isomorphism. For a finite product of targets, alternating maps identify boundedly with the product of the corresponding alternating-map spaces; under this identification, pullback acts componentwise. This proves closure under finite products.  $\square$

**Proposition 34** (Compatibility and maximal domains). *Fix a set of objects in the ambient product category. Form a graph whose vertices are those objects with analytic endomorphism action, joining two distinct vertices when both directed morphism-space actions are analytic. Full analytic domains are exactly the cliques in this graph. Every analytic domain extends to one maximal under inclusion, and every maximal domain contains every object of the universal-target core that belongs to the chosen set.*

*Proof.* The clique condition states exactly that every hom-space action in the full subcategory is analytic. The union of a chain of cliques is again a clique: any two of its vertices occur together in one member of the chain. Zorn's lemma gives a maximal extension. Proposition 33 permits adjoining every core object, so maximality forces each such object to be present.  $\square$

This description distinguishes existence of maximal restrictions from existence of a largest one. The counterexample below exhibits two vertices that are not joined. Thus no largest clique exists in any chosen ambient set containing both constructed objects. The set-of-objects qualification makes the use of Zorn's lemma explicit; no choice principle for a proper class of objects is being asserted.

The core is canonical, but its definition is a regularity condition rather than an intrinsic classification of targets. Main Theorem (2) supplies many of its objects. A different sufficient condition depends on the source and target together.

**Definition 35.** A pair  $(E, F)$  is *split in degree  $k$*  if the isometric inclusion

$$j_{E,F}: \text{Alt}^k(E; F) \longrightarrow \text{Mult}^k(E; F)$$

has a bounded linear retraction.

**Lemma 36.** *Every morphism-space action whose destination is a split pair is analytic. In particular, the full subcategory of split pairs is an analytic domain.*

*Proof.* Let  $(E, F)$  be split, with retraction  $R$ , and let the other object be  $(D, G)$ . For  $u: E \rightarrow D$  and  $v: G \rightarrow F$ , the action is

$$m \longmapsto R(v \circ m \circ (u, \dots, u)).$$

Before applying  $R$ , replace the  $k$  copies of  $u$  by independent variables. This gives a bounded  $(k+1)$ -linear lift of the joint action, with values in  $\mathcal{L}(\text{Alt}^k(D; G), \text{Mult}^k(E; F))$ . Postcomposition with  $R$  gives a lift with the required alternating codomain. No compatibility between the retractions chosen for different objects is needed.  $\square$

## 7.2. Two individually analytic objects with nonanalytic cross-action.

**Theorem 37** (No largest full analytic domain). *Let  $p$  be prime, let  $k \geq p$ , and equip  $K = \mathbb{F}_p(t)$  with its  $t$ -adic absolute value. There is no largest full analytic domain for*

$$\text{Alt}^k: \text{Vec}_K^{\text{op}} \times \text{Vec}_K \longrightarrow \text{Vec}_K.$$

The assertion remains true if the domains must be closed under isomorphisms, and if their objects are restricted to pairs of finite-dimensional, nonarchimedean normed  $K$ -spaces. In the latter case, isomorphism closure is understood inside that restricted ambient category.

More precisely, there are objects  $X_k = (E_k, G)$  and  $Y_k = (D_k, G)$  whose singleton full subcategories are analytic, such that  $Y_k$  is split and the action on morphisms from  $Y_k$  to  $X_k$  is not analytic.

Here finite-dimensional means finite *algebraic* dimension over the incomplete field  $K$ . These spaces need not admit continuous coordinates over  $K$ . The theorem does not give a counterexample over a complete field, or a counterexample with Banach objects.

**7.3. The rigid source and the split pair.** We use the following elementary completion principle. Let  $K$  be dense in a complete valued extension  $L$ , let  $(E_i)$  be a finite family of dense  $K$ -subspaces of Banach  $L$ -spaces  $H_i$ , and let  $Z$  be Banach over  $L$ . A bounded  $K$ -multilinear map  $\prod_i E_i \rightarrow Z$  extends uniquely to a bounded  $L$ -multilinear map  $\prod_i H_i \rightarrow Z$ , with the same norm. Indeed, approximate each input by vectors in  $E_i$ . A telescoping expansion and the multilinear bound make the resulting values Cauchy, independently of the approximations. The bound persists in the limit, and density of  $K$  in  $L$  gives  $L$ -linearity in each slot. In the applications below the extended codomain is the complete space  $L$  or  $A$ , while membership of values on the original inputs in the smaller spaces is retained.

Put  $L = \mathbb{F}_p((t))$ , the completion of  $K$ , and consider the normed  $L$ -algebra

$$A = L[\epsilon]/(\epsilon^p), \quad e_i = \epsilon^i \quad (0 \leq i < p),$$

with the maximum norm in this basis. Multiplication satisfies  $\|xy\| \leq \|x\| \|y\|$ . Choose elements  $a_0, \dots, a_{p-1} \in L$ , together with the finitely many auxiliary scalars  $\tau_w$  introduced below, algebraically independent over  $K$ . Such a choice is possible because  $K$  is countable and  $L$  is uncountable: the elements of  $L$  algebraic over any finitely generated extension of  $K$  form a countable set. Set

$$a = \sum_{i=0}^{p-1} a_i e_i, \quad E = K^p + Ka \subseteq A.$$

Every subspace used below carries the norm induced by  $A$  or  $L$ . In particular,  $E$  has  $K$ -dimension  $p+1$  and completion  $A$ .

**Lemma 38.** *Every bounded endomorphism of  $E$  is multiplication by a scalar in  $K$ , and every bounded linear map  $E \rightarrow K$  is zero.*

*Proof.* A bounded  $K$ -linear map extends to completions, and the extension is  $L$ -linear by density of  $K$  in  $L$ . If it preserves  $E$ , its columns, the images of the  $e_i$ , give a matrix  $M + ac^\top$ , with  $M \in M_p(K)$  and  $c \in K^p$ . Its value at  $a$  lies in  $K^p + Ka$ . Comparison of homogeneous degrees in the algebraically independent  $a_i$  shows first that  $a(c^\top a) = 0$ , hence  $c = 0$ , and then that  $M = sI$  for some  $s \in K$ .

Similarly, the extension of a bounded map  $E \rightarrow K$  is an  $L$ -linear functional on  $A$ . Its coefficients belong to  $K$ , since its values on the  $e_i$  do. Its value on  $a$  also lies in  $K$ , so algebraic independence makes all its coefficients zero.  $\square$

Multiplication by  $a$  has lower triangular Toeplitz matrix

$$(M_a)_{ij} = \begin{cases} a_{i-j}, & i \geq j, \\ 0, & i < j. \end{cases}$$

Define

$$D_0 = \text{span}_K\{e_i, \epsilon^i a : 0 \leq i < p\} + K a^2, \quad \mathcal{W} = \{e_i : 0 \leq i < p\} \cup \{e_i + e_j : 0 \leq i < j < p\},$$

and put

$$D = D_0 + \sum_{w \in \mathcal{W}} K \tau_w w, \quad G = \text{span}_K\{\det(d_1, \dots, d_p) : d_i \in D\} \subseteq L.$$

The determinant is taken in the basis  $(e_0, \dots, e_{p-1})$ . These are finite-dimensional  $K$ -spaces, with completions  $A$  and  $L$ :  $D$  contains  $K^p$ , and  $G$  contains 1. By multilinearity,  $G$  is spanned by determinants of the displayed finite generating set for  $D$ . We may therefore identify  $G$  with a finite-dimensional subspace of  $K[a_0, \dots, a_{p-1}, (\tau_w)_w]$ . Every element has degree at most one in each individual  $\tau_w$ , since a nonzero determinant cannot repeat the generator  $\tau_w w$ . Algebraic independence gives

$$(7.2) \quad \tau_w^2 G \cap G = \{0\} \quad (w \in \mathcal{W}).$$

**Lemma 39.** *Every bounded  $p$ -linear map  $D^p \rightarrow G$  is alternating. Thus  $(D, G)$  is split in degree  $p$ .*

*Proof.* Compose such a map  $m$  with  $G \hookrightarrow L$  and extend to a bounded  $L$ -multilinear map on  $A^p$ . Fix two argument positions. Put the same  $w \in \mathcal{W}$  in those positions and arbitrary standard basis vectors in the others, and call the value  $c \in G$ . Replacing both copies of  $w$  by  $\tau_w w \in D$  shows that  $\tau_w^2 c \in G$ , so (7.2) gives  $c = 0$ . Taking  $w = e_i$  gives vanishing on repeated basis vectors. Taking  $w = e_i + e_j$  then gives the signed interchange identity in the two positions. Expanding two equal arbitrary arguments in the basis proves that the completed map, and hence  $m$ , is alternating, including when  $p = 2$ . The inclusion of alternating maps into multilinear maps is therefore an isometric bijection, whose inverse is the desired retraction.  $\square$

**7.4. A missing coefficient.** The coefficient space of alternating maps on  $E$  is

$$C = \{\lambda \in G : a_i \lambda \in G \text{ for every } i\}.$$

Indeed, an alternating map extends to  $\lambda \det$  on the  $p$ -dimensional completion  $A$ . Determinants of tuples from  $E$  span  $\text{span}_K\{1, a_0, \dots, a_{p-1}\}$ : a determinant can contain at most one copy of  $a$ , and replacing one basis vector by  $a$  realizes each  $a_i$  up to sign. Consequently

$$(7.3) \quad \text{Alt}^p(E; G) = C \det.$$

This identification is isometric, since the determinant bound gives one inequality and evaluation on the standard basis gives the reverse inequality.

Algebraically set every  $\tau_w$  to zero. The image of  $G$  is

$$G_0 = \text{span}_K\{\det(d_1, \dots, d_p) : d_i \in D_0\}.$$

Put  $C_0 = \{\lambda \in G_0 : a_i \lambda \in G_0 \text{ for every } i\}$ . Specialization sends  $C$  into  $C_0$ ; no continuity of specialization is asserted. The displayed generators of  $D_0$  are homogeneous polynomial vectors of degrees zero, one, and two. Their determinants are homogeneous, so  $G_0$  is graded, and the defining conditions show that  $C_0$  is graded as well.

**Lemma 40.** *The homogeneous degree-one part of  $C_0$  is zero.*

*Proof.* The degree-two part of  $G_0$  is spanned by the  $2 \times 2$  minors of  $M_a$  and the coordinates of  $a^2$ . These correspond respectively to choosing two degree-one columns, or one degree-two

column, in a determinant of generators. On homogeneous quadratic polynomials define the  $K$ -linear map

$$\Phi(a_i a_j) = \begin{cases} T^{i+j}, & i+j \geq p-1, \\ 0, & i+j < p-1. \end{cases}$$

Every  $2 \times 2$  minor of  $M_a$  is killed by  $\Phi$ . When both product terms are nonzero, their index sums coincide. If only one is nonzero, write the rows as  $r < s$  and columns as  $i < j$ . The surviving term has index sum  $r + s - i - j$ , and the missing term implies  $r < j$ . Hence  $r + s - i - j \leq s - i - 1 \leq p - 2$ , so its image is again zero.

The coefficient of  $\epsilon^r$  in  $a^2$  is  $\sum_{i+j=r} a_i a_j$ . Its image is zero for  $r < p - 1$ ; for  $r = p - 1$  there are exactly  $p$  summands and its image is  $pT^{p-1} = 0$ . Thus  $\Phi$  kills the entire degree-two part of  $G_0$ . If  $\ell = \sum_j c_j a_j$  lies in the degree-one part of  $C_0$ , then  $a_{p-1}\ell \in G_0$ , whereas

$$\Phi(a_{p-1}\ell) = \sum_j c_j T^{p-1+j}.$$

The powers are distinct, so all  $c_j$  vanish. □

**7.5. The nonanalytic cross-action.** Set  $X = (E, G)$  and  $Y = (D, G)$ . Lemma 38 identifies  $\text{End}_K(E)$  isometrically with  $K$ , and the self-action of  $X$  is

$$(s, v) \mapsto s^p v_*.$$

This is a bounded polynomial, so the singleton full subcategory on  $X$  is analytic. The singleton on  $Y$  is analytic by Lemmas 39 and 36.

For  $x \in E$ , multiplication by  $x$  in  $A$  restricts to a bounded map  $M_x: E \rightarrow D$ . Indeed, writing  $x = b + ha$  with  $b \in K^p$  and  $h \in K$ , its values on the generators  $e_j, a$  lie in

$$K^p + \sum_i K \epsilon^i a + K a^2 \subseteq D.$$

Submultiplicativity gives a bounded linear family  $x \mapsto M_x$  from  $E$  to  $\mathcal{L}(E, D)$ . Its matrix is lower triangular with diagonal entry  $x_0$ , so  $\det M_x = x_0^p$ . In the cross-action from  $Y$  to  $X$ , fix the target morphism to  $\text{id}_G$ , evaluate at  $\det \in \text{Alt}^p(D; G)$ , and use (7.3). The resulting map is

$$(7.4) \quad f: E \longrightarrow C, \quad f(x) = x_0^p.$$

Its values lie in  $C$  because it was obtained by pulling back an alternating  $G$ -valued map.

Suppose that  $f$  were analytic at zero. Since  $f(sx) = s^p f(x)$ , uniqueness of power-series coefficients on scalar lines gives a bounded  $p$ -linear map  $B: E^p \rightarrow C$  with diagonal  $f$ . This is the homogeneous coefficient argument used in Proposition 8; it requires no factorial division or completeness. Compose  $B$  with  $C \hookrightarrow L$  and extend to an  $L$ -multilinear map on  $A^p$ . Put

$$b_i = B(e_0, \dots, e_0, e_i) \in C.$$

There are  $p - 1$  initial copies of  $e_0$ , and  $b_0 = f(e_0) = 1$ . Evaluation at  $a$  in the last position gives  $\sum_i a_i b_i \in C$ . Specializing all  $\tau_w$  to zero yields elements  $d_i \in C_0$  with

$$d_0 = 1, \quad \sum_i a_i d_i \in C_0.$$

The degree-one part of this last expression is  $\sum_i a_i (d_i)_0$ , where  $(d_i)_0 \in K$  is the constant term. Its  $a_0$ -coefficient equals one. This contradicts Lemma 40. Thus (7.4), and therefore the cross-action from  $Y$  to  $X$ , is not analytic.

**7.6. Higher degrees and the categorical conclusion.** For  $k = p + r$ , let

$$E_k = E \oplus K^r, \quad D_k = D \oplus K^r$$

with maximum norms. Their completions have dimension  $k$  over  $L$ . Determinants of tuples from  $E_k$  still span  $\text{span}_K\{1, a_0, \dots, a_{p-1}\}$ , so extraction of the top-degree coefficient identifies

$$(7.5) \quad \text{Alt}^k(E_k; G) = C \det_k$$

isometrically. In particular, the coefficient takes values in  $C$ , a fact needed to retain the preceding obstruction after increasing the degree.

By Lemma 38, the extension of every bounded endomorphism of  $E_k$  to its  $L$ -completion  $A \oplus L^r$  has block form

$$\begin{pmatrix} sI_p & B \\ 0 & T \end{pmatrix}, \quad s \in K, \quad T \in M_r(K).$$

The scalar  $s$  and entries of  $T$  are bounded linear functions of the endomorphism, as is seen by evaluation on the standard basis. Its determinant is  $s^p \det T$ , so its action on (7.5) is multiplication by this bounded polynomial. Including postcomposition shows that the singleton full subcategory on  $X_k = (E_k, G)$  is analytic.

The pair  $Y_k = (D_k, G)$  is split. Given  $m \in \text{Mult}^k(D_k; G)$ , restrict the first  $p$  arguments to  $D$  and fix the last  $r$  arguments at the standard auxiliary basis. The resulting  $p$ -linear map is alternating by Lemma 39. Pull it back along  $D_k \rightarrow D$  and take its exterior product with the standard volume form on  $K^r$ . The shuffle formula defines this exterior product without factorial division and gives a bounded linear map from  $\text{Mult}^k(D_k; G)$  to  $\text{Alt}^k(D_k; G)$ . It fixes every alternating map: after extension to the  $k$ -dimensional completion, both maps are determined by their value on the standard basis, where the shuffle formula has exactly one nonzero term. This is the required retraction. For  $r > 0$ , we do not need or assert that every  $k$ -linear map on  $D_k$  is alternating.

Finally, restrict the cross-action along the bounded affine family

$$E \longrightarrow \mathcal{L}(E_k, D_k), \quad x \longmapsto M_x \oplus I_r.$$

Evaluation at  $\det_k$  and coefficient extraction through (7.5) give exactly the nonanalytic map (7.4). Thus the cross-action from  $Y_k$  to  $X_k$  is not analytic.

A largest analytic full subcategory would have to contain both individually analytic singleton subcategories, and fullness would then require this cross-action to be analytic. This is impossible. If closure under isomorphisms is required, replace each singleton by its full isomorphism closure. Every action between two copies is transported from the self-action by bounded linear changes of coordinates, so both closures remain analytic. The same contradiction proves Theorem 37.

*Remark 41.* The theorem also shows that  $X_k$  cannot be adjoined to the entire split-pair category, since that category contains  $Y_k$ . It does not establish that the split-pair category has no other strict analytic extension. Nor does it resolve the largest-domain question for complete fields or Banach pairs. Completing this particular construction removes the obstruction: the sources become finite-dimensional  $L$ -spaces, the target becomes  $L$ , and the coordinate polynomial acquires a bounded multilinear lift.

## 8. A TENSOR CRITERION FOR ANALYTIC REFLECTION

The preceding parameter theorems suggest an intrinsic question about normed spaces: when does every analytic map into an ambient Banach space remain analytic into a closed subspace containing its values? This is stronger than the question for alternating-map families. A tensor

criterion characterizes the stronger property and clarifies why the fixed-degree argument for  $\ell^1$  goes further.

In this section  $K$  is complete and nontrivially normed, and  $P$  is Banach. For  $n \geq 1$ , let  $T_n(P)$  be the separated completed projective tensor power representing bounded  $n$ -linear maps from  $P^n$  to Banach spaces. We use the ordinary projective seminorm, defined by infima of sums of products of norms, and take its separated completion before invoking the universal property. Its canonical multilinear map has norm at most one. Define the closed subspace

$$\Delta_n(P) = \overline{\text{span}\{x^{\otimes n} : x \in P\}} \subseteq T_n(P).$$

A projection onto  $\Delta_n(P)$  will mean a bounded linear map  $R_n : T_n(P) \rightarrow \Delta_n(P)$  whose restriction to  $\Delta_n(P)$  is the identity.

**Proposition 42** (homogeneous reflection). *For a fixed  $n \geq 1$ , the following conditions are equivalent.*

- (1) *For every Banach space  $Z$ , closed subspace  $W \subseteq Z$ , and bounded  $n$ -linear map  $B : P^n \rightarrow Z$  with diagonal values in  $W$ , there is a bounded  $W$ -valued  $n$ -linear map with the same diagonal.*
- (2) *The subspace  $\Delta_n(P)$  is the range of a bounded projection in  $T_n(P)$ .*

*Proof.* Suppose first that a projection  $R_n$  exists. Linearize  $B$  to a bounded map  $\tilde{B} : T_n(P) \rightarrow Z$ . The diagonal hypothesis and closedness of  $W$  imply  $\tilde{B}(\Delta_n(P)) \subseteq W$ . Composing  $\tilde{B}|_{\Delta_n(P)}$  with  $R_n$  and the canonical tensor map gives the required multilinear representative, of norm at most  $\|B\|\|R_n\|$ . Conversely, take  $Z = T_n(P)$ ,  $W = \Delta_n(P)$ , and  $B$  equal to the canonical tensor map. Linearizing the representative supplied by the first condition gives a bounded map into  $\Delta_n(P)$  fixing every pure power. It fixes their linear span and, by continuity, their closed span, so it is a projection.  $\square$

**Definition 43.** The Banach parameter space  $P$  has *universal analytic reflection* if, for every Banach space  $Z$ , closed subspace  $W \subseteq Z$ , every open  $U \subseteq P$ , and every map  $f : U \rightarrow W$ , analyticity of the composite  $U \rightarrow W \hookrightarrow Z$  implies analyticity of  $f$ .

**Theorem 44** (analytic tensor criterion). *A Banach space  $P$  has universal analytic reflection if and only if there are projections  $R_n : T_n(P) \rightarrow \Delta_n(P)$  for all  $n \geq 1$  such that*

$$(8.1) \quad \limsup_{n \rightarrow \infty} \|R_n\|^{1/n} < \infty.$$

*Proof.* Assume first that the projections exist. Expand an ambient analytic map with values in  $W$  at a point. Lemma 19 puts every homogeneous diagonal in  $W$ . The construction in Proposition 42 replaces its coefficient  $p_n$  by a  $W$ -valued coefficient of norm at most  $\|p_n\|\|R_n\|$  and with the same diagonal. The exponential bound preserves a positive radius of convergence. The degree-zero coefficient is already in  $W$ , and the resulting expansion represents the given map.

Conversely, form the Banach spaces with supremum norms

$$Z = c_0(\{T_n(P)\}_{n \geq 1}), \quad W = c_0(\{\Delta_n(P)\}_{n \geq 1}) \subseteq Z.$$

Here  $c_0$  denotes families tending to zero in their respective Banach coordinate spaces;  $W$  is closed. On the open unit ball of  $P$ , define

$$f(x) = (x^{\otimes n})_{n \geq 1}.$$

Since  $\|x^{\otimes n}\| \leq \|x\|^n$ , this vector lies in  $W$ . As a  $Z$ -valued map it is analytic: its degree- $n$  coefficient is the canonical tensor map placed in coordinate  $n$ , and has norm at most one. This

gives an expansion at zero. To re-expand at  $a$  with  $\|a\| < 1$ , choose  $r > 0$  with  $\|a\| + r < 1$ . Expanding each tensor power in  $a + h$  gives translated coefficients, and the sum of all term bounds is at most  $\sum_{n \geq 1} (\|a\| + r)^n < \infty$ . Completeness of  $Z$  supplies the coefficients and the rearranged sum, proving analyticity throughout the unit ball. Universal reflection gives a  $W$ -valued expansion  $\sum_m q_m(x, \dots, x)$  near zero with  $\|q_m\| \leq C\rho^{-m}$  for some  $C, \rho > 0$ . Compose this expansion with the projection  $\pi_n: W \rightarrow \Delta_n(P)$  and compare on scalar lines with  $x \mapsto x^{\otimes n}$ . Lemma 7 gives

$$\pi_n q_n(x, \dots, x) = x^{\otimes n} \quad (x \in P).$$

Linearizing  $\pi_n q_n$  therefore gives a projection  $R_n$  onto  $\Delta_n(P)$  with  $\|R_n\| \leq \|q_n\| \leq C\rho^{-n}$ , proving (8.1).  $\square$

**8.1. Projection bounds and the special role of fixed degree.** For  $P = K^d$  with the maximum norm, the coordinate construction in Theorem 20, applied to the canonical tensor map, gives  $\|R_n\| \leq d^n$  for  $d \geq 1$ . A bounded linear change of coordinates introduces only a fixed distortion factor to the  $n$ -th power, so it does not change the exponential-growth condition.

For the ordinary sum-norm space  $P = \ell^1(I, K)$ , the projective tensor power is isometrically  $\ell^1(I^n, K)$ . Indeed, the coordinate tensor map has norm at most one, while expansion in coordinate tensors gives the reverse projective-norm estimate; completion then gives the identification. In these coordinates,  $\Delta_n(P)$  consists exactly of arrays constant on each permutation orbit in  $I^n$ . One inclusion follows because every pure power is constant on such orbits. For the other, finite-coordinate polynomial interpolation puts each finite orbit sum in the span of pure powers, and finite sums of orbit sums are dense in the space of orbit-constant  $\ell^1$  arrays.

Choose one representative of each orbit, send its basis vector to the sum of all basis vectors in that orbit, and send the other basis vectors in the orbit to zero. This gives a projection with norm at most  $n!$ . Over a nonarchimedean field and for infinite  $I$ , this estimate is optimal even among projections that do not preserve the individual orbit blocks.

**Proposition 45.** *If  $K$  is nonarchimedean and complete, and  $I$  is infinite, then*

$$\inf\{\|R\| : R: T_n(\ell^1(I, K)) \rightarrow \Delta_n(\ell^1(I, K)) \text{ is a bounded projection}\} = n!.$$

*Consequently  $\ell^1(I, K)$  does not have universal analytic reflection.*

*Proof.* The preceding construction gives the upper bound. For the lower bound, take an orbit  $\mathcal{O}$  of a tuple with  $n$  distinct indices; it has  $n!$  elements. For each  $w \in \mathcal{O}$ , let  $\lambda_w$  be the common coefficient on  $\mathcal{O}$  of  $R(e_w)$ . The orbit sum  $\sum_{w \in \mathcal{O}} e_w$  belongs to  $\Delta_n(P)$  and is fixed by  $R$ , hence

$$\sum_{w \in \mathcal{O}} \lambda_w = 1.$$

The nonarchimedean inequality implies that some  $|\lambda_w| \geq 1$ . For this  $w$ , the ordinary sum norm satisfies  $\|R(e_w)\|_1 \geq n!|\lambda_w| \geq n!$ . Thus  $\|R\| \geq n!$ . Finally,  $(n!)^{1/n}$  is unbounded, so Theorem 44 excludes universal analytic reflection.  $\square$

**Corollary 46** (Entire polynomial approximation of a nonanalytic smooth map). *Let  $K$  be a complete nonarchimedean nontrivially normed field and put  $P = \ell^1(\mathbb{N}, K)$ , with the ordinary sum norm. There are Banach spaces  $W \subseteq Z$ , with  $W$  closed, and a map  $f$  on the open unit ball of  $P$  such that  $f$  is analytic into  $Z$  and  $C^\infty$  into  $W$ , but is not analytic into  $W$  at zero. Moreover,  $f$  is a uniform limit on every smaller ball of entire  $W$ -valued polynomials.*

*Proof.* Take

$$Z = c_0(\{T_n(P)\}_{n \geq 1}), \quad W = c_0(\{\Delta_n(P)\}_{n \geq 1}), \quad f(x) = (x^{\otimes n})_{n \geq 1}.$$

The spaces are Banach and  $W$  is closed in  $Z$ . As in the proof of Theorem 44, the ambient expansion has coefficient norms at most one and defines an analytic map on the open unit ball. Smooth reflection through the closed isometric inclusion makes  $f$   $C^\infty$  as a map into  $W$ .

Suppose a  $W$ -valued expansion at zero has coefficients  $q_n$ . Uniqueness on scalar lines gives

$$\pi_n q_n(x, \dots, x) = x^{\otimes n},$$

where  $\pi_n: W \rightarrow \Delta_n(P)$  is the coordinate projection. Linearizing  $\pi_n q_n$  gives a projection  $T_n(P) \rightarrow \Delta_n(P)$  of norm at most  $\|q_n\|$ . Proposition 45 therefore gives  $\|q_n\| \geq n!$ . Since  $(n!)^{1/n}$  is unbounded, these coefficients give a formal multilinear series of radius zero, contradicting the assumed analytic expansion.

For each  $n$ , choose the sorted-orbit projection  $R_n: T_n(P) \rightarrow \Delta_n(P)$ . Its composition with the canonical tensor map is a bounded multilinear representative of  $x \mapsto x^{\otimes n}$  with values in  $\Delta_n(P)$ . Consequently the truncation

$$f_N(x) = (x, x^{\otimes 2}, \dots, x^{\otimes N}, 0, \dots)$$

is an entire  $W$ -valued polynomial. For  $0 < r < 1$  and  $\|x\| \leq r$ ,

$$\|f(x) - f_N(x)\|_W \leq r^{N+1},$$

which proves the asserted uniform convergence.  $\square$

This example is characteristic-free within the nonarchimedean setting. Every individual homogeneous term has an analytic lift, but their best multilinear bounds grow factorially. The fixed-degree argument for alternating families avoids this obstruction by using one outer degree throughout.

There is no conflict with Corollary 30: every alternating-map family on these parameter spaces is still admissible when its final output fiber is complete. The ambient alternating action has a fixed polynomial degree. Factoring a general analytic parameter map through a free  $\ell^1$  space therefore incurs a single factorial bound, rather than a sequence of factorial bounds across Taylor degrees. The special alternating-family property is consequently strictly weaker than universal analytic reflection.

For comparison, in the category of nonarchimedean normed spaces with the *nonarchimedean* projective tensor norm, the corresponding tensor power of  $c_0(I, K)$  is  $c_0(I^n, K)$  and the same sorted-orbit projection has norm at most one. This uses the infimum of maxima in the tensor seminorm, rather than the ordinary projective seminorm used in Theorem 44. It is the tensor counterpart of the norm-preserving coefficient construction in Theorem 26; the two tensor norms must not be interchanged.

Theorem 44 thus gives a complete answer for universal Banach-valued analytic reflection. It supplies a sufficient parameter criterion for alternating families, while Proposition 45 shows that it is not a necessary one. An intrinsic characterization of the parameter spaces for which every alternating-map family is admissible remains a separate question.

## 9. THE MULTIPLIER THEOREM OVER A FINITE FIELD

This section proves the algebraic obstruction used in the Laurent-series construction: over a finite field in which  $k! = 0$ , coordinatewise multiplication admits no multilinear lift satisfying a polarized diagonal identity and a uniform bound on exterior support. We give a roadmap of the complete construction, introduce the exterior support estimates, and then distinguish the diagonal identities: over a finite field equality at all scalar points need not be a polynomial identity.

**Roadmap of the Banach counterexample.** The proof separates into an algebraic obstruction, its realization by a normed space, and passage to the prescribed field. The analytic input is a lift criterion valid in every characteristic: if  $Q$  is analytic at any point, the degree- $k$  term of its power series has diagonal  $Q$ . One-variable uniqueness identifies this diagonal without dividing by  $k!$  (Proposition 8).

*The finite-field obstruction.* Let  $\kappa$  be finite and put  $E_0 = \kappa^{\mathbb{N}}$ . Theorem 55 rules out a multilinear lift of coordinatewise multiplication satisfying the polarized diagonal identity (Poll), with values in  $\Lambda_{\kappa}^k E_0$  when  $k! = 0$  and every value of the lift lies in an exterior power of a subspace of uniformly bounded dimension. Ramsey’s theorem first makes the coefficients depend only on the order and equality pattern of their indices. Signed sums over clusters of four indices cancel the terms not tied to the output. A staircase of clusters converts the support bound into invariance under adjacent transpositions; the polarized diagonal identity then forces  $k! \chi = 1$  for a common cluster value  $\chi$ , a contradiction.

*The Laurent series construction.* Over  $K_1 = \kappa((X))$ , take  $E_1 = \ell^{\infty}(\mathbb{N}, K_1)$  and let  $B$  be the completion of  $\Lambda_{K_1}^k E_1$  under its projective norm, defined using ordinary sums rather than maxima. The essential estimate is that the constant-coefficient map

$$\eta: B \longrightarrow \Lambda_{\kappa}^k E_0$$

is well defined and has uniformly bounded support dimension on bounded sets (Proposition 63). Evaluating a hypothetical lift on multiplication operators and on the universal wedge map, then applying  $\eta$ , produces exactly the finite-field lift excluded above. The coefficient map is  $\kappa$ -linear; no  $K_1$ -linearity is asserted or needed.

*Passage to the prescribed field.* Choose  $t \in K$  with  $0 < |t| < 1$ . The closure of  $\mathbb{F}_p(t)$  in the completion  $\widehat{K}$  is a spherically complete Laurent series field  $K_1 \cong \mathbb{F}_p((X))$ , with  $|X| = |t|$ . Ingleton’s Hahn–Banach theorem [4] gives a contracting  $K_1$ -linear projection  $\widehat{K} \rightarrow K_1$ . It makes the completed projective tensor products

$$E = E_1 \widehat{\otimes}_{\pi, K_1} \widehat{K}, \quad F = B \widehat{\otimes}_{\pi, K_1} \widehat{K}$$

contain  $E_1$  and  $B$  isometrically, and allows a lift over  $\widehat{K}$  to descend to one over  $K_1$ . Density handles an incomplete  $K$ . These are the explicit witnesses in Theorem 1. Finally, sums of unit wedges on disjoint blocks have norms growing linearly with the number of blocks. This growth persists in  $F$  and rules out every equivalent nonarchimedean norm (Proposition 79).

**9.1. Exterior powers, supports and determinant arrays.** Let  $L$  be a field,  $V$  an  $L$ -vector space and  $k \geq 1$ . We write  $\Lambda^k V = \Lambda_L^k V$  for the algebraic exterior power, in which  $x_1 \wedge \cdots \wedge x_k = 0$  whenever two  $x_i$  coincide. We use the standard facts: for a subspace  $W \subseteq V$  the map  $\Lambda^k W \rightarrow \Lambda^k V$  is injective, and we regard  $\Lambda^k W$  as a subspace of  $\Lambda^k V$ ; if  $u_1, \dots, u_d$  is a basis of a finite-dimensional space  $U$ , the wedges  $u_I := u_{i_1} \wedge \cdots \wedge u_{i_k}$ ,  $I = \{i_1 < \cdots < i_k\}$ , form a basis of  $\Lambda^k U$ ; and every  $\omega \in \Lambda^k V$  lies in  $\Lambda^k U$  for some finite-dimensional  $U \subseteq V$ .

**Definition 47.** The *support dimension* of  $\omega \in \Lambda^k V$  is

$$\text{sdim}(\omega) := \min\{\dim W : W \subseteq V \text{ finite-dimensional, } \omega \in \Lambda^k W\}.$$

The definition gives  $\text{sdim}(0) = 0$ ,  $\text{sdim}(c\omega) = \text{sdim}(\omega)$  for  $c \neq 0$ ,  $\text{sdim}(\omega + \omega') \leq \text{sdim}(\omega) + \text{sdim}(\omega')$  (use  $W + W'$ ), and  $\text{sdim}(\omega) \leq \dim \text{span } S$  if  $\omega$  is a sum of wedges of vectors from a finite set  $S$ . If  $V \subseteq V'$ , then  $\text{sdim}$  computed in  $V'$  is at most  $\text{sdim}$  computed in  $V$ .

For spaces of functions, exterior vectors can be detected by determinants of coordinate evaluations. These arrays also turn bounds on support dimension into ordinary matrix-rank

bounds. Let  $S$  be a set and  $V \subseteq L^S$  a space of functions on  $S$ . The *determinant array* is the linear map

$$\Omega: \Lambda^k V \rightarrow L^{S^k}, \quad \Omega_{y_1 \wedge \dots \wedge y_k}(c_1, \dots, c_k) := \det(y_b(c_a))_{a,b=1}^k.$$

It is well defined because the right side is multilinear and alternating in  $(y_1, \dots, y_k)$  (these are the columns). For each  $\omega$ , the function  $\Omega_\omega$  is alternating in  $(c_1, \dots, c_k)$  (the rows); in particular  $\Omega_\omega(c_{\sigma(1)}, \dots, c_{\sigma(k)}) = \text{sgn}(\sigma) \Omega_\omega(c_1, \dots, c_k)$ .

**Lemma 48.** *The determinant array  $\Omega: \Lambda^k V \rightarrow L^{S^k}$  is injective.*

*Proof.* Let  $\Omega_\omega = 0$  and choose a finite-dimensional  $U \subseteq V$  with  $\omega \in \Lambda^k U$ . The evaluations  $\text{ev}_s|_U$ ,  $s \in S$ , separate the points of  $U$  (a function in  $U$  vanishing at every  $s$  is 0), so they span the dual space  $U^*$ . Choose  $s_1, \dots, s_d \in S$  such that the  $\text{ev}_{s_i}|_U$  form a basis of  $U^*$ , and let  $u_1, \dots, u_d$  be the dual basis of  $U$ , so that  $u_j(s_i) = \delta_{ij}$ . For increasing  $k$ -subsets  $I, J$  of  $\{1, \dots, d\}$  we get  $\Omega_{u_I}(s_{j_1}, \dots, s_{j_k}) = \det(u_{i_b}(s_{j_a}))_{a,b} = \delta_{IJ}$ . Writing  $\omega = \sum_I c_I u_I$  therefore gives  $0 = \Omega_\omega(s_{j_1}, \dots, s_{j_k}) = c_J$  for every  $J$ , so  $\omega = 0$ .  $\square$

**Lemma 49** (support bounds flattening rank). *Let  $V \subseteq L^S$ ,  $\omega \in \Lambda^k V$  and  $\alpha \in \{1, \dots, k\}$ . Let  $R \subseteq S$  and  $R' \subseteq S^{\{1, \dots, k\} \setminus \{\alpha\}}$  be finite. The  $R \times R'$  matrix  $M$  with entries  $M(c_\alpha, c') := \Omega_\omega(c)$ , where  $c$  has  $c_\alpha$  in slot  $\alpha$  and the entries of  $c'$  in the other slots, has rank at most  $\text{sdim}(\omega)$ .*

*Proof.* Write  $\omega = \sum_j y_{j,1} \wedge \dots \wedge y_{j,k}$  with all  $y_{j,b}$  in a subspace  $W$  of dimension  $\text{sdim}(\omega)$ . Expanding each determinant along row  $\alpha$ ,  $\det(y_{j,b}(c_a))_{a,b} = \sum_b \pm y_{j,b}(c_\alpha) \cdot \mu_{j,b}(c')$ , where the minor  $\mu_{j,b}(c')$  does not involve  $c_\alpha$ . So for fixed  $c'$  the column  $c_\alpha \mapsto M(c_\alpha, c')$  is a linear combination of the restrictions  $y_{j,b}|_R$ . All columns lie in the restriction of  $W$  to  $R$ , a space of dimension at most  $\text{sdim}(\omega)$ .  $\square$

A second way to obtain lower bounds on support is to contract all but one exterior factor. The resulting vectors must lie in every subspace supporting the original exterior vector.

**Lemma 50** (contraction). *Let  $k \geq 2$  and  $\varphi = (\varphi_1, \dots, \varphi_{k-1}) \in (V^*)^{k-1}$ . There is a linear map  $c_\varphi: \Lambda^k V \rightarrow V$  with*

$$c_\varphi(x_1 \wedge \dots \wedge x_k) = \sum_{i=1}^k (-1)^{k+i} \det(\varphi_a(x_b))_{1 \leq a \leq k-1, b \neq i} x_i,$$

*and  $c_\varphi(\Lambda^k W) \subseteq W$  for every subspace  $W$ . Consequently  $\text{sdim}(\omega) \geq \dim \text{span}\{c_\varphi(\omega) : \varphi \in (V^*)^{k-1}\}$ .*

*Proof.* The right side is multilinear in  $(x_b)$ . For  $\psi \in V^*$ , applying  $\psi$  to it gives, by Laplace expansion along the last row, the determinant of the  $k \times k$  matrix with entries  $\varphi_a(x_b)$  in the rows  $a < k$  and  $\psi(x_b)$  in the last row; it vanishes when two columns coincide. Since  $V^*$  separates the points of  $V$ , the right side vanishes whenever  $x_a = x_b$  for some  $a \neq b$ , so it factors through  $\Lambda^k V$ . Each term is a multiple of some  $x_i$ , which gives  $c_\varphi(\Lambda^k W) \subseteq W$ . If  $\omega \in \Lambda^k W$  with  $\dim W = \text{sdim}(\omega)$ , all  $c_\varphi(\omega)$  lie in  $W$ .  $\square$

**Proposition 51** (canonical exterior support). *Let  $L$  be any field,  $V$  an  $L$ -vector space,  $k \geq 2$ , and  $\omega \in \Lambda^k V$ . Then*

$$S(\omega) := \text{span}\{c_\varphi(\omega) : \varphi \in (V^*)^{k-1}\}$$

*is the unique smallest subspace supporting  $\omega$ . In particular,  $\omega \in \Lambda^k S(\omega)$  and  $\text{sdim}(\omega) = \dim S(\omega)$ .*

*Proof.* Choose a finite-dimensional supporting subspace  $U \subseteq V$ . Lemma 50 gives  $S(\omega) \subseteq U$ , so  $S(\omega)$  is finite-dimensional. Choose a basis  $e_1, \dots, e_d$  of  $U$  whose first  $r$  vectors form a basis of  $S(\omega)$ , and extend its coordinate functionals algebraically to  $V$ . Write  $\omega = \sum_{|I|=k} a_I e_I$  in the exterior basis of  $U$ . If  $j \in I$  and  $j > r$ , contract against the  $k-1$  coordinate functionals indexed by  $I \setminus \{j\}$ . The  $e_j$ -coordinate of this contraction is  $\pm a_I$ : no other basis wedge contributes to that coordinate. The contraction belongs to  $S(\omega)$ , so this coordinate is zero. Hence  $a_I = 0$  whenever  $I$  meets  $\{r+1, \dots, d\}$ , proving  $\omega \in \Lambda^k S(\omega)$ . Every supporting subspace contains all the contractions by Lemma 50, and therefore contains  $S(\omega)$ . This proves minimality and the dimension formula. Only unit signs occur, so the argument is valid in every characteristic.  $\square$

The same conclusion holds in degree one: under  $\Lambda^1 V \simeq V$ , the empty-tuple contraction is the identity and  $S(\omega) = L\omega$ . In degree zero, every element of  $\Lambda^0 V = L$  is supported by the zero subspace, so its support dimension is zero; no contraction to  $V$  is needed.

For matrix calculations, choose any finite-dimensional supporting subspace  $U$  and a basis of it. Form the finite matrix whose columns are the coordinates of the contractions against  $(k-1)$ -tuples of coordinate functionals, extended to  $V$ . Arbitrary functionals restrict to linear combinations of these coordinates on  $U$ ; multilinearity in the covectors therefore shows that the column span is exactly  $S(\omega)$ . The matrix consequently has rank  $\text{sdim}(\omega)$ . This uses ordinary matrix rank, and entails no choice of a rank convention for an infinite matrix.

**9.2. Lifts and three diagonal identities.** Fix a field  $L$ , an integer  $k \geq 1$ ,  $L$ -vector spaces  $A$  and  $V$ , and a linear map  $D: A \rightarrow \text{End}_L(V)$ ,  $a \mapsto D_a$ . A *lift of  $D$  in degree  $k$*  is a  $2k$ -linear map  $\Psi: A^k \times V^k \rightarrow \Lambda^k V$  over  $L$  that is alternating in the  $V$ -slots; its diagonal identity will be specified separately. For every  $n \geq 1$  write  $[n] := \{1, \dots, n\}$ , and for  $\alpha \in \mathbb{N}^m$  put  $|\alpha| := \sum_i \alpha_i$ . Given  $m \geq 1$ , a map  $f: [k] \rightarrow [m]$ , a family  $b = (b_1, \dots, b_m) \in A^m$  and  $\lambda \in L^m$ , put

$$b_f := (b_{f(1)}, \dots, b_{f(k)}), \quad \text{type } f := (|f^{-1}(1)|, \dots, |f^{-1}(m)|) \in \mathbb{N}^m, \quad \lambda_f := \prod_{j=1}^k \lambda_{f(j)} = \lambda^{\text{type } f}.$$

The *pointwise identity* is

$$(Pw) \quad \Psi(a, \dots, a; x_1, \dots, x_k) = D_a x_1 \wedge \dots \wedge D_a x_k \quad \text{for all } a \in A, x \in V^k.$$

The *polarized identity* is: for all  $m \geq 1$ ,  $b \in A^m$ ,  $\alpha \in \mathbb{N}^m$  with  $|\alpha| = k$ , and  $x \in V^k$ ,

$$(Pol) \quad \sum_{\text{type } f = \alpha} \Psi(b_f; x) = \sum_{\text{type } f = \alpha} D_{b_{f(1)}} x_1 \wedge \dots \wedge D_{b_{f(k)}} x_k.$$

Its *multilinear instance* is: for all  $b_1, \dots, b_k \in A$  and  $x \in V^k$ ,

$$(Pol1) \quad \sum_{\sigma \in S_k} \Psi(b_{\sigma(1)}, \dots, b_{\sigma(k)}; x) = \sum_{\sigma \in S_k} D_{b_{\sigma(1)}} x_1 \wedge \dots \wedge D_{b_{\sigma(k)}} x_k.$$

Here  $\sum_{\text{type } f = \alpha}$  runs over the maps  $f: [k] \rightarrow [m]$  of type  $\alpha$ . (Pol1) is the instance  $m = k$ ,  $\alpha = (1, \dots, 1)$  of (Pol), since the maps of type  $(1, \dots, 1)$  are the permutations. For  $b \in A^m$  and  $x \in V^k$  put  $a_\lambda := \sum_i \lambda_i b_i$  and

$$\begin{aligned} \Delta_{b,x}(\lambda) &:= \Psi(a_\lambda, \dots, a_\lambda; x) - D_{a_\lambda} x_1 \wedge \dots \wedge D_{a_\lambda} x_k = \sum_{|\alpha|=k} \lambda^\alpha \delta_\alpha(b, x), \\ \delta_\alpha(b, x) &:= \sum_{\text{type } f = \alpha} \left[ \Psi(b_f; x) - D_{b_{f(1)}} x_1 \wedge \dots \wedge D_{b_{f(k)}} x_k \right]. \end{aligned}$$

Multilinearity of  $\Psi$  in the  $A$ -slots, linearity of  $D$  and multilinearity of the wedge give the displayed expansion. Thus  $\Delta_{b,x}$  is a homogeneous polynomial of degree  $k$  with coefficients in

$\Lambda^k V$ . Identity (Pol) says that every coefficient  $\delta_\alpha(b, x)$  is zero. Identity (Pw) says instead that  $\Delta_{b,x}$  vanishes at every point of  $L^m$ ; conversely, this pointwise vanishing gives (Pw) by taking  $m = 1$  and  $\lambda = 1$ . The following two identity principles identify the cases in which pointwise vanishing forces all coefficients to vanish.

**Lemma 52** (identity principle). *Let  $\mathbb{K}$  be an infinite field,  $Y$  a  $\mathbb{K}$ -vector space and  $g(\lambda) = \sum_\alpha \lambda^\alpha y_\alpha$  a polynomial in  $\lambda \in \mathbb{K}^m$  with finitely many nonzero coefficients  $y_\alpha \in Y$ . If  $g(\lambda) = 0$  for all  $\lambda \in \mathbb{K}^m$ , then every  $y_\alpha = 0$ .*

*Proof.* The finitely many coefficients  $y_\alpha$  span a finite-dimensional subspace of  $Y$ . Applying its coordinate functionals reduces the claim to scalar polynomials. For these, induction on  $m$  reduces to the fact that a nonzero polynomial in one variable has only finitely many roots.  $\square$

**Lemma 53** (homogeneous forms over a finite field). *Let  $q$  be a prime power,  $L' \supseteq \mathbb{F}_q$  a field,  $1 \leq e \leq q$ , and  $g \in L'[\lambda_1, \dots, \lambda_m]$  homogeneous of degree  $e$  with  $g(\lambda) = 0$  for every  $\lambda \in \mathbb{F}_q^m$ . Then  $g = 0$ .*

*Proof.* Call a monomial  $\lambda^\beta$  *reduced* if every  $\beta_i \leq q - 1$ . The indicator function of a point  $a \in \mathbb{F}_q^m$  is  $\prod_i (1 - (\lambda_i - a_i)^{q-1})$ , which expands into reduced monomials. So the  $q^m$  reduced monomials span the  $q^m$ -dimensional  $L'$ -space of functions  $\mathbb{F}_q^m \rightarrow L'$ , and they are linearly independent as functions. A monomial of degree  $e \leq q$  that is not reduced is  $\lambda_i^q$  (and then  $e = q \geq 2$ ); as a function on  $\mathbb{F}_q^m$  it agrees with the reduced monomial  $\lambda_i$ , of degree  $1 < e$ . If  $e = q$ , replace each  $\lambda_i^q$  in  $g$  by  $\lambda_i$ . These degree-one images are distinct, and homogeneity separates them from all the remaining monomials, which still have degree  $q$ . The replacement therefore sends the monomials of  $g$  injectively to reduced monomials without changing the associated function. Linear independence of the reduced monomials forces every coefficient to vanish. If  $e < q$ , no replacement is needed.  $\square$

**Proposition 54** (pointwise versus polarized). *Let  $\Psi$  be a lift of  $D$  in degree  $k$  over a field  $L$ .*

- (1) (Pol)  $\Rightarrow$  (Pw)  $\Rightarrow$  (Pol1), over every field.
- (2) If  $L$  is infinite, or if  $L = \mathbb{F}_q$  and  $k \leq q$ , then (Pw)  $\Rightarrow$  (Pol).
- (3) If  $L = \mathbb{F}_q$  and  $k \geq q + 1$ , then (Pw) does not imply (Pol) in general.
- (4) If  $k! = 0$  in  $L$ , then (Pol1) does not imply (Pw) in general.

*Proof.* (1) *Inclusion-exclusion.* The implication (Pol)  $\Rightarrow$  (Pw) is the case  $m = 1$  and  $\alpha = (k)$ . To prove (Pw)  $\Rightarrow$  (Pol1), fix  $b_1, \dots, b_k \in A$  and  $x \in V^k$ , and put  $b_S := \sum_{i \in S} b_i$  for  $S \subseteq [k]$ . Multilinearity gives

$$\begin{aligned} \sum_{S \subseteq [k]} (-1)^{k-|S|} \Psi(b_S, \dots, b_S; x) &= \sum_S (-1)^{k-|S|} \sum_{f: [k] \rightarrow S} \Psi(b_f; x) \\ &= \sum_{f: [k] \rightarrow [k]} \Psi(b_f; x) \sum_{S \supseteq f([k])} (-1)^{k-|S|} = \sum_{f \text{ bijective}} \Psi(b_f; x), \end{aligned}$$

because  $\sum_{S \supseteq T} (-1)^{k-|S|} = (1 - 1)^{k-|T|}$  is 1 if  $T = [k]$  and 0 otherwise. The same computation applies to the multilinear map  $(a_1, \dots, a_k) \mapsto D_{a_1} x_1 \wedge \dots \wedge D_{a_k} x_k$ , whose diagonal is the right side of (Pw). Evaluating (Pw) at the  $2^k$  subset sums  $b_S$  therefore gives (Pol1). The computation is valid in every characteristic and involves no division by  $k!$ .

(2) *The identity principles.* The coefficients of  $\Delta_{b,x}$  span a finite-dimensional subspace of  $\Lambda^k V$ ; in a basis of it each coordinate of  $\Delta_{b,x}$  is a scalar form of degree  $k$  vanishing on  $L^m$ . It is zero by Lemma 52 if  $L$  is infinite, and by Lemma 53 if  $L = \mathbb{F}_q$  and  $k \leq q$ .

(3) *Pointwise vanishing without full polarization.* Let  $\dim A \geq 3$ , with  $b_1, b_2, b_3 \in A$  and functionals  $g_1, g_2, g_3$  on  $A$  such that  $g_i(b_j) = \delta_{ij}$ , and let  $y_1, \dots, y_k \in V$  be linearly independent. Put

$$\theta(a_1, \dots, a_k) := [g_1(a_1) \cdots g_1(a_q) g_2(a_{q+1}) - g_1(a_1) g_2(a_2) \cdots g_2(a_{q+1})] \cdot g_3(a_{q+2}) \cdots g_3(a_k)$$

(for  $k = q + 1$  the  $g_3$ -factor is empty and  $\dim A \geq 2$  suffices), and  $\Theta(a; x) := \theta(a) x_1 \wedge \cdots \wedge x_k$ . Then  $\Theta$  is a lift of  $D = 0$ . It satisfies (Pw), because  $\theta(a, \dots, a) = (g_1(a)^q g_2(a) - g_1(a) g_2(a)^q) g_3(a)^{k-q-1} = 0$  since  $c^q = c$  for  $c \in \mathbb{F}_q$ . It violates (Pol) for  $\alpha = (q, 1, k - q - 1)$  and  $b = (b_1, b_2, b_3)$ : the first product contributes to  $\sum_{\text{type } f = \alpha} \theta(b_f)$  exactly for  $f = (1, \dots, 1, 2, 3, \dots, 3)$ , with value 1; the second product would need  $f(2) = \cdots = f(q + 1) = 2$ , that is, at least  $q \geq 2$  values equal to 2, which is impossible for type  $\alpha$ . So the coefficient  $\delta_\alpha(b, y)$  of  $\Theta$  is  $y_1 \wedge \cdots \wedge y_k \neq 0$ . When  $k = q + 1$ , the third label and its functional can be omitted throughout, with  $\alpha = (q, 1)$ .

(4) *The multilinear identity without the pointwise identity.* Take  $\bar{a} \in A$  and a functional  $g_1$  with  $g_1(\bar{a}) = 1$ , and  $y_1, \dots, y_k$  linearly independent. The lift  $\Theta'(a; x) := g_1(a_1) \cdots g_1(a_k) x_1 \wedge \cdots \wedge x_k$  of  $D = 0$  satisfies (Pol1), since  $\sum_\sigma \Theta'(b_\sigma; x) = k! g_1(b_1) \cdots g_1(b_k) x_1 \wedge \cdots \wedge x_k = 0$  for all  $b_1, \dots, b_k \in A$  and  $x \in V^k$ , but  $\Theta'(\bar{a}, \dots, \bar{a}; y) = y_1 \wedge \cdots \wedge y_k \neq 0$ .  $\square$

The finite-field obstruction below requires only (Pol1), which follows from either (Pw) or (Pol). The distinction between the latter two remains important: over  $\mathbb{F}_q$  they need not agree when  $k > q$ . The lift constructed in §10 nevertheless satisfies the full identity (Pol) in every degree: the identity is polarized over an infinite Laurent-series field before constant coefficients are taken (Lemma 64).

**9.3. Statement and proof strategy.** For the proof of the theorem, assume that  $L$  is a *finite* field with  $k! = 0$  in  $L$ , that is,  $\text{char } L = p \leq k$ ; in particular  $k \geq 2$ . Put  $V := L^{\mathbb{N}}$  and let  $V_{\text{fin}} := L^{(\mathbb{N})} \subseteq V$  be the subspace of finitely supported sequences. For  $u, x \in V$  we write  $ux$  for the coordinatewise product; so  $D_u x = ux$  is the multiplier family.

**Theorem 55** (multipliers over a finite field). *Let  $L$  be a finite field and  $k \geq 1$  with  $k! = 0$  in  $L$ .*

- (1) *There is no  $2k$ -linear map  $\Psi: V_{\text{fin}}^k \times V_{\text{fin}}^k \rightarrow \Lambda_L^k V$  over  $L$  such that*
  - (a)  *$\Psi$  is antisymmetric in the last  $k$  slots;*
  - (b)  *$\Psi$  satisfies (Pol1) for  $D_u x = ux$ : for all  $u_1, \dots, u_k, x_1, \dots, x_k \in V_{\text{fin}}$ ,*

$$\sum_{\sigma \in S_k} \Psi(u_{\sigma(1)}, \dots, u_{\sigma(k)}; x_1, \dots, x_k) = \sum_{\sigma \in S_k} (u_{\sigma(1)} x_1) \wedge \cdots \wedge (u_{\sigma(k)} x_k);$$

- (c) *for some  $d \in \mathbb{N}$ ,  $\text{sdim}(\Psi(u; x)) \leq d$  for all  $u, x \in V_{\text{fin}}^k$  with entries in  $\{0, \pm 1\}$ .*
- (2) *Let  $E_0 = V$  or  $E_0 = V_{\text{fin}}$ . There is no  $2k$ -linear map  $\Psi: E_0^k \times E_0^k \rightarrow \Lambda_L^k E_0$  over  $L$  that is antisymmetric in the last  $k$  slots, satisfies (Pol1) for all finitely supported  $u, x$ , and has  $\text{sdim}(\Psi(u; x)) \leq d$  for all  $u, x \in E_0^k$ , for some  $d$ .*

*Reduction to (1).* Given a map as in (2), restrict its inputs to  $V_{\text{fin}}^k \times V_{\text{fin}}^k$  and compose its output with the injective map  $\Lambda^k E_0 \rightarrow \Lambda^k V$ . This preserves antisymmetry and (Pol1). Support dimension can only decrease under the inclusion of the ambient space (§9.1), so the resulting map satisfies (a)–(c).

We therefore prove (1). Suppose that  $\Psi$  satisfies its hypotheses, and fix the bound  $d$  in (c). Write  $e_i \in V_{\text{fin}}$  for the unit vectors,  $e_a := (e_{a_1}, \dots, e_{a_k})$  for  $a \in \mathbb{N}^k$ , and  $\Omega$  for the determinant array of §9.1 with  $S = \mathbb{N}$ .

The proof has three steps. Ramsey's theorem first makes the coefficients of  $\Omega\Psi$  depend only on the relative order and equalities among their indices. Special weights on four-point

clusters then cancel every contribution from an occupied cluster containing no output index. This cancellation produces a two-valued staircase matrix from one exterior vector. Its bounded support forces the two values to agree, and hence makes all cluster order patterns have a common value. Finally, antisymmetry and (Pol1) say that the sum of these  $k!$  equal values is 1, contradicting  $k! = 0$  in  $L$ .

**9.4. Order-homogeneity from Ramsey's theorem.** We use Ramsey's theorem [9] in its infinite form: *for every  $n \geq 1$  and every coloring of the  $n$ -element subsets of  $\mathbb{N}$  with finitely many colors there is an infinite  $H \subseteq \mathbb{N}$  all of whose  $n$ -element subsets have the same color.*

For a tuple  $z \in \mathbb{N}^N$  with distinct values  $y_1 < \dots < y_r$ , its *pattern* is the map  $i \mapsto j$  determined by  $z_i = y_j$ . In particular, repeated entries are retained in the pattern. Two tuples  $z$  and  $z'$  have the same pattern if and only if  $z_i < z_j \Leftrightarrow z'_i < z'_j$  and  $z_i = z_j \Leftrightarrow z'_i = z'_j$  for all  $i, j$ .

**Lemma 56** (pattern homogeneity). *For every  $N \geq 1$  and every function  $T: \mathbb{N}^N \rightarrow L$  there is an infinite  $H \subseteq \mathbb{N}$  such that  $T(z) = T(z')$  whenever  $z, z' \in H^N$  have the same pattern.*

*Proof.* Color an  $N$ -element set  $s = \{s_1 < \dots < s_N\} \subseteq \mathbb{N}$  by the vector

$$(T(s_{\beta(1)}, \dots, s_{\beta(N)}))_{\beta: [N] \rightarrow [N]} \in L^{N^N}.$$

The coloring records all maps  $\beta$ , including those that repeat indices. It has finitely many values because  $L$  is finite; this is the only use of the finiteness of  $L$  in the obstruction proof. Ramsey's theorem gives an infinite  $H$  on which the color is constant, say equal to  $(t_\beta)_\beta$ . Let  $z \in H^N$  have value set  $\{y_1 < \dots < y_r\}$ . Since  $H$  is unbounded, add  $N - r$  elements of  $H$  above  $y_r$  to obtain an  $N$ -element set  $s \subseteq H$ . Padding above the largest value leaves the ranks unchanged:  $s_j = y_j$  for  $j \leq r$ , and  $z = (s_{\beta(1)}, \dots, s_{\beta(N)})$ , where  $\beta(i)$  is the rank of  $z_i$  in  $\{y_1, \dots, y_r\}$ . The map  $\beta$  depends only on the pattern of  $z$ , and  $T(z) = t_\beta$ .  $\square$

*Coefficients.* For  $(a; y; c) \in \mathbb{N}^k \times \mathbb{N}^k \times \mathbb{N}^k$  put

$$T(a; y; c) := \Omega_{\Psi(e_a; e_y)}(c_1, \dots, c_k) \in L.$$

Lemma 56 with  $N = 3k$  gives an infinite set  $H = \{h_0 < h_1 < h_2 < \dots\}$  on which  $T$  depends only on the pattern. For  $u_j, v_j \in V_{\text{fin}}$  supported in  $H$ , multilinearity of  $\Psi$  and linearity of  $\Omega$  give

$$(9.1) \quad \Omega_{\Psi(u; v)}(c) = \sum_{a, y \in H^k} \prod_{j=1}^k u_j(a_j) \prod_{j=1}^k v_j(y_j) \cdot T(a; y; c).$$

Although the sum is indexed by  $H^k \times H^k$ , only finitely many summands are nonzero, because all the input vectors have finite support. Every output tuple  $c$  used below also lies in  $H^k$ , so pattern homogeneity applies to all  $3k$  indices in (9.1).

**9.5. Clusters and the cluster value.** A *cluster* is a set  $C = \{h_n, h_{n+1}, h_{n+2}, h_{n+3}\}$  of four consecutive elements of  $H$ ; we write  $C(i) := h_{n+i-1}$  for  $i = 1, \dots, 4$ . Since a cluster is an interval of  $H$ , two disjoint clusters are *comparable*: every element of one is smaller than every element of the other; we write  $C < C'$ . Distinct clusters need not be adjacent in  $H$ .

We choose two weight vectors, one for each kind of input slot:

$$\begin{aligned} s &= (s_1, s_2, s_3, s_4) := (1, 0, -1, 0) && \text{on the operator slots,} \\ w &= (w_1, w_2, w_3, w_4) := (1, -1, 1, -1) && \text{on the vector slots.} \end{aligned}$$

Over  $\mathbb{Z}$ , hence over every field,

$$(W) \quad \sum_i s_i = \sum_i w_i = \sum_i s_i w_i = \sum_{i < i'} s_i w_{i'} = \sum_{i > i'} s_i w_{i'} = 0, \quad s_1 w_1 = 1.$$

Indeed  $\sum_i s_i = 1 - 1$ ,  $\sum_i w_i = 1 - 1 + 1 - 1$ ,  $s_1 w_1 = 1$ , and

$$\begin{aligned} \sum_i s_i w_i &= s_1 w_1 + s_3 w_3 = 1 - 1 = 0, \\ \sum_{i < i'} s_i w_{i'} &= s_1(w_2 + w_3 + w_4) + s_3 w_4 = -1 + 1 = 0, \\ \sum_{i > i'} s_i w_{i'} &= s_3(w_1 + w_2) = -(1 - 1) = 0. \end{aligned}$$

The first two zero sums will cancel a cluster occupied by only one input index. The other three will cancel a cluster occupied by an operator index and a vector index, according to their relative order. For a cluster  $C$  put

$$s_C := \sum_{i=1}^4 s_i e_{C(i)}, \quad w_C := \sum_{i=1}^4 w_i e_{C(i)}.$$

These are vectors of  $V_{\text{fin}}$  with entries in  $\{0, \pm 1\}$ , and  $s_C w_C = e_{C(1)} - e_{C(3)}$ .

*The cluster value.* For pairwise disjoint clusters  $C_1, \dots, C_k$ , one for each slot, define

$$\begin{aligned} \chi(C_1, \dots, C_k) &:= \Omega_{\Psi(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})}(C_1(1), \dots, C_k(1)) \\ &= \sum_{i, i' \in [4]^k} \prod_{j=1}^k s_{i_j} w_{i'_j} \cdot T(C_1(i_1), \dots, C_k(i_k); C_1(i'_1), \dots, C_k(i'_k); C_1(1), \dots, C_k(1)). \end{aligned}$$

Let  $\tau \in S_k$  be the *order pattern* of  $(C_1, \dots, C_k)$ :  $\tau(j)$  is the position of  $C_j$  among  $C_1, \dots, C_k$ . The pattern of every  $3k$ -tuple in the sum is determined by  $(i, i')$  and  $\tau$ : entries in different clusters compare as the clusters do, and the entries  $C_j(i_j)$ ,  $C_j(i'_j)$ ,  $C_j(1)$  in the same cluster compare as their indices do, including equalities. Pattern homogeneity therefore shows that the displayed sum depends only on  $\tau$ . Every permutation occurs as the order pattern of some disjoint clusters in  $H$ ; we denote the resulting scalar by  $\chi(\tau)$ .

**Lemma 57** (cancellation of output-free clusters). *Let  $\mathcal{C}_1, \dots, \mathcal{C}_k$  be finite families of clusters that are pairwise disjoint as sets of clusters, such that all clusters in  $\bigcup_j \mathcal{C}_j$  are pairwise disjoint. Put  $u_j := \sum_{C \in \mathcal{C}_j} s_C$  and  $v_j := \sum_{C \in \mathcal{C}_j} w_C$ . Let  $C_l \in \mathcal{C}_l$  for  $l = 1, \dots, k$  and  $c := (C_1(1), \dots, C_k(1))$ . Then  $\Omega_{\Psi(u; v)}(c) = \chi(C_1, \dots, C_k)$ .*

*Proof.* We show that every cluster assignment except the one matching the output clusters has total contribution zero. Expand by (9.1). Since the clusters are disjoint, a nonzero term has, for each slot  $j$ , an operator index  $a_j = C_j^{\text{op}}(i_j)$  and a vector index  $y_j = C_j^{\text{vec}}(i'_j)$  with  $C_j^{\text{op}}, C_j^{\text{vec}} \in \mathcal{C}_j$ , and weight  $\prod_j s_{i_j} w_{i'_j}$ . Group the terms according to the *cluster assignment*  $(C_j^{\text{op}}, C_j^{\text{vec}})_j$ .

Call an occupied cluster *output-free* if it is none of  $C_1, \dots, C_k$ . Suppose a group has such a cluster  $\Gamma$ . Disjointness of the families assigns  $\Gamma$  to exactly one slot  $j$ , so the only tuple entries it can contain are  $a_j$  and  $y_j$ . Disjointness of the clusters excludes every output index from  $\Gamma$ .

Fix the positions of all indices outside  $\Gamma$  and sum over the positions inside it. Since  $\Gamma$  consists of consecutive elements of  $H$ , every outside entry lies either below the whole cluster or above it. Moving an index within  $\Gamma$  cannot change any comparison with an outside entry. By pattern homogeneity, the coefficient  $T$  therefore depends on the internal positions only as follows:

- (i) if only  $a_j \in \Gamma$ , not at all, and the partial sum is  $(\sum_i s_i) \cdot T = 0$ ;
- (ii) if only  $y_j \in \Gamma$ , not at all, and the partial sum is  $(\sum_i w_i) \cdot T = 0$ ;

- (iii) if both lie in  $\Gamma$ , only through whether  $i_j < i'_j$ ,  $i_j = i'_j$  or  $i_j > i'_j$ ; with the corresponding values  $T_<, T_=: T_>$  of  $T$  the partial sum is

$$T_< \sum_{i < i'} s_i w_{i'} + T_=: \sum_i s_i w_i + T_> \sum_{i > i'} s_i w_{i'} = 0.$$

In each case the weights of the other slots are a common factor, and the displayed partial sum vanishes by (W). Thus every assignment with an output-free cluster contributes zero. In any surviving assignment, every occupied cluster must be an output cluster. The only output cluster belonging to  $\mathcal{C}_j$  is  $C_j$ , which forces  $C_j^{\text{op}} = C_j^{\text{vec}} = C_j$  for every  $j$ . The contribution of this one assignment is exactly  $\chi(C_1, \dots, C_k)$ .  $\square$

**9.6. The staircase and the diagonal.** We now use the support bound to compare two order patterns that differ by an adjacent exchange. The key is to realize arbitrarily large staircase matrices as flattenings of a single value of  $\Psi$ .

**Lemma 58** (staircase). *Let  $\tau \in S_k$ , let  $\alpha \neq \beta$  be slots with  $\tau(\alpha) = \ell$  and  $\tau(\beta) = \ell + 1$ , and let  $\tau'$  be  $\tau$  with these two positions exchanged, so  $\tau'(\alpha) = \ell + 1$ ,  $\tau'(\beta) = \ell$  and  $\tau'(\gamma) = \tau(\gamma)$  otherwise. Then  $\chi(\tau) = \chi(\tau')$ .*

*Proof.* Let  $m \geq 1$ . Choose pairwise disjoint clusters, in increasing order in  $H$ : first a cluster  $C_\gamma$  for each slot  $\gamma$  with  $\tau(\gamma) < \ell$ , ordered as  $\tau$  prescribes; then  $2m$  clusters  $A_1 < B_1 < A_2 < B_2 < \dots < A_m < B_m$ ; then a cluster  $C_\gamma$  for each slot  $\gamma$  with  $\tau(\gamma) > \ell + 1$ , ordered as  $\tau$  prescribes. Put  $\mathcal{C}_\alpha := \{A_1, \dots, A_m\}$ ,  $\mathcal{C}_\beta := \{B_1, \dots, B_m\}$  and  $\mathcal{C}_\gamma := \{C_\gamma\}$  for the other slots, and let  $u, v$  be as in Lemma 57 and  $\omega := \Psi(u; v)$ . Disjointness of the clusters ensures that the entries of  $u$  and  $v$  still lie in  $\{0, \pm 1\}$ , so  $\text{sdim}(\omega) \leq d$  by hypothesis (c) of Theorem 55(1). For this fixed  $m$ , keep  $u, v$  and hence  $\omega$  fixed while varying only the output indices.

Consider the  $m \times m$  matrix  $M$  with rows indexed by  $c_\alpha = A_n(1)$  and columns by  $c_\beta = B_{n'}(1)$  ( $n, n' = 1, \dots, m$ ), all other output indices being fixed at  $c_\gamma = C_\gamma(1)$ , and entries  $M_{nn'} := \Omega_\omega(c)$ . By Lemma 57, applied with the output clusters  $A_n \in \mathcal{C}_\alpha$ ,  $B_{n'} \in \mathcal{C}_\beta$  and  $C_\gamma \in \mathcal{C}_\gamma$ ,  $M_{nn'}$  is the cluster value of the tuple with  $A_n$  in slot  $\alpha$ ,  $B_{n'}$  in slot  $\beta$ , and  $C_\gamma$  in every other slot. The interlacing order gives  $A_n < B_{n'}$  exactly when  $n \leq n'$ . The tuple therefore has pattern  $\tau$  on and above the matrix diagonal, and pattern  $\tau'$  below it.

Put  $\delta := \chi(\tau) - \chi(\tau')$ , let  $\mathbf{1} \in L^m$  be the all-ones column, and let  $U$  be the upper unitriangular matrix with  $U_{nn'} = 1$  for  $n \leq n'$  and  $U_{nn'} = 0$  otherwise. Then

$$M = \begin{pmatrix} \chi(\tau) & \chi(\tau) & \cdots & \chi(\tau) \\ \chi(\tau') & \chi(\tau) & \cdots & \chi(\tau) \\ \vdots & \ddots & \ddots & \vdots \\ \chi(\tau') & \cdots & \chi(\tau') & \chi(\tau) \end{pmatrix} = \chi(\tau') \mathbf{1}\mathbf{1}^\top + \delta U,$$

The rows vary only slot  $\alpha$ ; each column fixes all the other slots, including the chosen value in slot  $\beta$ . Thus  $M$  is a submatrix of the flattening of this one array  $\Omega_\omega$  along slot  $\alpha$ . Lemma 49 gives  $\text{rank } M \leq \text{sdim}(\omega) \leq d$ .

If  $\delta \neq 0$ , the matrix  $\delta U$  is invertible, whereas the constant matrix  $\chi(\tau') \mathbf{1}\mathbf{1}^\top$  has rank at most one. Applying rank subadditivity to  $\delta U = M - \chi(\tau') \mathbf{1}\mathbf{1}^\top$  gives

$$m = \text{rank}(\delta U) \leq \text{rank } M + 1 \leq d + 1.$$

The construction is available for every  $m \geq 1$ ; choosing  $m = d + 2$  is a contradiction. Hence  $\delta = 0$ , as required.  $\square$

The support bound has now forced equality between adjacent order patterns. The remaining identity comes from (Poll); it does not use the support bound.

**Lemma 59** (the diagonal).  $\sum_{\tau \in S_k} \chi(\tau) = 1$ .

*Proof.* Take clusters  $C_1 < \dots < C_k$ , and put  $u_j := s_{C_j}$ ,  $v_j := w_{C_j}$  and  $c := (C_1(1), \dots, C_k(1))$ . Apply  $\omega \mapsto \Omega_\omega(c)$  to both sides of (Pol1), with the operator vectors  $u_1, \dots, u_k$  and the vector arguments  $v_1, \dots, v_k$ .

*Right side.* If  $\sigma \neq \text{id}$ , some slot  $j$  has  $\sigma(j) \neq j$ . Disjoint supports then give  $u_{\sigma(j)}v_j = 0$ , so the corresponding wedge vanishes. Only  $\sigma = \text{id}$  survives. It gives  $\Omega_{(u_1v_1) \wedge \dots \wedge (u_kv_k)}(c) = \det((u_bv_b)(C_a(1)))_{a,b}$ . Since  $u_bv_b = e_{C_b(1)} - e_{C_b(3)}$  and  $C_a(1) \neq C_b(3)$  for all  $a, b$ , this is the determinant of the identity matrix, so the right side is 1.

*Left side.* Fix  $\sigma \in S_k$  and write  $u_\sigma := (u_{\sigma(1)}, \dots, u_{\sigma(k)})$  and  $v_\sigma := (v_{\sigma(1)}, \dots, v_{\sigma(k)})$ . To identify the summand with a cluster value, we must permute both the vector arguments and the output coordinates to match the operator arguments. Each permutation contributes the same sign. More precisely, antisymmetry of  $\Psi$  in the vector slots gives  $\Psi(u_\sigma; v) = \text{sgn}(\sigma)\Psi(u_\sigma; v_\sigma)$ . For every  $\omega \in \Lambda^k V$  the function  $\Omega_\omega$  is alternating in its arguments, so  $\Omega_\omega(c) = \text{sgn}(\sigma)\Omega_\omega(c_{\sigma(1)}, \dots, c_{\sigma(k)})$ ; we apply this with  $\omega = \Psi(u_\sigma; v_\sigma)$ . Since  $\text{sgn}(\sigma)^2 = 1$ ,

$$\begin{aligned} \Omega_{\Psi(u_\sigma; v)}(c) &= \text{sgn}(\sigma)\Omega_{\Psi(u_\sigma; v_\sigma)}(c) = \text{sgn}(\sigma)^2\Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}) \\ &= \Omega_{\Psi(u_\sigma; v_\sigma)}(c_{\sigma(1)}, \dots, c_{\sigma(k)}). \end{aligned}$$

In the last expression slot  $j$  has operator  $s_{C_{\sigma(j)}}$ , vector  $w_{C_{\sigma(j)}}$  and output  $C_{\sigma(j)}(1)$ , so it equals  $\chi(C_{\sigma(1)}, \dots, C_{\sigma(k)})$  by definition. Its order pattern is  $\sigma$ , since  $C_{\sigma(j)}$  is the  $\sigma(j)$ -th cluster in increasing order. Thus the convention for patterns gives  $\sigma$ , not  $\sigma^{-1}$ . No cluster cancellation is needed in this calculation.

As  $\sigma$  runs over  $S_k$ , each order pattern occurs exactly once. Comparing the two sides gives

$$\sum_{\sigma \in S_k} \Omega_{\Psi(u_\sigma; v)}(c) = \sum_{\tau \in S_k} \chi(\tau) = 1. \quad \square$$

*Proof of Theorem 55.* Exchanging adjacent ranks acts on an order pattern by  $\tau \mapsto (\ell \ell+1) \circ \tau$ ,  $1 \leq \ell < k$ . These transpositions generate  $S_k$ , and Lemma 58 makes  $\chi$  invariant under each exchange. Hence  $\chi$  is constant; write its value as  $\chi_0$ . Lemma 59 now gives  $k!\chi_0 = 1$  in  $L$ , contradicting  $k! = 0$ . This proves (1), and the preceding reduction gives (2).  $\square$

*Remark 60* (where the hypotheses enter). Ramsey homogeneity is the only step in the obstruction proof that uses the finiteness of  $L$ . Multilinearity then gives the cluster expansion and cancellation. The support bound is used in Lemma 58 only at the finitely many test inputs obtained by taking  $m = d + 2$ ; all have entries in  $\{0, \pm 1\}$ . Antisymmetry and (Pol1) enter in Lemma 59, which does not use that bound. Finally,  $k! = 0$  is used only to turn the diagonal identity into a contradiction.

The inclusion–exclusion argument in Proposition 54(1) uses only multilinearity in the operator slots. The theorem therefore remains true with the pointwise identity (Pw) at  $L$ -points in place of (Pol1), for this family of multipliers. In characteristic 2 antisymmetry is the same as symmetry, and the two signs in Lemma 59 are trivial.

*Remark 61* (when  $k!$  is invertible). The vanishing of  $k!$  is essential. If instead  $k! \neq 0$  in  $L$ , the *normalized alternatization*

$$\Psi_{\text{alt}}(u; x) := \frac{1}{k!} \sum_{\sigma \in S_k} (u_{\sigma(1)}x_1) \wedge \dots \wedge (u_{\sigma(k)}x_k)$$

is alternating in  $x$  and satisfies (Pol1). Every summand is supported in the span of the  $k^2$  vectors  $u_i x_j$ , so  $\text{sdim}(\Psi_{\text{alt}}(u; x)) \leq k^2$ . The preceding homogeneity, cancellation, staircase and

diagonal arguments apply, but now yield  $\chi \equiv 1/k!$ : for pairwise disjoint clusters only  $\sigma = \text{id}$  survives in  $\Psi_{\text{alt}}(s_{C_1}, \dots, s_{C_k}; w_{C_1}, \dots, w_{C_k})$ , and the determinant at  $(C_1(1), \dots, C_k(1))$  is that of the identity. The final equation  $k! \cdot (1/k!) = 1$  is then consistent. So the obstruction is exactly the vanishing of  $k!$ , not positive characteristic as such. For example, at  $(p, k) = (5, 3)$  every step is consistent, since  $3! = 6 = 1$  in  $\mathbb{F}_5$  and  $\chi_0 = 1$ .

## 10. THE BANACH EXTERIOR TARGET OVER A LAURENT SERIES FIELD

We now realize the finite-field obstruction in a Banach-space setting. The target is the completion of an exterior power with its ordinary projective norm. Its determinant arrays admit a constant-coefficient map into the *algebraic* exterior power over the coefficient field, with support dimension controlled by the Banach norm. This estimate turns a hypothetical bounded lift into the uniformly bounded-support map ruled out by Theorem 55. The passage to the completion is the key point: constant coefficients stabilize exactly once the approximation error is less than one.

**10.1. Setting.** Throughout this section  $k \geq 1$ ,  $\kappa$  is a field,  $r \in (0, 1)$ , and  $K_1 := \kappa((X))$  is the Laurent series field with absolute value  $|a| := r^{\text{ord } a}$  ( $|0| = 0$ ). This field is complete, nontrivially normed and nonarchimedean, and  $|c| = 1$  for every  $c \in \kappa^\times$ . Write  $\text{coeff}_n(a) \in \kappa$  for the  $X^n$ -coefficient of  $a \in K_1$ . For a set  $S$  and  $y \in \ell^\infty(S, K_1)$  we take the constant coefficient coordinatewise:

$$\text{coeff}_0(y) := (\text{coeff}_0(y(s)))_{s \in S} \in \kappa^S.$$

The resulting *constant-coefficient map*  $\text{coeff}_0$  is  $\kappa$ -linear but not  $K_1$ -linear. A nonzero Laurent series of absolute value less than one has order at least one. Consequently,

$$(10.1) \quad \|y\|_\infty < 1 \implies \text{coeff}_0(y) = 0, \quad \text{hence} \quad \|y - y'\|_\infty < 1 \implies \text{coeff}_0(y) = \text{coeff}_0(y').$$

Put  $E_1 := \ell^\infty(\mathbb{N}, K_1)$  with the sup norm, a nonarchimedean  $K_1$ -Banach space, and let  $E_0 := \kappa^\mathbb{N} \subseteq E_1$  be the  $\kappa$ -subspace of sequences with entries in  $\kappa$ . Every nonzero  $z \in E_0$  has  $\|z\|_\infty = 1$ . For  $x \in E_1$  and  $n \in \mathbb{Z}$  let  $x_{[n]} := (\text{coeff}_n(x(i)))_{i \in \mathbb{N}} \in E_0$  be the  $n$ -th Laurent component. For  $x \neq 0$  the set  $\{\text{ord } x(i) : x(i) \neq 0\}$  is bounded below, because  $r^{\text{ord } x(i)} \leq \|x\|_\infty$ , so it has a minimum  $\nu(x)$ ; then  $\|x\|_\infty = r^{\nu(x)}$  and  $x_{[n]} = 0$  for  $n < \nu(x)$ . Finally put

$$M_r := \max_{l \in \mathbb{N}} (l+1)r^l.$$

The maximum exists because  $(l+1)r^l \rightarrow 0$ . This constant will control the number of Laurent components that can contribute to a constant coefficient. For  $r \leq \frac{1}{2}$  one has  $(l+1)r^l \leq (l+1)2^{-l} \leq 1$ , so  $M_r = 1$ .

**10.2. The projective exterior norm.** On  $\Lambda := \Lambda_{K_1}^k E_1$  put

$$\|\omega\|_\pi := \inf \left\{ \sum_j \prod_{i=1}^k \|x_{j,i}\|_\infty : \omega = \sum_j x_{j,1} \wedge \cdots \wedge x_{j,k} \right\},$$

where the infimum is over finite decompositions and the cost is an ordinary sum. Let  $\Omega^{K_1} : \Lambda \rightarrow \ell^\infty(\mathbb{N}^k, K_1)$  be the determinant array of §9.1 with  $L = K_1$ ,  $S = \mathbb{N}$  and  $V = E_1$ , so that  $\Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)(i_1, \dots, i_k) = \det(x_b(i_a))_{a,b}$ ; its values are bounded by the next lemma.

**Lemma 62.**  $\|\cdot\|_\pi$  is a  $K_1$ -norm on  $\Lambda$  with  $\|x_1 \wedge \cdots \wedge x_k\|_\pi \leq \prod_i \|x_i\|_\infty$ , and  $\Omega^{K_1}$  is  $K_1$ -linear and injective with  $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$ . Let  $B$  be the completion of  $(\Lambda, \|\cdot\|_\pi)$ , a  $K_1$ -Banach space,  $J: B \rightarrow \ell^\infty(\mathbb{N}^k, K_1)$  the continuous extension of  $\Omega^{K_1}$  (so  $\|J\| \leq 1$ ), and  $W_B: E_1^k \rightarrow B$ ,  $W_B(x) := x_1 \wedge \cdots \wedge x_k$ . Then  $W_B \in \text{Alt}_{K_1}^k(E_1; B)$ ,  $\|W_B\| \leq 1$ , and  $J(W_B(x)) = \Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)$ .

*Proof.* For a pure wedge, each entry of the determinant array is a signed sum of products  $x_1(i_{\sigma(1)}) \cdots x_k(i_{\sigma(k)})$ . Each product has absolute value at most  $\prod_b \|x_b\|_\infty$ , so the nonarchimedean inequality in  $K_1$  gives the same bound for the determinant. For any finite decomposition of  $\omega$ , it follows that

$$\|\Omega^{K_1}(\omega)\|_\infty \leq \max_j \prod_i \|x_{j,i}\|_\infty \leq \sum_j \prod_i \|x_{j,i}\|_\infty.$$

Thus the array is bounded, and taking the infimum gives  $\|\Omega^{K_1}(\omega)\|_\infty \leq \|\omega\|_\pi$ . The map  $\Omega^{K_1}$  is linear and injective by Lemma 48, with  $L = K_1$ ,  $S = \mathbb{N}$  and  $V = E_1$ . In particular,  $\|\omega\|_\pi > 0$  whenever  $\omega \neq 0$ .

The remaining norm laws follow directly from decompositions. Concatenation gives the triangle inequality. Since  $k \geq 1$ , a scalar can be absorbed into the first factor of each wedge; for  $\lambda \neq 0$  this gives  $\|\lambda\omega\|_\pi \leq |\lambda| \|\omega\|_\pi$ , and applying the same inequality to  $\lambda^{-1}$  gives equality. The pure-wedge bound follows by using a single summand.

The array space  $\ell^\infty(\mathbb{N}^k, K_1)$  is complete, so  $\Omega^{K_1}$  extends to the contraction  $J$  on  $B$ . The wedge map into  $B$  is  $K_1$ -multilinear and vanishes whenever two arguments coincide. The pure-wedge bound makes it continuous with norm at most one, and the asserted identity with  $J$  holds on its values by construction.  $\square$

For  $k \geq 2$ ,  $B$  is not nonarchimedean (Proposition 79).

**10.3. The support estimate.** We next show that the constant coefficient of every completed determinant array is represented by a finite algebraic exterior vector. The quantitative bound comes from counting the Laurent components needed as wedge *factors*; the number of wedge summands may be much larger.

Let  $\Omega^\kappa: \Lambda_\kappa^k E_0 \rightarrow \kappa^{\mathbb{N}^k}$  be the determinant array over  $\kappa$ ; it is injective by Lemma 48 with  $L = \kappa$  and  $V = E_0$ . For  $y_1, \dots, y_k \in E_0$  we have  $\Omega^{K_1}(y_1 \wedge \cdots \wedge y_k) = \Omega^\kappa(y_1 \wedge \cdots \wedge y_k)$ : the determinants are the same, with entries in  $\kappa \subseteq K_1$ . Support dimensions of elements of  $\Lambda_\kappa^k E_0$  are computed in  $E_0$  over  $\kappa$ .

**Proposition 63** (support estimate). *For every  $b \in B$  there is a unique  $\eta(b) \in \Lambda_\kappa^k E_0$  with  $\Omega^\kappa(\eta(b)) = \text{coeff}_0(Jb)$ . The map  $\eta$  is  $\kappa$ -linear, and*

$$(10.2) \quad \text{sdim}(\eta(b)) \leq kM_r \|b\|_B \quad \text{for all } b \in B.$$

*Proof. Step 1: one decomposable.* Let  $x_1, \dots, x_k \in E_1$ . If some  $x_b = 0$ , the wedge is zero and we take  $\eta = 0$ . Otherwise put  $\nu_b := \nu(x_b)$  and  $l := -\sum_b \nu_b$ , so that  $\prod_b \|x_b\|_\infty = r^{-l}$ . The possible exponent tuples contributing to a constant coefficient form the set

$$\mathcal{N} := \left\{ n \in \mathbb{Z}^k : n_b \geq \nu_b \text{ for all } b, \sum_b n_b = 0 \right\}.$$

For such a tuple, the nonnegative integers  $n_b - \nu_b$  sum to  $l$ . Hence  $\mathcal{N}$  is empty if  $l < 0$ ; if  $l \geq 0$ , every coordinate satisfies  $\nu_b \leq n_b \leq \nu_b + l$ , so  $\mathcal{N}$  is finite.

Fix  $(i_1, \dots, i_k) \in \mathbb{N}^k$  and expand each entry as  $x_b(i_a) = \sum_{n \geq \nu_b} \text{coeff}_n(x_b(i_a)) X^n$ . In the determinant expansion

$$\det(x_b(i_a))_{a,b} = \sum_\sigma \text{sgn}(\sigma) \prod_b x_b(i_{\sigma(b)}),$$

the constant coefficient of each product is its finite coefficient convolution over  $\mathcal{N}$ . Exchanging this finite sum with the permutation sum gives

$$\text{coeff}_0 \det(x_b(i_a))_{a,b} = \sum_{n \in \mathcal{N}} \det(x_{b,[n_b]}(i_a))_{a,b}.$$

The same finite set  $\mathcal{N}$  works for every array coordinate. Thus  $\text{coeff}_0(\Omega^{K_1}(x_1 \wedge \cdots \wedge x_k)) = \Omega^\kappa(\eta)$ , where

$$\eta := \sum_{n \in \mathcal{N}} x_{1,[n_1]} \wedge \cdots \wedge x_{k,[n_k]} \in \Lambda_\kappa^k E_0.$$

This is a finite sum in the algebraic exterior power. When  $l < 0$  it is zero. When  $l \geq 0$ , all its wedge factors belong to the set  $\{x_{b,[n]} : 1 \leq b \leq k, \nu_b \leq n \leq \nu_b + l\}$ , which has at most  $k(l+1)$  elements. Taking their span gives

$$\text{sdim}(\eta) \leq k(l+1) = k(l+1)r^l \cdot r^{-l} \leq kM_r \prod_b \|x_b\|_\infty.$$

The final bound therefore holds in both cases.

*Step 2: an element of  $\Lambda$ .* Write  $\omega = \sum_j w_j$ , where  $w_j = x_{j,1} \wedge \cdots \wedge x_{j,k}$ , and take  $\eta_j$  from Step 1. Additivity gives

$$\text{coeff}_0(\Omega^{K_1}(\omega)) = \sum_j \text{coeff}_0(\Omega^{K_1}(w_j)) = \Omega^\kappa\left(\sum_j \eta_j\right).$$

Injectivity of  $\Omega^\kappa$  shows that  $\eta(\omega) := \sum_j \eta_j$  is independent of the chosen decomposition. Subadditivity of support dimension now yields

$$\text{sdim}(\eta(\omega)) \leq \sum_j \text{sdim}(\eta_j) \leq kM_r \sum_j \prod_i \|x_{j,i}\|_\infty.$$

The left side is fixed while the decomposition varies. Taking the infimum therefore gives  $\text{sdim}(\eta(\omega)) \leq kM_r \|\omega\|_\pi$ , without requiring a minimizing decomposition.

*Step 3: completion.* Choose  $\omega_n \in \Lambda$  with  $\|b - \omega_n\|_B \rightarrow 0$ . Since  $J$  is a contraction,

$$\|Jb - \Omega^{K_1}(\omega_n)\|_\infty \leq \|b - \omega_n\|_B < 1 \quad (n \geq n_0).$$

Equation (10.1) therefore gives the *exact* identities

$$\text{coeff}_0(Jb) = \text{coeff}_0(\Omega^{K_1}(\omega_n)) = \Omega^\kappa(\eta(\omega_n)) \quad (n \geq n_0).$$

By injectivity of  $\Omega^\kappa$ , the algebraic vectors  $\eta(\omega_n)$  are eventually equal. Define  $\eta(b)$  to be this common value. Its array is  $\text{coeff}_0(Jb)$ , so the same injectivity also proves independence of the approximating sequence and uniqueness. In particular, no limit in  $\Lambda_\kappa^k E_0$  is involved.

For every  $n \geq n_0$  the fixed vector  $\eta(b)$  satisfies  $\text{sdim}(\eta(b)) \leq kM_r \|\omega_n\|_\pi$ . Since  $\|\omega_n\|_\pi \rightarrow \|b\|_B$ , this gives (10.2). Finally, the array characterization proves  $\kappa$ -linearity:  $J$  is  $K_1$ -linear,  $\text{coeff}_0$  is  $\kappa$ -linear, and  $\Omega^\kappa$  is linear and injective.  $\square$

**10.4. The coefficient lift.** The support estimate converts norm control into an algebraic obstruction. We apply it to a hypothetical lift, evaluated on coordinatewise multiplication operators and on the universal wedge map  $W_B$ .

Suppose that  $P$  is a bounded  $k$ -linear lift of  $Q_{E_1, E_1, B}^{K_1}$  over  $K_1$ . For  $a \in E_1$  let  $D_a \in \mathcal{L}(E_1, E_1)$  be coordinatewise multiplication,  $D_a x = ax$ . The map  $a \mapsto D_a$  is  $K_1$ -linear and  $\|D_a\| \leq \|a\|_\infty$ . Define

$$\begin{aligned} \Phi(a_1, \dots, a_k; x_1, \dots, x_k) &:= P(D_{a_1}, \dots, D_{a_k})(W_B)(x_1, \dots, x_k) \in B, \\ \Psi &:= \eta \circ \Phi|_{E_0^{2k}} : E_0^k \times E_0^k \rightarrow \Lambda_\kappa^k E_0. \end{aligned}$$

Since  $\Omega^\kappa \circ \eta = \text{coeff}_0 \circ J$ , the map  $\Psi$  is  $\text{coeff}_0 \circ J \circ \Phi|_{E_0^{2k}}$ , read in  $\Lambda_\kappa^k E_0$  through the injective map  $\Omega^\kappa$ .

**Lemma 64.** *The map  $\Psi$  has the following properties.*

- ( $\Psi 1$ )  $\Psi$  is  $2k$ -linear over  $\kappa$ .
- ( $\Psi 2$ )  $\Psi$  is alternating in the last  $k$  slots.
- ( $\Psi 3$ )  $\text{sdim}(\Psi(z)) \leq d := \lfloor kM_r \|P\| \rfloor$  for all  $z \in E_0^{2k}$ .
- ( $\Psi 4$ )  $\Psi$  satisfies (**Pol**) for the multiplier family  $D^\kappa: E_0 \rightarrow \text{End}_\kappa(E_0)$ ,  $D_a^\kappa x = ax$ : for all  $m \geq 1$ ,  $b_1, \dots, b_m \in E_0$ ,  $\alpha \in \mathbb{N}^m$  with  $|\alpha| = k$ , and  $x \in E_0^k$ ,

$$\sum_{\text{type } f = \alpha} \Psi(b_f; x) = \sum_{\text{type } f = \alpha} (b_{f(1)}x_1) \wedge \cdots \wedge (b_{f(k)}x_k) \quad \text{in } \Lambda_\kappa^k E_0.$$

In particular  $\Psi$  satisfies (**Pol1**).

*Proof.* The map  $\Phi$  is  $K_1$ -multilinear, so its restriction to  $E_0^{2k}$  is  $\kappa$ -multilinear. Composing with the  $\kappa$ -linear map  $\eta$  proves ( $\Psi 1$ ). Since  $P(D_{a_1}, \dots, D_{a_k})(W_B)$  belongs to  $\text{Alt}_{K_1}^k(E_1; B)$ , it vanishes when two of the last  $k$  arguments coincide; the same holds after applying  $\eta$ , proving ( $\Psi 2$ ).

For ( $\Psi 3$ ), all inputs from  $E_0$  have norm at most one. Thus

$$\|\Phi(a; x)\| \leq \|P\| \prod_i \|D_{a_i}\| \|W_B\| \prod_i \|x_i\|_\infty \leq \|P\|.$$

Apply (10.2) and use that support dimension is an integer to obtain the stated bound by  $d$ .

To prove ( $\Psi 4$ ), we first compare coefficients over the infinite field  $K_1$ , while the values still lie in  $B$ . Fix  $b_1, \dots, b_m \in E_0$  and  $x \in E_0^k$ . For  $\lambda \in K_1^m$  put  $a_\lambda := \sum_i \lambda_i b_i \in E_1$ . The diagonal identity for the lift gives

$$\Phi(a_\lambda, \dots, a_\lambda; x) = P(D_{a_\lambda}, \dots, D_{a_\lambda})(W_B)(x) = Q(D_{a_\lambda})(W_B)(x) = W_B(a_\lambda x_1, \dots, a_\lambda x_k).$$

Expanding both sides by multilinearity gives  $\sum_\alpha \lambda^\alpha (L_\alpha - R_\alpha) = 0$  for all  $\lambda \in K_1^m$ , where

$$L_\alpha := \sum_{\text{type } f = \alpha} \Phi(b_f; x) \in B, \quad R_\alpha := \sum_{\text{type } f = \alpha} W_B(b_{f(1)}x_1, \dots, b_{f(k)}x_k) \in B.$$

The field  $K_1$  is infinite even when  $\kappa$  is finite. Lemma 52, applied in the  $K_1$ -vector space  $B$ , therefore yields  $L_\alpha = R_\alpha$  for every  $\alpha$ .

We can now apply  $\eta$ . Additivity gives  $\eta(L_\alpha) = \sum_{\text{type } f = \alpha} \Psi(b_f; x)$ . To identify the other side, take  $y_1, \dots, y_k \in E_0$ . By Lemma 62 and the determinant-array agreement preceding Proposition 63,

$$J(W_B(y)) = \Omega^{K_1}(y_1 \wedge \cdots \wedge y_k) = \Omega^\kappa(y_1 \wedge \cdots \wedge y_k).$$

This array has entries in  $\kappa$ , so taking its constant coefficient leaves it unchanged. Uniqueness in Proposition 63 then gives  $\eta(W_B(y)) = y_1 \wedge \cdots \wedge y_k \in \Lambda_\kappa^k E_0$ . Using  $y_j = b_{f(j)}x_j$  in each summand of  $R_\alpha$  proves

$$\eta(R_\alpha) = \sum_{\text{type } f = \alpha} (b_{f(1)}x_1) \wedge \cdots \wedge (b_{f(k)}x_k),$$

which is the required identity.  $\square$

The order of these operations is essential for the full identity ( $\Psi 4$ ). Evaluating the diagonal only at  $\kappa$ -points would give (**Pw**) there, which for  $\kappa = \mathbb{F}_q$  and  $k > q$  is strictly weaker than (**Pol**) (Proposition 54(3)). The proof below uses only the consequence (**Pol1**).

### 10.5. No bounded lift over a Laurent series field.

**Theorem 65.** *Let  $\kappa$  be a finite field of characteristic  $p$ , let  $k \geq p$  and  $r \in (0, 1)$ , and let  $K_1 = \kappa((X))$  with  $|X| = r$ . Let  $E_1 = \ell^\infty(\mathbb{N}, K_1)$ , and let  $B$  be the completion of  $\Lambda_{K_1}^k E_1$  under the projective exterior norm. Then  $Q_{E_1, E_1, B}^{K_1}$  admits no bounded  $k$ -linear lift over  $K_1$ . Consequently it is analytic at no point of  $\mathcal{L}(E_1, E_1)$ .*

*Proof.* Suppose  $P$  exists, and form  $\Psi$  as in §10.4. By Lemma 64,  $\Psi: E_0^k \times E_0^k \rightarrow \Lambda_\kappa^k E_0$  with  $E_0 = \kappa^\mathbb{N}$  is  $2k$ -linear over  $\kappa$ , alternating and hence antisymmetric in the last  $k$  slots, satisfies (Pol1), and has  $\text{sdim}(\Psi(z)) \leq d$  for all inputs  $z$ . Moreover  $k! = 0$  in  $\kappa$ , since  $k \geq p = \text{char } \kappa$ . This contradicts Theorem 55(2) with  $L = \kappa$  and  $E_0 = \kappa^\mathbb{N}$ . The last sentence follows from Proposition 8.  $\square$

*Remark 66.* Lemmas 62 and 64 and Proposition 63 hold for every field  $\kappa$ , and nothing in them uses a normalization of  $r$ : the constant  $M_r$  makes the argument uniform in  $r \in (0, 1)$ . Only the final appeal to Theorem 55 needs  $\kappa$  finite. We do not need, and do not discuss, the analogue of Theorem 65 for infinite  $\kappa$  with this particular pair  $(E_1, B)$ . For an infinite  $\kappa$  of characteristic  $p$  the field  $\kappa((X))$  is covered by Theorem 1, with the pair obtained by ascent from its subfield  $\mathbb{F}_p((X))$ .

## 11. ASCENT TO AN ARBITRARY BASE FIELD

To pass from a Laurent series field to a prescribed field, we work first inside its completion. A suitable Laurent subfield is spherically complete, so Ingleson's theorem supplies a contractive scalar retraction. The ordinary projective tensor norm then gives isometric scalar extensions of the source and target, together with a retraction onto the original Banach target. Operators and alternating forms extend compatibly with these embeddings. As a result, any lift after scalar extension descends to the Laurent subfield, and nonexistence of a lift ascends. The final density argument removes the completeness assumption on the prescribed field.

**11.1. The Laurent subfield.** We use spherical completeness in the sense of §2: every nonempty family of pairwise-meeting closed balls has a common point. For the field case of the following lemma, see Schneider [13, Lemmas 1.3 and 1.6].

**Lemma 67** (discrete distances). *Let  $Y$  be a complete nonarchimedean space whose distances lie in  $r^\mathbb{Z} \cup \{0\}$  for some  $r \in (0, 1)$ . Then  $Y$  is spherically complete.*

*Proof.* Write  $\mathcal{D}(a, \rho) = \{y \in Y : d(y, a) \leq \rho\}$ . In a nonarchimedean space, every point of a ball is a center. Consequently, whenever two closed balls meet, the one with the smaller radius lies in the other. The discrete distance assumption also gives

$$\mathcal{D}(a, \rho) = \mathcal{D}(a, \rho^b), \quad \rho^b := \max\{s \in r^\mathbb{Z} \cup \{0\} : s \leq \rho\}.$$

We therefore replace every radius by a value in  $r^\mathbb{Z} \cup \{0\}$ .

Let  $(\mathcal{D}(a_i, \rho_i))_{i \in I}$  be a nonempty family of pairwise meeting closed balls. If its radii have a least element  $\rho_{i_0}$ , then  $\mathcal{D}(a_{i_0}, \rho_{i_0})$  lies in every ball of the family, so  $a_{i_0}$  is a common point. Otherwise, discreteness implies that all the radii are positive and that their infimum is zero. Choose  $i_1, i_2, \dots$  so that  $\rho_{i_n}$  decreases strictly to zero. The corresponding balls are nested, and

$$d(a_{i_{n'}}, a_{i_n}) \leq \rho_{i_n} \quad (n' \geq n).$$

Their centers form a Cauchy sequence. By completeness it has a limit  $a$ , and closedness puts  $a$  in every ball of the sequence. For any  $i \in I$ , choose  $n$  with  $\rho_{i_n} \leq \rho_i$ . The  $n$ -th ball then lies in  $\mathcal{D}(a_i, \rho_i)$ , so  $a$  belongs to every ball of the original family.  $\square$

*Remark 68* (the hypothesis (K2) of [7]). The construction of [7] assumes that the base field satisfies the hypothesis (K2) there: there are closed balls  $\mathcal{D}(c_n, r_n)$  with  $|c_{n+1} - c_n| \leq r_n$ ,  $r_0 > r_1 > \dots$  and  $\bigcap_n \mathcal{D}(c_n, r_n) = \emptyset$ . A nonarchimedean field (indeed any nonarchimedean space) satisfies (K2) if and only if it is not spherically complete. Indeed, the balls in (K2) pairwise meet, since  $|c_n - c_m| \leq \max_{m \leq j < n} |c_{j+1} - c_j| \leq r_m$  for  $m \leq n$ . Conversely, let a family of pairwise meeting closed balls have no common point. A ball of least radius in the family would lie in all the others (as in the proof of Lemma 67), and its center would be a common point; so the radii have no least element, and every radius in the family exceeds their infimum. A sequence of balls from the family whose radii decrease strictly to that infimum is then nested, so it satisfies  $|c_{n+1} - c_n| \leq r_n$ , and it has empty intersection: a common point of the sequence would lie in every ball of the family, because every ball of the family contains a ball of the sequence. This remark is used only in §1 and §14.

**Lemma 69** (Laurent subfield). *Let  $K'$  be a complete nontrivially normed field of characteristic  $p$ , and let  $t \in K'$  with  $0 < |t| = r < 1$ .*

- (1)  *$K'$  is nonarchimedean, and  $|c| = 1$  for every  $c \in \mathbb{F}_p^\times$ .*
- (2) *For  $a_m, \dots, a_M \in \mathbb{F}_p$ , not all zero,  $|\sum_{i=m}^M a_i t^i| = r^{\min\{i: a_i \neq 0\}}$ .*
- (3) *Give  $\mathbb{F}_p((X))$  the absolute value with  $|X| = r$ . There is a continuous isometric ring homomorphism  $\text{ev}_t: \mathbb{F}_p((X)) \rightarrow K'$  such that  $\text{ev}_t(\sum_{i \geq m} a_i X^i) = \sum_{i \geq m} a_i t^i$ . Its image  $K_1$  is a closed subfield of  $K'$ ; it is the closure of  $\mathbb{F}_p(t)$ , and it is isometrically isomorphic to  $\mathbb{F}_p((X))$ .*
- (4)  *$K_1$  is spherically complete.*

*Proof.* (1) Lemma 6 makes  $K'$  nonarchimedean. For  $c \in \mathbb{F}_p^\times$ , the identity  $c^{p-1} = 1$  gives  $|c| = 1$ .

(2) The nonzero terms  $a_i t^i$  have pairwise distinct absolute values  $r^i$ . In a nonarchimedean field,  $|x + y| = |x|$  whenever  $|y| < |x|$ . Repeatedly applying this identity shows that the sum has the absolute value of its largest term, namely  $r^{\min\{i: a_i \neq 0\}}$ .

(3) Evaluation on Laurent polynomials defines a ring homomorphism  $\text{ev}_t: \mathbb{F}_p[X, X^{-1}] \rightarrow K'$ , and (2) shows that  $|\text{ev}_t(g)| = r^{\text{ord } g} = |g|$ . Laurent polynomials are dense in  $\mathbb{F}_p((X))$ , since truncations converge. Completeness of  $K'$  therefore gives a unique continuous extension to  $\mathbb{F}_p((X))$ . Passing to limits preserves the isometry and the ring operations. It also gives the stated evaluation formula, because the partial sums of  $\sum_{i \geq m} a_i X^i$  map to the partial sums of  $\sum_{i \geq m} a_i t^i$ .

The image is complete as the isometric image of a complete space, hence closed in  $K'$ . It is a subfield: for  $g \neq 0$ , the identity  $\text{ev}_t(g)\text{ev}_t(g^{-1}) = 1$  provides the inverse. This subfield contains  $\mathbb{F}_p(t)$ , while the Laurent polynomials  $\mathbb{F}_p[t, t^{-1}] \subseteq \mathbb{F}_p(t)$  are dense in it. The image is consequently exactly the closure of  $\mathbb{F}_p(t)$ .

(4) By (3),  $K_1$  is complete and has value group  $|K_1^\times| = r^{\mathbb{Z}}$ . It is nonarchimedean by (1), so its distances lie in  $r^{\mathbb{Z}} \cup \{0\}$ . Lemma 67 gives spherical completeness.  $\square$

**11.2. Ingleton's theorem and its consequences.** Throughout the rest of this section we fix fields  $K_1 \subseteq K'$  with the following properties.

- (H1)  $K_1$ , with the restricted absolute value, is nontrivially normed, complete and spherically complete.
- (H2)  $K'$  is nonarchimedean.

By multiplication,  $K'$  is a nonarchimedean normed  $K_1$ -space. In our application  $K' = \widehat{K}$  and  $K_1$  is the field of Lemma 69.

**Theorem 70** (Ingleton [4]; see also [11]). *Let  $K_1$  be a spherically complete nonarchimedean normed field,  $V$  a  $K_1$ -vector space with a nonarchimedean norm,  $V_0 \subseteq V$  a subspace,  $C \geq 0$ , and  $T_0: V_0 \rightarrow K_1$  linear with  $|T_0 v| \leq C\|v\|$  for  $v \in V_0$ . Then  $T_0$  extends to a linear map  $T: V \rightarrow K_1$  with  $|Tx| \leq C\|x\|$  for all  $x \in V$ .*

Schneider [13, Prop. 9.2] gives the more general formulation with a dominating nonarchimedean seminorm.

We will use Theorem 70 on the nonarchimedean  $K_1$ -space  $K'$  in two ways: to obtain the scalar retraction below, and to extend finite-dimensional coordinate functionals in the proof of positivity of the projective tensor norm. Both applications have codomain  $K_1$ . The spaces  $E_1$  and  $F_1$  introduced later need not be nonarchimedean.

**Corollary 71** (Ingleton projection). *There is a  $K_1$ -linear map  $\varpi: K' \rightarrow K_1$  with  $\varpi|_{K_1} = \text{id}$  and  $|\varpi(\lambda)| \leq |\lambda|$  for all  $\lambda \in K'$ .*

*Proof.* Theorem 70 with  $V = K'$ ,  $V_0 = K_1$ ,  $T_0 = \text{id}$  and  $C = 1$ ; its hypotheses hold by (H1) and (H2).  $\square$

**Lemma 72** (extension). *Let  $\mathbb{K} \in \{K_1, K'\}$  and let  $C \geq 0$ .*

- (1) *Let  $V_0$  be a dense  $\mathbb{K}$ -subspace of a normed  $\mathbb{K}$ -space  $V$ , let  $Y$  be a  $\mathbb{K}$ -Banach space and  $T_0: V_0 \rightarrow Y$  a  $\mathbb{K}$ -linear map with  $\|T_0 x\| \leq C\|x\|$ . Then  $T_0$  extends uniquely to a continuous map  $T: V \rightarrow Y$ , which is  $\mathbb{K}$ -linear with  $\|Tx\| \leq C\|x\|$ .*
- (2) *Let  $V_1, \dots, V_k$  be normed  $K'$ -spaces with dense subspaces  $V_i^0$ ,  $Y$  a  $K'$ -Banach space, and  $M: \prod_i V_i^0 \rightarrow Y$  a  $K'$ -multilinear map with  $\|M(d)\| \leq C \prod_i \|d_i\|$ . Then  $M$  extends uniquely to a continuous  $K'$ -multilinear map  $\overline{M}: \prod_i V_i \rightarrow Y$  with  $\|\overline{M}\| \leq C$ .*

*Proof.* (1) The bound makes  $T_0$  Lipschitz, hence uniformly continuous. Density of  $V_0$  and completeness of  $Y$  give a unique continuous extension. Linearity and the norm bound pass to limits by continuity of addition, scalar multiplication and the norm.

(2) The case  $k = 0$  is immediate. For  $k \geq 1$ , multilinearity gives the telescoping estimate

$$\|M(d) - M(d')\| \leq C \rho^{k-1} \sum_i \|d_i - d'_i\|$$

whenever  $\|d_i\|, \|d'_i\| \leq \rho$  for every  $i$ , with  $\rho > 0$ . For any  $v \in \prod_i V_i$ , choose tuples  $d^{(n)} \in \prod_i V_i^0$  converging to  $v$ . Their coordinates are uniformly bounded, so the estimate makes  $M(d^{(n)})$  Cauchy in  $Y$ . Define  $\overline{M}(v)$  to be its limit. The same estimate, applied to two approximating sequences, proves that this definition is independent of the choices and is continuous on bounded sets. Multilinearity and the product norm bound pass to limits. Finally, density of  $\prod_i V_i^0$  proves uniqueness.  $\square$

**11.3. Projective base change.** The next construction extends scalars while retaining the original norm and a contractive retraction whenever the original space is complete. The scalar retraction from Corollary 71 is the essential ingredient; the construction applies to arbitrary normed spaces.

For a normed  $K_1$ -space  $V$  let  $V_{K'} := V \otimes_{K_1} K'$ , a  $K'$ -vector space through the second factor, and put

$$\|u\|_\pi := \inf \left\{ \sum_j \|x_j\| |\lambda_j| : u = \sum_j x_j \otimes \lambda_j \right\},$$

the infimum over finite decompositions, with an ordinary sum. For a bounded  $K_1$ -linear  $\psi: K' \rightarrow K_1$  let  $T_\psi: V_{K'} \rightarrow V$  be the  $K_1$ -linear map with  $T_\psi(x \otimes \lambda) = \psi(\lambda)x$ ; it exists

because  $(x, \lambda) \mapsto \psi(\lambda)x$  is  $K_1$ -bilinear. Let  $\iota_V: V \rightarrow V_{K'}, x \mapsto x \otimes 1$ ; it is  $K_1$ -linear, since  $cx \otimes 1 = x \otimes c = c \cdot (x \otimes 1)$  for  $c \in K_1$ .

**Lemma 73** (projective base change). *Let  $V$  be a normed  $K_1$ -space; no nonarchimedean hypothesis is made on  $V$ .*

- (1)  $\|T_\psi u\| \leq \|\psi\| \|u\|_\pi$  for every bounded  $K_1$ -linear  $\psi: K' \rightarrow K_1$ .
- (2)  $\|\cdot\|_\pi$  is a  $K'$ -norm on  $V_{K'}$ .
- (3)  $\iota_V$  is isometric.
- (4) Let  $V \widehat{\otimes}_\pi K'$  be the completion of  $(V_{K'}, \|\cdot\|_\pi)$ . If  $V$  is complete, then  $T_\varpi$ , with  $\varpi$  as in Corollary 71, extends to a  $K_1$ -linear map  $\Pi_V: V \widehat{\otimes}_\pi K' \rightarrow V$  with  $\|\Pi_V\| \leq 1$  and  $\Pi_V \circ \iota_V = \text{id}_V$ .

*Proof.* (1) For every decomposition  $u = \sum_j x_j \otimes \lambda_j$ ,

$$\|T_\psi u\| = \left\| \sum_j \psi(\lambda_j) x_j \right\| \leq \sum_j \|\psi\| \|\lambda_j\| \|x_j\|.$$

Taking the infimum over decompositions proves the contraction estimate.

(2) Concatenating decompositions gives the triangle inequality. Multiplying their second factors by  $\mu \in K'$  gives  $\|\mu u\|_\pi \leq |\mu| \|u\|_\pi$ ; for  $\mu \neq 0$ , applying the same inequality with  $\mu^{-1}$  gives equality. The zero-scalar case is immediate.

It remains to prove positivity. Let  $u \neq 0$ . Choose a finite tensor representation and rewrite its second factors in a basis of their  $K_1$ -span. This gives

$$u = \sum_{j=1}^n x_j \otimes \lambda_j$$

with  $\lambda_1, \dots, \lambda_n$  linearly independent over  $K_1$ . Some  $x_{j_0}$  is nonzero, since otherwise  $u = 0$ . On the finite-dimensional subspace  $U := \text{span}_{K_1}\{\lambda_1, \dots, \lambda_n\} \subseteq K'$ , take the coordinate functional  $\psi_0(\lambda_j) = \delta_{jj_0}$ . It is bounded by some  $C \geq 0$  by Lemma 12, using completeness of  $K_1$  in (H1). Apply Theorem 70 to the nonarchimedean  $K_1$ -space  $K'$  to extend it to  $\psi: K' \rightarrow K_1$  with  $\|\psi\| \leq C$ . Then  $T_\psi u = x_{j_0}$ , so (1) gives

$$0 < \|x_{j_0}\| \leq C \|u\|_\pi.$$

Thus  $\|u\|_\pi > 0$ . This argument detects a nonzero vector coefficient by a functional on the *scalar* factor  $K'$ ; it requires no separating continuous dual on  $V$ .

(3) The elementary tensor representation gives  $\|x \otimes 1\|_\pi \leq \|x\|$ . For the reverse inequality, use  $\varpi(1) = 1$  and  $\|\varpi\| \leq 1$  in (1):

$$\|x\| = \|T_\varpi(x \otimes 1)\| \leq \|x \otimes 1\|_\pi.$$

(4) When  $V$  is complete, the contraction  $T_\varpi$  on the dense tensor subspace extends by Lemma 72(1), with  $\mathbb{K} = K_1$ , to a  $K_1$ -linear contraction  $\Pi_V$  with values in  $V$ . On the embedded copy of  $V$  it satisfies  $\Pi_V \iota_V x = T_\varpi(x \otimes 1) = \varpi(1)x = x$ .  $\square$

We now fix the spaces to which the descent argument will be applied. Let  $E_1$  be a normed  $K_1$ -space and  $F_1$  a  $K_1$ -Banach space, and put

$$E := E_1 \widehat{\otimes}_\pi K', \quad F := F_1 \widehat{\otimes}_\pi K',$$

the completions of  $((E_1)_{K'}, \|\cdot\|_\pi)$  and  $((F_1)_{K'}, \|\cdot\|_\pi)$ . They are  $K'$ -Banach spaces;  $\iota_E: E_1 \rightarrow E$  and  $\iota_F: F_1 \rightarrow F$  are  $K_1$ -linear isometries, and  $\Pi_F: F \rightarrow F_1$  is  $K_1$ -linear with  $\|\Pi_F\| \leq 1$  and  $\Pi_F \circ \iota_F = \text{id}_{F_1}$  (Lemma 73).

**Lemma 74** (operators). *For  $f \in \mathcal{L}_{K_1}(E_1, E_1)$  the map  $f \otimes \text{id}$  on  $(E_1)_{K'}$  is  $K'$ -linear with  $\|(f \otimes \text{id})u\|_\pi \leq \|f\| \|u\|_\pi$ . Its extension  $f_{K'} \in \mathcal{L}_{K'}(E, E)$  satisfies  $\|f_{K'}\| \leq \|f\|$  and  $f_{K'} \circ \iota_E = \iota_E \circ f$ , and  $f \mapsto f_{K'}$  is  $K_1$ -linear.*

*Proof.* For any decomposition  $u = \sum_j x_j \otimes \lambda_j$ , the image under  $f \otimes \text{id}$  has the decomposition  $\sum_j f(x_j) \otimes \lambda_j$ , whose cost is at most  $\|f\| \sum_j \|x_j\| \|\lambda_j\|$ . Taking the infimum proves the norm bound on the algebraic tensor product. Since  $E$  is complete, Lemma 72(1), with  $\mathbb{K} = K'$ , gives the continuous extension  $f_{K'}$  with the same bound. Its action on  $x \otimes 1$  proves the embedding identity. Additivity and  $K_1$ -homogeneity in  $f$  hold on elementary tensors, hence on their span, and then on  $E$  by continuity.  $\square$

**Lemma 75** (forms). *For  $m \in \text{Alt}_{K_1}^k(E_1; F_1)$  there is  $m_{K'} \in \text{Alt}_{K'}^k(E; F)$  with  $\|m_{K'}\| \leq \|m\|$  and*

$$m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = \iota_F(m(x_1, \dots, x_k)) \quad (x_i \in E_1),$$

*and  $m \mapsto m_{K'}$  is  $K_1$ -linear.*

*Proof.* We first define the form on algebraic tensors, then establish a bound independent of their decompositions, and finally pass to the completions. Fix a  $K_1$ -basis  $(\lambda_\beta)_{\beta \in \mathfrak{B}}$  of  $K'$ . Every  $u \in (E_1)_{K'}$  has a unique finite expansion  $u = \sum_\beta u^\beta \otimes \lambda_\beta$ , with  $u^\beta \in E_1$ , and each coordinate map  $u \mapsto u^\beta$  is  $K_1$ -linear. Define

$$m^{\text{alg}}(u_1, \dots, u_k) := \sum_{\beta_1, \dots, \beta_k} m(u_1^{\beta_1}, \dots, u_k^{\beta_k}) \otimes \lambda_{\beta_1} \cdots \lambda_{\beta_k} \in (F_1)_{K'}.$$

The sum is finite, since each input has finitely many nonzero coordinates.

- (i) Linearity of the coordinate maps and multilinearity of  $m$  make  $m^{\text{alg}}$   $K_1$ -multilinear.
- (ii) On elementary tensors, the definition gives

$$m^{\text{alg}}(x_1 \otimes \mu_1, \dots, x_k \otimes \mu_k) = m(x_1, \dots, x_k) \otimes \mu_1 \cdots \mu_k \quad (x_i \in E_1, \mu_i \in K').$$

Indeed, writing  $\mu_i = \sum_\beta c_{i\beta} \lambda_\beta$  with  $c_{i\beta} \in K_1$  gives  $x_i \otimes \mu_i = \sum_\beta (c_{i\beta} x_i) \otimes \lambda_\beta$ . Substitution in the defining sum yields

$$\sum_{\beta_1, \dots, \beta_k} \left( \prod_i c_{i\beta_i} \right) m(x) \otimes \prod_i \lambda_{\beta_i} = m(x) \otimes \prod_i \left( \sum_\beta c_{i\beta} \lambda_\beta \right),$$

which is the asserted formula.

(iii) The map  $m^{\text{alg}}$  is  $K'$ -multilinear. By additivity in each slot, it suffices to check  $K'$ -homogeneity when every argument is an elementary tensor; there it follows from (ii). The same formula determines the map on all tuples of algebraic tensors, so the initial basis choice does not affect it.

- (iv) For arbitrary decompositions  $u_i = \sum_j x_{ij} \otimes \mu_{ij}$ , multilinearity and (ii) give

$$m^{\text{alg}}(u) = \sum_{j_1, \dots, j_k} m(x_{1j_1}, \dots, x_{kj_k}) \otimes \prod_i \mu_{ij_i}.$$

The cost of this tensor decomposition is at most  $\|m\| \prod_i (\sum_j \|x_{ij}\| \|\mu_{ij}\|)$ . Taking the infimum separately over the decompositions in each slot proves

$$\|m^{\text{alg}}(u_1, \dots, u_k)\|_\pi \leq \|m\| \prod_i \|u_i\|_\pi.$$

(v) To prove strong alternation, suppose two distinct slots  $a, b$  both contain  $u = \sum_j x_j \otimes \mu_j$ . Use this *same* decomposition in both slots and expand all arguments into elementary tensors. Terms using the same index  $j$  in slots  $a$  and  $b$  vanish because  $m$  has a repeated argument. For

$j \neq j'$ , the term using indices  $(j, j')$  in these slots pairs with the term using  $(j', j)$ . Their scalar factors agree, while the values of  $m$  are negatives of one another, so the pair sums to zero. Thus  $m^{\text{alg}}$  vanishes whenever any two arguments coincide. Pair cancellation remains valid in characteristic 2; merely checking  $m^{\text{alg}}(u, \dots, u) = 0$  would not establish strong alternation for  $k \geq 3$ .

(vi) Regard  $m^{\text{alg}}$  as taking values in the completion  $F$ . By (iii), (iv) and Lemma 72(2), it extends to a continuous  $K'$ -multilinear map  $m_{K'}: E^k \rightarrow F$  with  $\|m_{K'}\| \leq \|m\|$ . Strong alternation also passes to the limit: if  $v_a = v_b$ , approximate this common vector by one sequence in  $(E_1)_{K'}$ , and approximate the other entries separately. Every approximating tuple has equal entries in slots  $a, b$ , so (v) and continuity give  $m_{K'}(v) = 0$ .

(vii) Taking every scalar factor equal to one in (ii) gives

$$m_{K'}(\iota_E x_1, \dots, \iota_E x_k) = m^{\text{alg}}(x_1 \otimes 1, \dots, x_k \otimes 1) = m(x) \otimes 1 = \iota_F(m(x)).$$

The defining formula is  $K_1$ -linear in  $m$ . These linear identities pass to the continuous extensions by uniqueness, proving  $K_1$ -linearity of  $m \mapsto m_{K'}$ .  $\square$

**Proposition 76** (descent of a lift). *Assume (H1) and (H2). Let  $E_1$  be a normed  $K_1$ -space,  $F_1$  a  $K_1$ -Banach space, and  $E, F$  as above. If  $Q_{E,E,F}^{K'}$  admits a bounded  $k$ -linear lift  $P$  over  $K'$ , then  $Q_{E_1,E_1,F_1}^{K_1}$  admits a bounded  $k$ -linear lift  $P_1$  over  $K_1$  with  $\|P_1\| \leq \|P\|$ . Equivalently, by Proposition 8 over  $K_1$  and over  $K'$ : if  $Q_{E_1,E_1,F_1}^{K_1}$  is analytic at no point, then  $Q_{E,E,F}^{K'}$  is analytic at no point either.*

*Proof.* Extend the operator and form inputs to  $K'$ , apply the given lift, and retract the value to  $F_1$ . Explicitly, for  $f_1, \dots, f_k \in \mathcal{L}_{K_1}(E_1, E_1)$ ,  $m \in \text{Alt}_{K_1}^k(E_1; F_1)$  and  $x \in E_1^k$ , set

$$P_1(f_1, \dots, f_k)(m)(x_1, \dots, x_k) := \Pi_F \left( P((f_1)_{K'}, \dots, (f_k)_{K'})(m_{K'})(\iota_E x_1, \dots, \iota_E x_k) \right).$$

We check the norm bound and then the diagonal identity.

For fixed  $f$  and  $m$ , the map  $P((f_i)_{K'})(m_{K'})$  is a continuous alternating  $K'$ -multilinear form on  $E$ . Precomposing each slot with the  $K_1$ -linear isometry  $\iota_E$  and postcomposing with  $\Pi_F$  gives a  $K_1$ -multilinear map on  $E_1$ , still vanishing whenever two arguments coincide. Lemmas 73–75 give

$$\|P_1(f)(m)\| \leq \|\Pi_F\| \|P\| \prod_i \|(f_i)_{K'}\| \|m_{K'}\| \leq \|P\| \prod_i \|f_i\| \|m\|.$$

In particular,  $P_1(f)(m) \in \text{Alt}_{K_1}^k(E_1; F_1)$ . The form extension is  $K_1$ -linear in  $m$ , the operator extension is  $K_1$ -linear in each  $f_i$ , and  $P$  is  $K'$ -multilinear with values in  $K'$ -linear operators. Consequently  $P_1$  is  $K_1$ -linear in  $m$  and  $K_1$ -multilinear in  $(f_1, \dots, f_k)$ . The displayed estimate proves boundedness at both levels and yields  $\|P_1\| \leq \|P\|$ .

On the diagonal, use  $P(g, \dots, g) = Q^{K'}(g)$  with  $g = f_{K'}$ , followed by the embedding identities in Lemmas 74 and 75 and the retraction identity  $\Pi_F \iota_F = \text{id}$ :

$$\begin{aligned} P_1(f, \dots, f)(m)(x) &= \Pi_F(m_{K'}(f_{K'} \iota_E x_1, \dots, f_{K'} \iota_E x_k)) \\ &= \Pi_F(m_{K'}(\iota_E f x_1, \dots, \iota_E f x_k)) \\ &= \Pi_F(\iota_F(m(f x_1, \dots, f x_k))) \\ &= m(f x_1, \dots, f x_k) = Q^{K_1}(f)(m)(x). \end{aligned} \quad \square$$

*Remark 77.* Proposition 76 is characteristic-free and uses no hypothesis on  $k$ . It transports negative results upward only, from  $K_1$  to  $K'$ . Ingleton's theorem is used only on  $K'$ , with no nonarchimedean norm hypothesis on  $F_1$ . The completeness of  $F_1$  is used essentially: it is what makes the retraction  $\Pi_F$  land in  $F_1$ .

**11.4. Incomplete base fields.** The preceding construction takes place over the completion  $\widehat{K}$  of the prescribed field. To return to  $K$ , we restrict scalars on the resulting Banach spaces. Density of  $K$  in  $\widehat{K}$  forces every continuous  $K$ -multilinear map between these spaces to be  $\widehat{K}$ -multilinear, so a lift cannot appear under this restriction.

**Lemma 78.** *Let  $K$  be a nontrivially normed field with completion  $\widehat{K}$ , and let  $E, E', F$  be  $\widehat{K}$ -Banach spaces, regarded as  $K$ -Banach spaces by restriction of scalars (with the same norms). Then  $\mathcal{L}_K(E, E') = \mathcal{L}_{\widehat{K}}(E, E')$ ,  $\text{Alt}_K^k(E; F) = \text{Alt}_{\widehat{K}}^k(E; F)$  and  $\text{Alt}_K^k(E'; F) = \text{Alt}_{\widehat{K}}^k(E'; F)$  with the same norms,  $Q_{E, E', F}^K = Q_{E, E', F}^{\widehat{K}}$ , and every bounded  $k$ -linear lift of  $Q_{E, E', F}^K$  over  $K$  is a bounded  $k$ -linear lift of  $Q_{E, E', F}^{\widehat{K}}$  over  $\widehat{K}$ .*

*Proof.* Let  $g$  be a continuous  $K$ -linear map between normed spaces that already carry compatible  $\widehat{K}$ -scalar actions. For  $\lambda \in \widehat{K}$ , choose  $\lambda_n \in K$  with  $\lambda_n \rightarrow \lambda$ . Since  $\|\lambda_n x - \lambda x\| = |\lambda_n - \lambda| \|x\|$ , continuity gives

$$g(\lambda x) = \lim_n g(\lambda_n x) = \lim_n \lambda_n g(x) = \lambda g(x).$$

Thus  $g$  is  $\widehat{K}$ -linear. Applying the same argument with all but one slot fixed shows that every continuous  $K$ -multilinear map is  $\widehat{K}$ -multilinear. Alternation is the same equal-input vanishing condition over either field.

The spaces of bounded maps therefore coincide as sets. Their operator norms agree as well, since the norms are defined by the same pointwise bounds on the same vectors. They are  $\widehat{K}$ -Banach spaces, and  $Q_{E, E', F}^K$  and  $Q_{E, E', F}^{\widehat{K}}$  agree by their defining precomposition formula. Finally, apply the slotwise density argument once more to a bounded  $k$ -linear lift  $P$  between these map spaces. It becomes  $\widehat{K}$ -multilinear with the same norm and the same diagonal identity, and hence is a lift over  $\widehat{K}$ .  $\square$

## 12. THE SHARP BANACH COUNTEREXAMPLE AND CLASSIFICATION

The preceding sections give the obstruction to a bounded lift. We first establish the remaining property of the target, then assemble the proof of Theorem 1 and deduce Corollary 2.

### 12.1. The target has no equivalent nonarchimedean norm.

**Proposition 79.** *Let  $\kappa$  be any field,  $k \geq 2$ ,  $r \in (0, 1)$ , and let  $K_1 = \kappa((X))$ ,  $E_1$ ,  $E_0$  and  $B$  be as in §10. Let  $e_i \in E_0$  be the unit vectors and  $\omega_j := e_{kj+1} \wedge \cdots \wedge e_{kj+k} \in B$  ( $j \geq 0$ ), wedges on pairwise disjoint blocks of indices. Then  $\|\omega_j\|_B = 1$  and*

$$\frac{N}{M_r} \leq \left\| \sum_{j < N} \omega_j \right\|_B \leq N \quad (N \geq 1).$$

*Hence the norm of  $B$  is not nonarchimedean, and  $B$  admits no equivalent nonarchimedean norm. The same holds for  $F = B \widehat{\otimes}_\pi K'$  as in §11, for any  $K' \supseteq K_1$  satisfying (H1) and (H2), regarded as a normed space over  $K'$  or over any subfield of  $K'$ .*

*Proof.* Each  $\omega_j$  is a wedge of unit vectors, so  $\|\omega_j\|_B \leq 1$  by the defining projective norm. Evaluation of its determinant array at the block indices gives  $\det I = 1$ , hence  $\|\omega_j\|_B \geq \|J\omega_j\|_\infty \geq 1$ . The triangle inequality now gives the upper bound  $N$ .

For the lower bound, write  $\xi_N$  for the sum of the same block wedges in  $\Lambda_\kappa^k E_0$ , and put  $\omega := \sum_{j < N} \omega_j \in B$ . Its array  $J\omega$  is the constant Laurent array  $\Omega^\kappa(\xi_N)$ . Thus  $\text{coeff}_0(J\omega) = \Omega^\kappa(\xi_N)$ , and uniqueness in Proposition 63 gives  $\eta(\omega) = \xi_N$ .

We apply Lemma 50 to  $\xi_N$ . Fix one of the first  $N$  blocks and an index  $i_0$  in it. Take for  $\varphi$  the  $k - 1$  coordinate functionals corresponding to the other indices of that block. On every other block all these functionals vanish, so its contraction is zero; here  $k \geq 2$  ensures that the minors have positive size. On the chosen block, the coefficient of  $e_{i_0}$  is  $\pm 1$ , while every other cofactor contains the zero column obtained by evaluating the functionals at  $e_{i_0}$ . Consequently  $c_\varphi(\xi_N) = \pm e_{i_0}$ . These contractions span all  $kN$  unit vectors in the first  $N$  blocks, so

$$kN \leq \text{sdim}(\xi_N) = \text{sdim}(\eta(\omega)) \leq kM_r \|\omega\|_B$$

by (10.2). This proves the lower bound.

If  $\|\cdot\|_u$  were an equivalent nonarchimedean norm, choose positive constants  $c, c'$  with  $\|\cdot\|_B \leq c\|\cdot\|_u \leq c'\|\cdot\|_B$ . Then

$$\frac{N}{M_r} \leq \left\| \sum_{j < N} \omega_j \right\|_B \leq c \max_{j < N} \|\omega_j\|_u \leq c'$$

for every  $N \geq 1$ , a contradiction. In particular, the original norm is not nonarchimedean. This argument uses only the nonarchimedean inequality for  $\|\cdot\|_u$ , not its scalar homogeneity.

Finally,  $\iota_F: B \rightarrow F$  is a  $K_1$ -linear isometry by Lemma 73(3). The images  $\iota_F \omega_j$  satisfy the same unit-norm and sum-growth estimates, so the argument applies to  $F$  as well. Restriction to a subfield leaves these estimates unchanged.  $\square$

If  $r \leq \frac{1}{2}$ , then  $M_r = 1$  and  $\|\sum_{j < N} \omega_j\|_B = N$  exactly.

## 12.2. Proof of Theorem 1.

*Proof.* Let  $K$  be a nontrivially normed field of characteristic  $p > 0$ , and let  $k \geq p$ .

- (1) *Choose a Laurent subfield.* By Lemma 6,  $K$  is nonarchimedean. Its completion  $\widehat{K}$  is a complete, nontrivially normed, nonarchimedean field of characteristic  $p$ . Choose  $t \in K$  with  $0 < |t| = r < 1$ , and let  $K_1$  be the closure of  $\mathbb{F}_p(t)$  in  $\widehat{K}$ . Lemma 69 identifies  $K_1$  isometrically with  $\mathbb{F}_p((X))$ , with  $|X| = r$ , by  $\text{ev}_t$ . In particular,  $K_1$  is complete and spherically complete.
- (2) *Obtain the obstruction over  $K_1$ .* Put  $E_1 := \ell^\infty(\mathbb{N}, K_1)$ , and let  $B$  be the completion of  $\Lambda_{K_1}^k E_1$  under the projective exterior norm. Transport of scalars along  $\text{ev}_t$  preserves the norms, exterior powers and completions in this construction. It also identifies the operator and alternating-map spaces, their precomposition maps, and bounded lifts. Theorem 65, with  $\kappa = \mathbb{F}_p$ , therefore shows that  $Q_{E_1, E_1, B}^{K_1}$  has no bounded  $k$ -linear lift.
- (3) *Ascend to  $\widehat{K}$ .* The inclusion  $K_1 \subseteq \widehat{K}$  satisfies (H1) and (H2), and  $B$  is complete. Proposition 76, with  $K' = \widehat{K}$ , gives Banach spaces

$$E := E_1 \widehat{\otimes}_\pi \widehat{K}, \quad F := B \widehat{\otimes}_\pi \widehat{K}$$

for which  $Q_{E, E, F}^{\widehat{K}}$  has no bounded  $k$ -linear lift.

- (4) *Descend to  $K$ .* Regard the same  $E$  and  $F$  as  $K$ -Banach spaces. By Lemma 78, a bounded lift of  $Q_{E, E, F}^{\widehat{K}}$  would extend to a bounded lift of  $Q_{E, E, F}^{\widehat{K}}$ , which is impossible. Proposition 8 applies over the possibly incomplete field  $K$ :  $Q_{E, E, F}^{\widehat{K}}$  is analytic at no point, and  $\text{ContDiffAt } K \ \omega$  fails at every point. Reindexing as in §3 gives the same conclusions, with the same spaces  $E, F$ , for every finite index set with  $k$  elements.
- (5) *Exclude an equivalent nonarchimedean target norm.* Since  $k \geq p \geq 2$ , Proposition 79 applies to  $B$  and  $F$ , after the scalar identification in step (2). Its conclusion persists when  $F$  is regarded over  $K$ .

$\square$

*Remark 80.* If  $K_1 = \widehat{K}$ , as when  $K = \mathbb{F}_p((u))$  and  $t = u$ , the completed tensor products identify isometrically with  $E_1$  and  $B$ : use Lemma 73(3) and  $x \otimes \lambda = \lambda x \otimes 1$ . Thus  $E$  is nonarchimedean in this case. For  $K = \mathbb{F}_q((u))$ ,  $q = p^n$ , with its  $u$ -adic absolute value, one can instead apply Theorem 65 directly with  $\kappa = \mathbb{F}_q$  and  $X = u$ , obtaining  $E = \ell^\infty(\mathbb{N}, K)$  and  $F = B$ . The proof over a general base does not require, or assert, that  $E$  is nonarchimedean.

### 12.3. Proof of Corollary 2.

*Proof.* If  $k! \neq 0$  in  $K$ , Proposition 10 gives a finite power series at every point for arbitrary normed  $E, E', F$ , including degree zero. If  $k! = 0$  in  $K$ , the field has positive prime characteristic  $p \leq k$ . Theorem 1 gives the required Banach counterexample. Together these prove the classification and both strengthened conclusions.  $\square$

## 13. COMPLEMENTS AND OPEN PROBLEMS

We apply the Main Theorem to finite-dimensional targets and to targets over discretely valued fields, then give an explicit lift from an orthogonal basis. The section ends with comparisons and questions about nonarchimedean targets.

### Finite-dimensional targets.

*Remark 81* (finite-dimensional targets). Let  $K$  be a complete nontrivially normed field, let  $E, E'$  be normed  $K$ -spaces, and let  $F$  be finite-dimensional. If  $F \neq 0$ , then, at every  $f_0 \in \mathcal{L}(E, E')$ ,

$$Q_{E, E', F}^K \text{ is analytic at } f_0 \iff Q_{E, E', K}^K \text{ is analytic at } f_0.$$

If  $F = 0$ , precomposition is constant and hence analytic.

Indeed, choose a basis  $v_1, \dots, v_d$  of  $F$  with coordinate functionals  $\lambda_1, \dots, \lambda_d$ . These functionals are continuous because  $K$  is complete. For a vector  $v$  and a scalar-valued form  $a$ , write  $va$  for the form  $x \mapsto a(x)v$ . Writing  $Q_F := Q_{E, E', F}^K$  and  $Q_K := Q_{E, E', K}^K$ , we have

$$Q_F(f)(m) = \sum_{a=1}^d v_a Q_K(f)(\lambda_a \circ m).$$

The right-hand side is a fixed continuous linear function of the operator  $Q_K(f)$ , so analyticity of  $Q_K$  implies that of  $Q_F$ . Conversely, choose  $v \in F$  and  $\lambda \in \mathcal{L}(F, K)$  with  $\lambda(v) = 1$ . For  $a \in \text{Alt}^k(E'; K)$ ,

$$Q_K(f)(a) = \lambda \circ Q_F(f)(va).$$

This expresses  $Q_K$  as a fixed continuous linear function of  $Q_F$  and proves the converse. Thus finite-dimensionality of the target reduces the question to a scalar target. This reduction gives no assertion of analyticity over a nonspherically complete base.

If  $K$  is nonarchimedean and spherically complete, there is a positive conclusion for every finite-dimensional  $F$ , with any norm. The coordinate norm  $\|y\|_\infty := \max_a |\lambda_a(y)|$  is equivalent to the given norm. With this norm,  $F$  is spherically complete: a closed ball in  $K^d$  is a product of coordinate balls, and spherical completeness of  $K$  gives a point in each coordinate intersection. The Main Theorem therefore applies. Returning to the original norm preserves analyticity, since it changes the norms on the alternating-map and operator spaces only by fixed comparison factors.

### Targets over discretely valued bases.

**Proposition 82** (targets over discretely valued bases). *Let  $K$  be a nonarchimedean normed field with  $|K^\times| = r^\mathbb{Z}$  for some  $r \in (0, 1)$ , complete or not; for instance  $K = \mathbb{F}_q((t))$  or  $K = \kappa((t))$ . Let  $E, E'$  be normed  $K$ -spaces and  $F$  a  $K$ -Banach space that admits an equivalent nonarchimedean norm. Then  $Q_{E, E', F}^K$  is analytic at every point.*

*Proof. Round the norm.* Let  $\|\cdot\|_u$  be an equivalent nonarchimedean norm on  $F$ . For  $y \neq 0$ , round  $\|y\|_u$  up to the least element  $\|y\|_u^\sharp$  of  $r^\mathbb{Z}$  above it, and put  $\|0\|_u^\sharp := 0$ . Then

$$\|y\|_u \leq \|y\|_u^\sharp \leq r^{-1}\|y\|_u.$$

Rounding is monotone and commutes with multiplication by elements of  $r^\mathbb{Z}$ . Since  $|\lambda| \in r^\mathbb{Z}$  for  $\lambda \in K^\times$ , it preserves scalar homogeneity and the nonarchimedean inequality. Thus  $\|\cdot\|_u^\sharp$  is an equivalent nonarchimedean norm, and  $F' := (F, \|\cdot\|_u^\sharp)$  is complete with distances in  $r^\mathbb{Z} \cup \{0\}$ .

*Apply the Main Theorem.* Lemma 67 makes  $F'$  spherically complete. Part (1) in characteristic zero, and part (2) in positive characteristic, therefore give analyticity of  $Q' := Q_{E, E', F'}^K$  at every point.

*Return to the original norm.* Equivalent target norms give the same spaces of bounded alternating maps, with equivalent operator norms. Consequently

$$\mathcal{L}(\text{Alt}^k(E'; F), \text{Alt}^k(E; F)) \quad \text{and} \quad \mathcal{L}(\text{Alt}^k(E'; F'), \text{Alt}^k(E; F'))$$

are the same vector space with equivalent norms, and  $Q = Q'$  under this identification. A power series for  $Q'$  is therefore a power series for  $Q$ : coefficient norms change by at most a fixed factor, and convergence is unchanged.  $\square$

Thus, over a discretely valued base, every Banach counterexample must have a target admitting no equivalent nonarchimedean norm. Proposition 79 establishes this property for our target over every base.

*Remark 83* (a nonspherical target with analytic precomposition). Spherical completeness of the given target norm is not necessary for analyticity, even in degrees at least the characteristic. Let  $K = \mathbb{F}_p((t))$  with  $|t| = 1/2$ , and let  $F = c_0(\mathbb{N}, K)$ , the space of sequences tending to zero. Put

$$w_j = 1 + \frac{1}{j+2}, \quad \|x\|_w = \sup_{j \geq 0} w_j |x_j|.$$

This is a nonarchimedean norm, and  $\|x\|_\infty \leq \|x\|_w \leq \frac{3}{2}\|x\|_\infty$ . Thus  $F$  is Banach in either norm. The usual supremum norm has nonzero values in  $2^\mathbb{Z}$ , since every nonzero null sequence attains its maximum absolute value. Lemma 67 makes it spherically complete. Main Theorem (2) therefore gives analytic precomposition in every finite degree with the supremum norm; the change of norm argument in Proposition 82 gives the same conclusion for  $\|\cdot\|_w$ , for all normed  $E, E'$ .

Nevertheless, the weighted norm is not spherically complete. Write  $e_j$  for the coordinate vectors and  $a_n = \sum_{j < n} e_j$ . The closed balls  $B_n = \overline{B}_w(a_n, w_n)$  are nested:  $w_{n+1} < w_n$  and  $\|a_{n+1} - a_n\|_w = w_n$ . If  $x$  belonged to every  $B_n$ , then, for each  $j$  and any  $n > j$ ,

$$|x_j - 1| \leq \frac{w_n}{w_j} < 1,$$

so  $|x_j| = 1$  for every  $j$ , contrary to  $x_j \rightarrow 0$ . Hence  $\bigcap_n B_n = \emptyset$ . This example separates spherical completeness of a particular norm from analyticity, which is invariant under equivalent norms. It still admits an equivalent nonarchimedean, spherically complete norm.

*Remark 84* (necessary, but not sufficient). Completeness of the target is essential in Proposition 82: it is used through Lemma 67. Even for complete targets, the absence of an equivalent nonarchimedean norm is not sufficient for a counterexample. Take the degree-3 space  $B$  of §10.2 over  $K_1 = \mathbb{F}_5((X))$ . It admits no equivalent nonarchimedean norm by Proposition 79, which requires only  $k \geq 2$ , but  $Q_{E_1, E_1, B}^{K_1}$  is analytic by Corollary 2(1), since  $3! = 1$  in  $\mathbb{F}_5$ .

The factorial hypothesis is also essential to the finite-field obstruction. Applying the construction of §10.4 to this bounded lift gives a map  $\Psi$  with all the properties of Lemma 64, which is valid in every degree. Thus  $\Psi$  satisfies every hypothesis of Theorem 55(2), with  $L = \mathbb{F}_5$ , except  $k! = 0$ ; compare Remark 61.

### An explicit lift from an orthogonal basis.

*Remark 85* (an explicit lift). Let  $K$  be nonarchimedean, let  $E$  have an orthogonal basis  $(e_i)_{i \in I}$ , and let  $F$  be complete and nonarchimedean. Here a basis means that every  $x \in E$  has an unconditional expansion  $x = \sum_i e_i^*(x)e_i$ , with  $\|\sum_i c_i e_i\| = \sup_i |c_i| \|e_i\|$ . Fix a total order on  $I$  and put

$$P(f_1, \dots, f_k)(m)(x^1, \dots, x^k) := \sum_{s_1 < \dots < s_k} \det(e_{s_a}^*(x^b))_{a,b} m(f_1 e_{s_1}, \dots, f_k e_{s_k}).$$

For each  $x$ , the weighted coordinates  $|e_i^*(x)| \|e_i\|$  tend to zero off finite subsets of  $I$ . Expanding the determinant bounds a summand by a fixed constant times a finite sum of products of these null families, one coordinate from each input. Such products tend to zero off finite subsets of the tuple space. Hence, for every  $\varepsilon > 0$ , only finitely many summands have norm at least  $\varepsilon$ , and the family is summable in the complete nonarchimedean space  $F$ .

Orthogonality gives  $|e_s^*(x)| \|e_s\| \leq \|x\|$ . The nonarchimedean determinant estimate therefore bounds every summand by  $\|m\| \prod_a \|f_a\| \prod_b \|x^b\|$ ; the nonarchimedean inequality in  $F$  then gives  $\|P\| \leq 1$ . The formula is multilinear in  $f$  and linear in  $m$ , and its output is alternating in  $x$  because the determinant is alternating in its columns. Finally, finite coordinate expansions give

$$m(fx^1, \dots, fx^k) = \sum_{s_1 < \dots < s_k} \det(e_{s_a}^*(x^b))_{a,b} m(fe_{s_1}, \dots, fe_{s_k});$$

the general identity follows by continuity along finite coordinate projections. Thus  $P(f, \dots, f) = Q(f)$ . For  $k = 0$ , the formula has one empty-index term, with determinant 1.

This *sorted lift* works in every characteristic and degree, without division by  $k!$  or spherical completeness. It is not used in the main proof. Over a complete discretely valued field, every nonarchimedean Banach space is topologically isomorphic to a space  $c_0(I)$  of null families [10, §1, Prop. 1 and its Corollaire] (see also [13, Prop. 10.1]), and hence has such a basis for an equivalent norm.

### Nonarchimedean counterexamples.

Comparison with characteristic 2. For  $p = k = 2$ , Theorem 1 gives a second counterexample over every nontrivially normed field of characteristic 2. Unlike the construction of [7], it applies to spherically complete bases such as  $\mathbb{F}_2((t))$ . The distinction lies in the target: that of [7] is nonarchimedean, whereas ours admits no equivalent nonarchimedean norm. Proposition 82, using Main Theorem (2), shows that a complete counterexample target over  $\mathbb{F}_2((t))$  cannot carry an equivalent nonarchimedean norm.

The proofs also use different mechanisms. The characteristic-2 argument combines a parity identity for determinants—the determinant of an alternating matrix is the square of its Pfaffian—with the failure of Hahn–Banach over a base that is not spherically complete, using  $\ell^\infty$  to assemble the construction. Here the finite-field theorem is combined with the constant-coefficient

map of a Laurent series field, and Ingleton's theorem is applied over a spherically complete subfield. In characteristic 2, Theorem 55 needs no signs (Remark 60).

**Open problems.** The distinction between restrictions on objects and restrictions on analytic families leaves several different classification problems. They are substantive only in characteristic  $p > 0$  and degree  $k \geq p$ ; when  $k! \neq 0$ , the alternating-map bifunctor is already analytic on the whole product category. For  $C^\infty$  smoothness, no restriction is needed in any characteristic.

- (1) *Intrinsic characterization of universal targets.* Determine  $T_k(K)$  in terms of the topological linear structure of its objects, rather than by restating analyticity or the multilinear lift criterion. In particular, characterize its Banach objects and the targets that belong to  $T_k(K)$  in every finite degree. An equivalent nonarchimedean, spherically complete norm is sufficient, by Main Theorem (2) and bounded-isomorphism invariance. Remark 83 shows that spherical completeness of the given norm is not necessary; it does not decide whether such an equivalent norm is necessary. Proposition 33 shows that this target classification determines a canonical analytic domain compatible with every other analytic domain.
- (2) *Analytic domains over complete fields and among Banach pairs.* For a prescribed complete field  $K$ , does there exist a largest full analytic domain, closed under isomorphisms, in  $\text{Vec}_K^{\text{op}} \times \text{Vec}_K$ ? Ask the same question in  $\text{Ban}_K^{\text{op}} \times \text{Ban}_K$ , also when  $K$  itself is incomplete. Theorem 37 answers the unrestricted existence question negatively over  $\mathbb{F}_p(t)$ , using finite-dimensional but incomplete spaces. Its construction does not decide either of these restricted questions. Where no largest domain exists, seek structural criteria for compatibility between objects and descriptions of full domains maximal under inclusion.
- (3) *Extensions of the split-pair category.* Which objects can be adjoined to the full category of pairs for which  $\text{Alt}^k(E; F) \hookrightarrow \text{Mult}^k(E; F)$  has a bounded linear retraction? The universal-target core can always be adjoined, by Proposition 33. The objects  $X_k$  of Theorem 37, despite having analytic self-action, cannot be adjoined. A useful answer should give sufficient and necessary conditions on the spaces themselves, rather than merely requiring analyticity of all the cross-actions.
- (4) *Parameter spaces for analytic families.* Characterize normed parameter spaces  $P$  for which every operator-valued analytic family with parameters in an open subset of  $P$  induces an analytic family on alternating-map spaces. The hypotheses on the target fibers should be recorded explicitly: completeness and nonarchimedean norms enter some of the positive results. General reflection of analyticity through closed linear embeddings is a sufficient condition, but it is stronger than the property needed for this particular finite-degree construction. The ordinary  $\ell^1$  parameter theorem makes this distinction concrete.
- (5) *Countable index.* Let  $K$  be complete, not spherically complete, of characteristic  $p \geq 3$ , and let  $k \geq p$ . Is there a complete nonarchimedean  $F$  such that  $Q_{E,E,F}^K$  is not analytic for  $E = \ell^\infty(\mathbb{N}, K)$ ? For  $p = 2$ , the answer is yes: degree two is the theorem of [7], and the higher degrees follow by the wedging argument of [7, Remark 18], formalized under hypothesis (K2) of that paper. Remark 86 proposes a counterexample with a larger index set; that extension and the countable case remain open.
- (6) *Densely valued spherically complete bases.* Let  $K$  be spherically complete, densely valued and of characteristic  $p$ , for example  $\mathbb{F}_p((t^\mathbb{Q}))$ , and let  $k \geq p$ . Is  $Q_{E,E',F}^K$  analytic for all Banach  $E, E'$  and every complete nonarchimedean  $F$ ? Main Theorem (2) gives a positive answer for spherically complete  $F$ . The counterexample of Theorem 1 has

a target with no equivalent nonarchimedean norm and therefore does not answer this question.

### Further approaches.

*Remark 86* (nonarchimedean targets over bases that are not spherically complete; proposed extension). The following statement is a proposed extension; it has not been proved here and is not used elsewhere in this paper. Let  $k \geq 2$ , and let  $K$  be a complete nontrivially normed field, not spherically complete, with  $k! = 0$  in  $K$ . Then  $K$  is nonarchimedean by Lemma 6. For a cardinal  $\mu$  put  $\beth_0(\mu) := \mu$  and  $\beth_{n+1}(\mu) := 2^{\beth_n(\mu)}$ , and let  $\mu^+$  be the successor cardinal of  $\mu$ . Let  $I$  be a set of cardinality at least  $\beth_{3k-1}(|K|)^+$ , put  $E := \ell^\infty(I, K)$ , and let  $F_I$  be the closure in  $\ell^\infty(I^k, K)$  of the span of the determinant arrays  $(i_1, \dots, i_k) \mapsto \det(x_b(i_a))_{a,b}$ ,  $x \in E^k$ . Then  $E$  and  $F_I$  are nonarchimedean  $K$ -Banach spaces, and  $Q_{E,E,F_I}^K$  is analytic at no point. The proposed proof uses a partition argument with the Erdős–Rado theorem in place of Ramsey’s theorem. For  $k = 2$  in characteristic 2 the countable index set  $\mathbb{N}$  suffices [7].

An alternative algebraic approach.

*Remark 87* (another possible approach). The following optional sketch suggests an alternative route to the input needed in §10. It requires an additional infinite-field obstruction, which is not proved here. The contraction-rank identity needed for the base-change step is Proposition 51. No other argument in this paper depends on the proposed route.

*The infinite-field input.* The first required statement is the analogue of Theorem 55 over infinite fields  $L$  with  $k! = 0$  in  $L$ , for lifts that satisfy (Pw) and have uniformly bounded support, of an arbitrary linear family  $D: A \rightarrow \text{End}_L(V)$  whose image modulo the finite-rank operators is infinite-dimensional.

*Base change and minors.* Let  $\Psi$  be a lift over  $\mathbb{F}_q$  whose values have support dimension at most  $d$ . After the finite base change  $\mathbb{F}_q \subseteq \mathbb{F}_{q^N}$ , the values of the base change of  $\Psi$  at  $\mathbb{F}_{q^N}$ -points have support dimension at most  $d_N := N^{2k}d$ , since each input has only  $N$  pieces over  $\mathbb{F}_q$ ; choose  $N$  with  $q^N > d_N + 1$ . By Proposition 51, support dimension equals the rank of a finite contraction matrix. Now consider any finite set of independent scalar parameters in each of the  $2k$  input slots. The resulting polynomial family has finitely many exterior-vector coefficients; the sum of their supports is one finite-dimensional space supporting every value, over every scalar extension. Choose a basis of that space to form the contraction matrix. By multilinearity, each matrix entry has degree at most one in each scalar parameter. Its  $(d_N + 1)$ -minors therefore have degree at most  $d_N + 1 < q^N$  in each variable. They vanish at every  $\mathbb{F}_{q^N}$ -point, so they are zero polynomials and remain zero over the algebraic closure  $\overline{\mathbb{F}}_q$ . The resulting support bound would allow the proposed obstruction to be applied over this infinite field.

*Compatibility requirements.* In addition to the proposed infinite-field obstruction, the route requires the following properties of  $\Psi$ .

- (1) The full identity (Pol), not only (Pol1): for  $k > q$  the identity (Pw) at  $\mathbb{F}_q$ -points does not survive base change (Proposition 54(3)), whereas (Pol) gives (Pw) at every point after base change.
- (2) Alternation, as required by the unproved input above. Both alternation and antisymmetry survive scalar extension, but in characteristic 2 antisymmetry does not imply alternation.
- (3) An infinite-dimensional image modulo finite rank. For the multipliers  $D_a x = ax$  on  $L^{\mathbb{N}}$  this holds because the multipliers by the indicators of pairwise disjoint infinite sets are linearly independent modulo finite rank.

The lift of §10 satisfies (Pol) and is alternating (Lemma 64), and its multiplier family is the one in (3), with  $L = \kappa$ .

#### 14. FORMALIZATION

The original operator proof snapshot is recorded in the private repository [12], at commit 5e9c3cf, using Lean 4.34.0-rc2 and Mathlib revision 2b73d9821d29. At that revision, the proof library was built without admitted proofs, and its transitive axiom check reported only `propext`, `Classical.choice` and `Quot.sound`. The repository’s `THEOREM_MAP.md` maps the paper’s statements to the Lean declarations.

The principal declarations, in the namespace `AlternatingAnalytic`, are

- `exists_banach_counterexample_full`, for Theorem 1, including incomplete base fields, the same Banach witnesses for every finite index type of cardinality  $k$ , no bounded lift, nowhere analyticity, nowhere  $C^\omega$ , and no equivalent nonarchimedean norm on the target;
- `factorial_ne_zero_iff_allBanachPrecompositionAnalytic`, for both directions of Corollary 2;
- `finiteField_multiplier_theorem`, for both parts of Theorem 55, with the support bound required only on the stated finitely supported test inputs in its first part.

The supporting development covers the  $\kappa$ -linear coefficient map and support estimate (Proposition 63), all four properties of the residue lift (Lemma 64), the Laurent counterexample, and ascent and descent to a prescribed field. It also includes the lift criterion, the sharp polarization distinctions, the operator form of Main Theorem (2), Proposition 82, and the sorted lift of Remark 85. Spherical completeness uses the development’s own class, with the definition in §2. The formalization used Codex with Autoform and incorporated previously formalized ingredients with their provenance retained.

Proposition 16 reproduces the argument of [8], whose formal proof is included in the pinned Mathlib dependency. The finite-coordinate proof (Proposition 11 and Corollary 13) and the finite-target reduction (Remark 81) are supplementary informal results; they are not claimed as separate formalized declarations in this development. The weighted  $c_0$  example of Remark 83 is likewise an additional informal corollary; its sufficient analyticity criterion is formalized, but the example is not a separate Lean declaration. Parts (1) and (2) of the categorical Main Theorem follow mathematically from these operator results and bounded bilinear postcomposition. The categorical packaging itself is not a separate Lean declaration.

The extension formalization covers Lemma 19, Theorem 20 and Corollary 22, including point-wise, open-domain and supplied continuous-coordinate versions. The family proof constructs the ambient  $(k + 1)$ -linear representative with norm at most one; it retains every characteristic and degree and arbitrary normed fibers, without completeness assumptions. Remark 21 is also formalized: in maximum-norm coordinates, a nonarchimedean ambient norm permits reflection with equal diagonals, no increase in coefficient norms and no loss of the represented ball, including infinite radius.

The checked development further includes the joint action and its functorial laws, the core admissible-family laws and graph characterization of Proposition 18, Proposition 24 with its Banach counterexample at the identity, and the canonical-support identity of Proposition 51. The degree-zero and degree-one support cases and finite-ambient contraction-matrix formula are included. These proofs have passed Lean verification without admitted proofs. Statement-review results and declaration-level scope are recorded in the repository theorem map.

The family assertion of Main Theorem (3) is thus formalized; its vector-bundle construction and operator-valued bundle morphisms remain pending, as does Main Theorem (4). Other

pending results include the  $c_0$  and ordinary  $\ell^1$  parameter theorems, the tensor criterion and approximation example, quotient parameters, universal targets and maximal domains, the shear's bundle interpretation, and contraction-matrix transport from a finite supporting subspace into an arbitrary ambient space. The theorem map retains the complete inventory. These additions do not alter the three existing comparator challenge statements.

The characteristic-2 counterexample [7] has separate provenance. Its formalization proves non-analyticity in degree 2 at the origin, with a wedging extension under (K2) to every degree at least 2 and every base point: `Round24Transfer.char2_not_analyticAt_of_hasEmptyNestedBalls`. That construction is not re-proved or re-audited here.

The proposed alternative route of Remark 87, the proposed large-index construction of Remark 86, the open problems, and Serre's external classification cited in Remark 85 are excluded from the proved results. The repository's comparator challenge contains two short contradiction corollaries of Theorem 1 and the operator form of Main Theorem (2) for every regularity order. Their solutions use the corresponding library theorems. The Banach and spherical-completeness hypotheses are given by transparent definitions in Mathlib terms. Intentional placeholders occur only in the challenge file, not in the proof library or solution. Current challenge checks use Kim Morrison's comparator, with the configuration recorded in the repository.

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